20261026 Approximate Continuum Ratio between Multiple Pointings

Although i dont have the pointing information, we use the `VARIANCE` maps at different wavelengths to roughly segment each galaxy into miltiple pointings. This works well if the pointing pattern is clear but fails when pointings are messy like NGC4264, NGC4321 and NGC4383. Anyway, then i take the continuum flux ratios of these pairs of neighouring pointings to inspect if we still have some large scale sine-like variations along wavelength, so that we may increase `MDEG` in `ppxf` fitting to mitigate the pointing issue in our final mass-to-light ratio/stellar mass map.

So here i show how i determine the pointing and overlapping regions of each galaxy, and then show the continuum ratio vs wavelength of each pair of neighbouring pointings (both 7000 and 8900 runs), and finally the SFH QC maps (7000 run only).

Below is what i think there are clear sine-like residual between two neighbouring pointings and so we may need to redo the `SFH` module in 7000 run with `MDEG` increase from 12 to 16:

1. NGC4216: R6/R7, the top two pointings.
2. NGC4351
3. NGC4380: R3/R6, R5/R6
4. NGC4450: R1/R4, R2/R5, R4/R5
5. NGC4457: R2/R4, R3/R4
6. NGC4501: R4/R8, R7/R8
7. NGC4535: R1/R4, R4/R5
8. NGC4654: R3/R4, R5/R6, R7/R8
9. NGC4689: R2/R3, R3/R6, R5/R6 show clear sine-like variations, and they are indeed fixed by refitting with `MDEG=16`.

Also some galaxies I am unsure:

1. NGC4192,
2. NGC4254,
3. NGC4298 (SFH QC maps seem fine, but ratio vs wavelength show sine-like

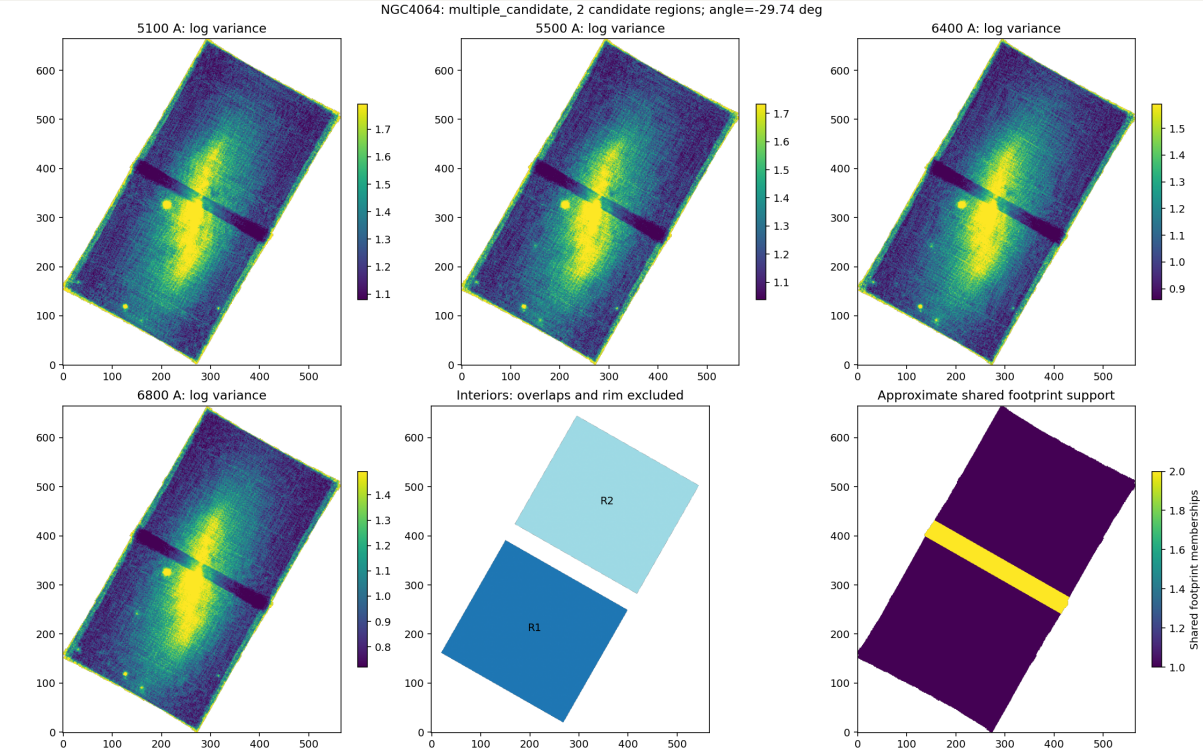
variation in R1/R2, R3/R5 and R5/R6),

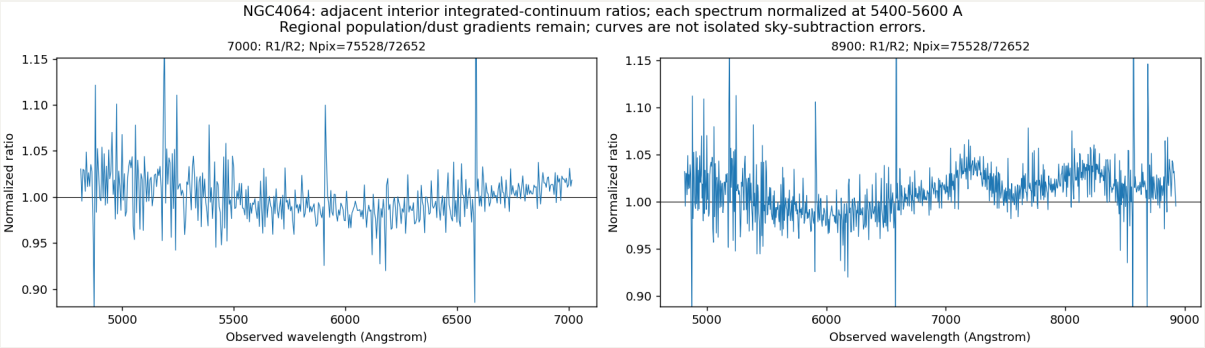
4. NGC4383,
5. NGC4567_8: the top left R3, maybe worth to try
6. NGC4569: bottom right R6, maybe worth to try
7. NGC4698

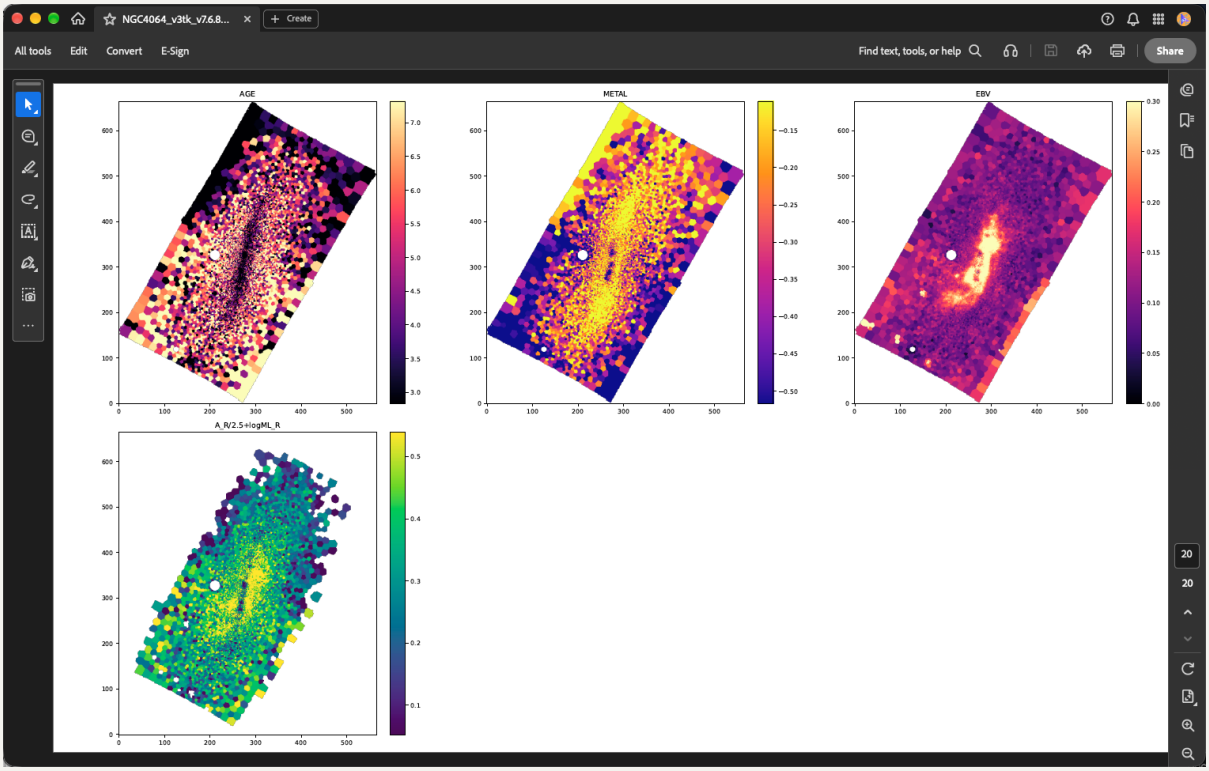
And for the resf of galaxies, I am happy with their current `nGIST` products:

1. NGC4064,
2. NGC4189,
3. NGC4222,
4. NGC4293,
5. NGC4294,
6. NGC4302,
7. NGC4321,
8. NGC4330,
9. NGC4388,
10. NGC4394,
11. NGC4396 (although R2/R3 seems a bit sine-like vacation, the QC maps look fine for me),
12. NGC4402,
13. NGC4405,
14. NGC4419 (although seems pointing residual in QC maps, the continuum ratio does not show sine-like variation so increasing `MDEG` does not expect to fix this issue),
15. NGC4424,
16. NGC4522,
17. NGC4580,
18. NGC4607.

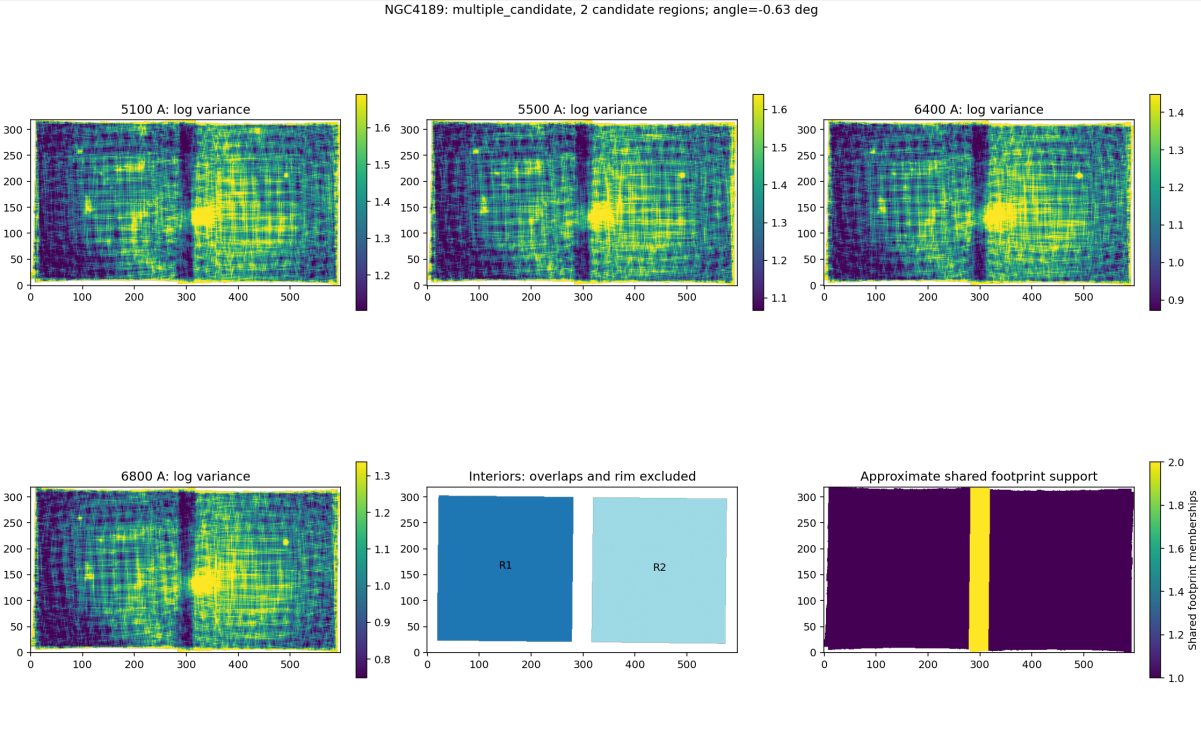
NGC4064 (fine)

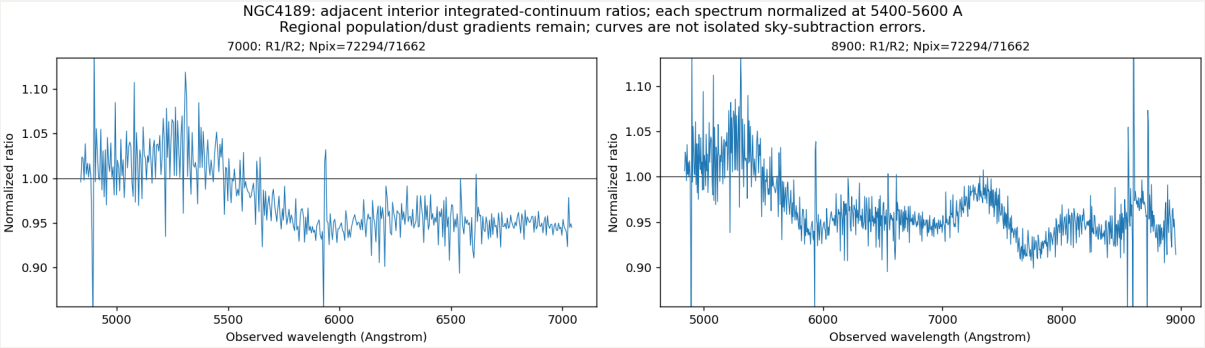

NGC4064: multiple_candidate, 2 candidate regions; angle=-29.74 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R2
R1
Approximate shared footprint support
Shared footprint memberships

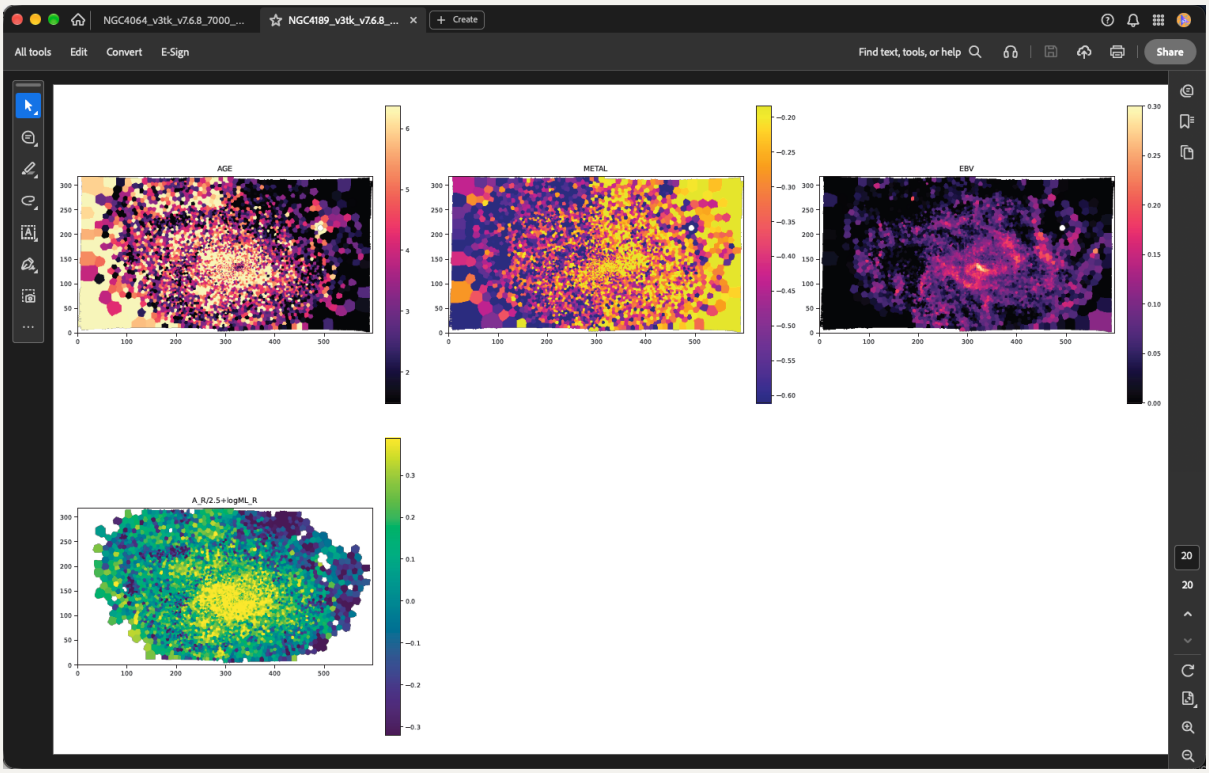

NGC4064: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=75528/72652
8900: R1/R2; Npix=75528/72652
Normalized ratio
Observed wavelength (Angstrom)


NGC4064_v3tk_v7.6.8...
Create
All tools
Edit
Convert
E-Sign
Find text, tools, or help
Share
AGE
METAL
EBV
A_R/2.5+logML_R

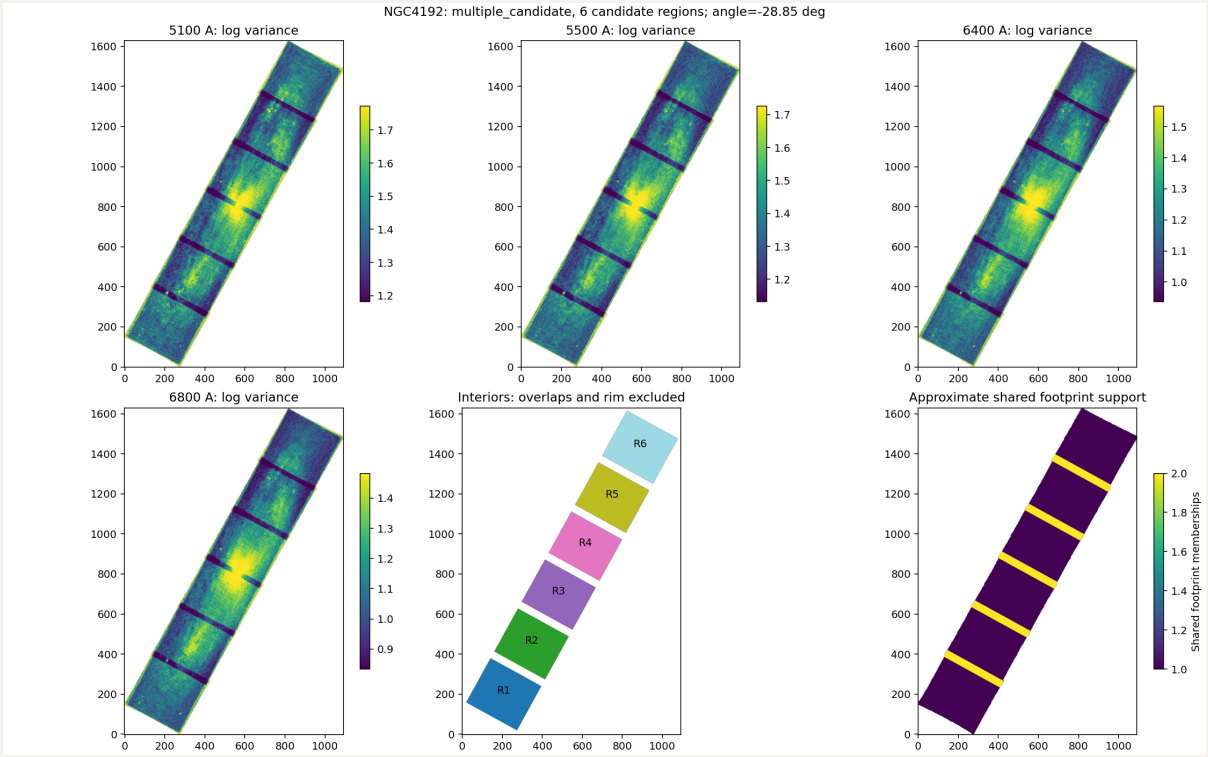
NGC4189 (fine)

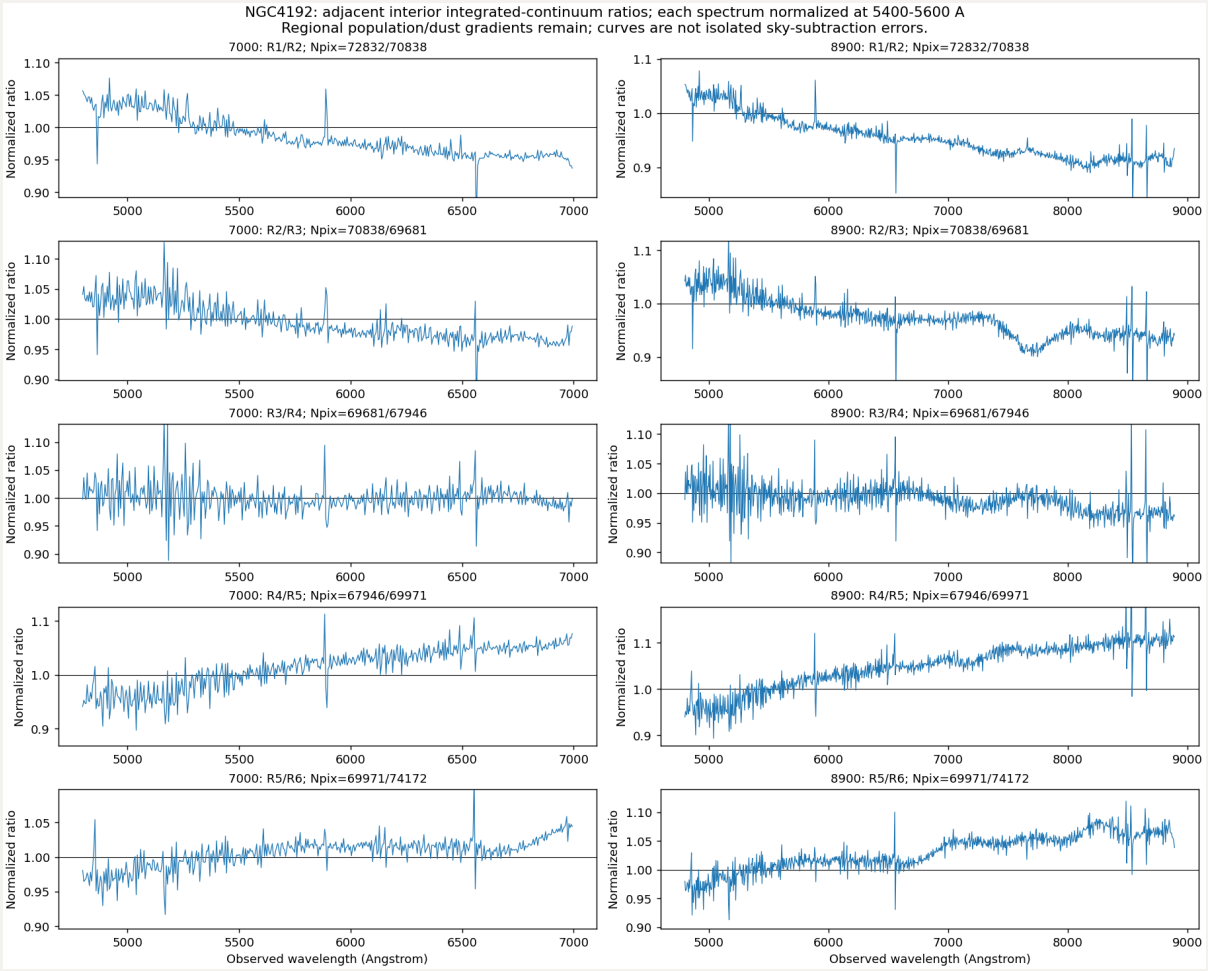


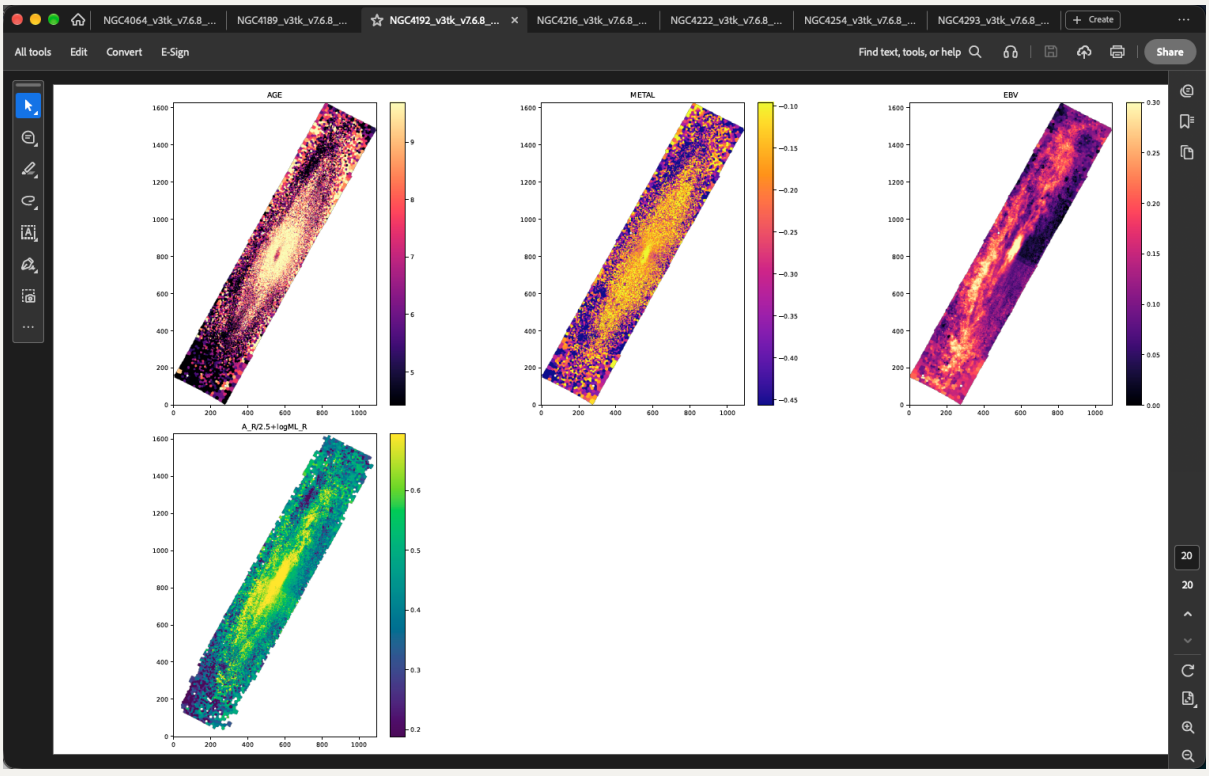




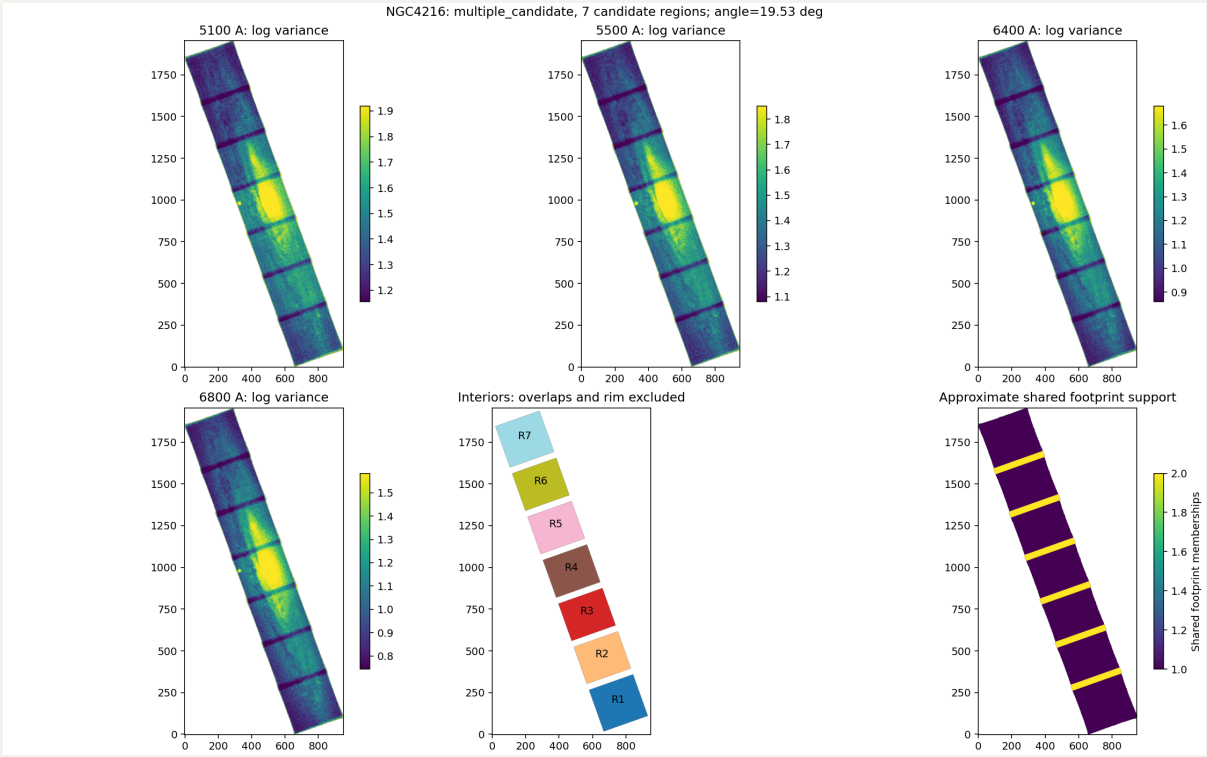
NGC4192 (maybe)

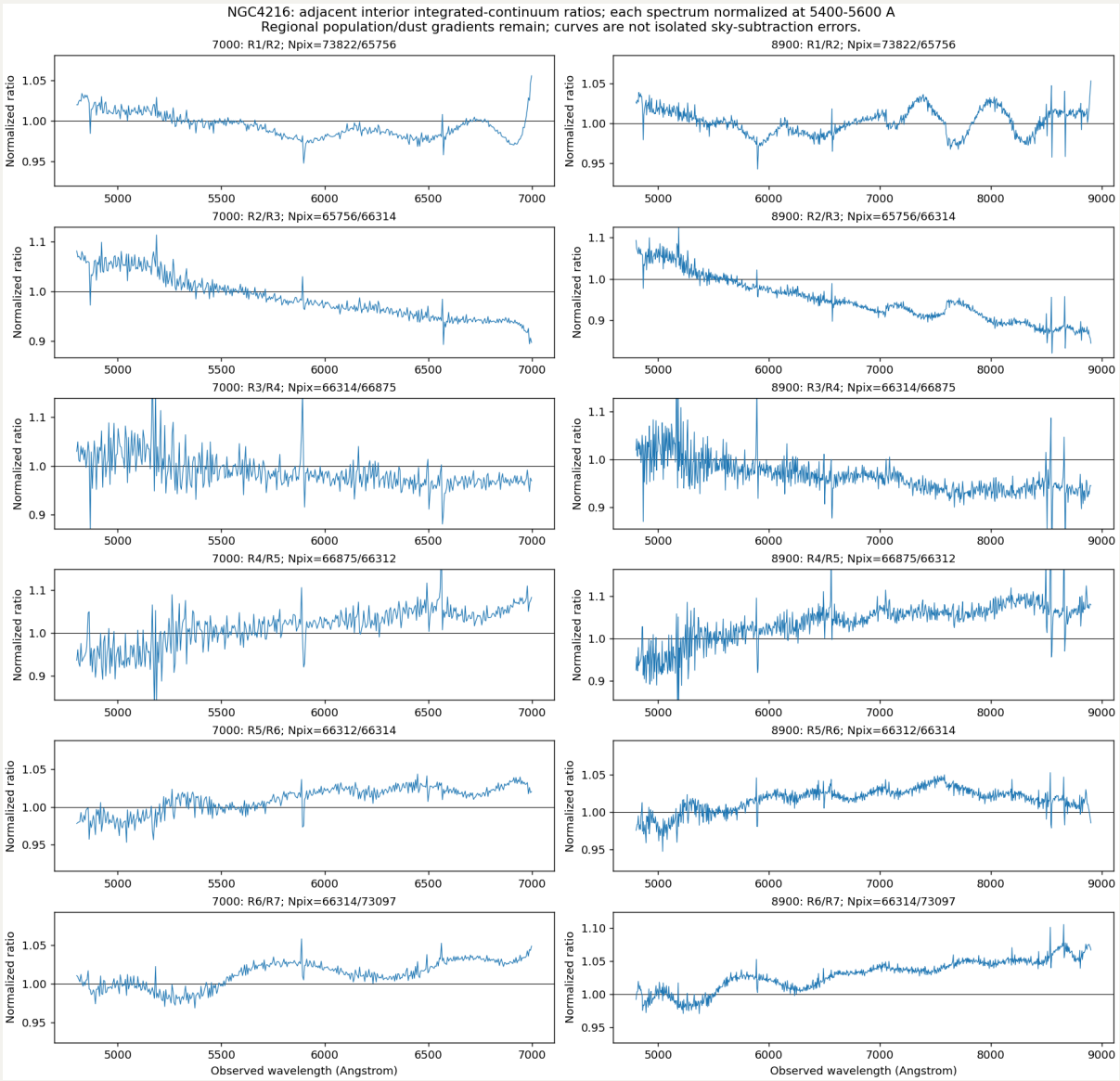


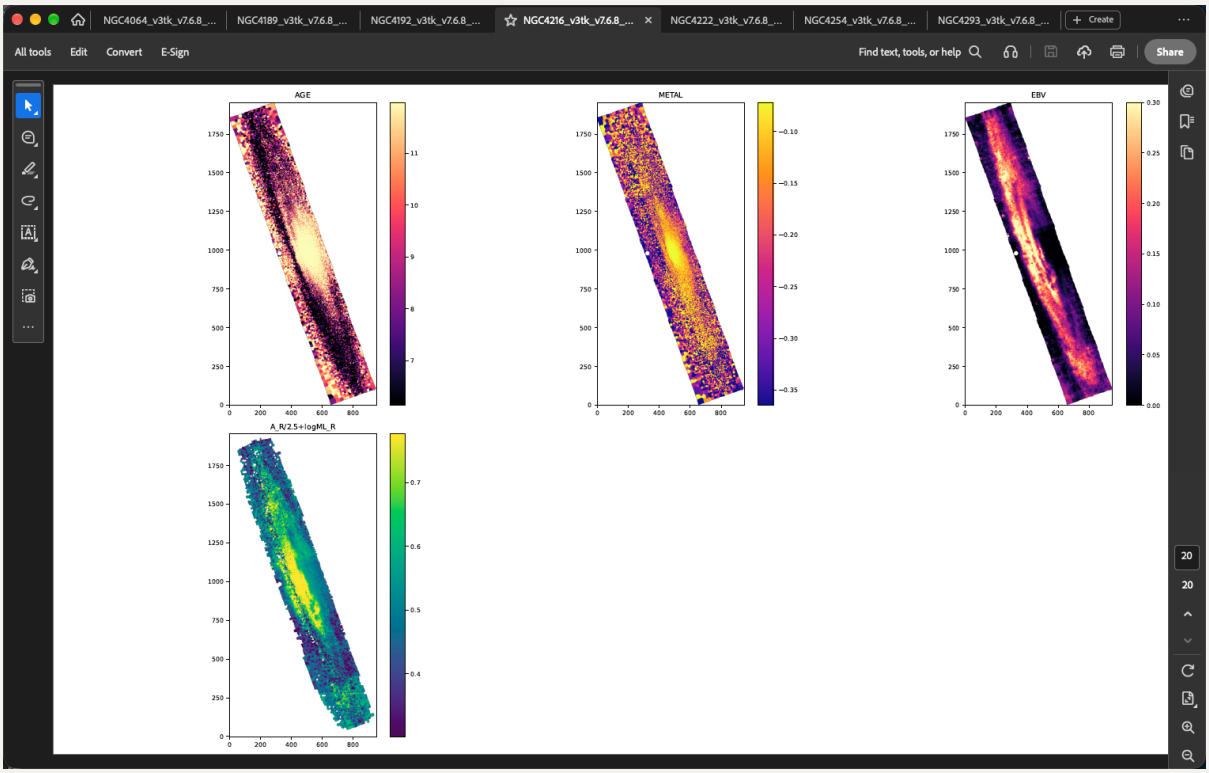




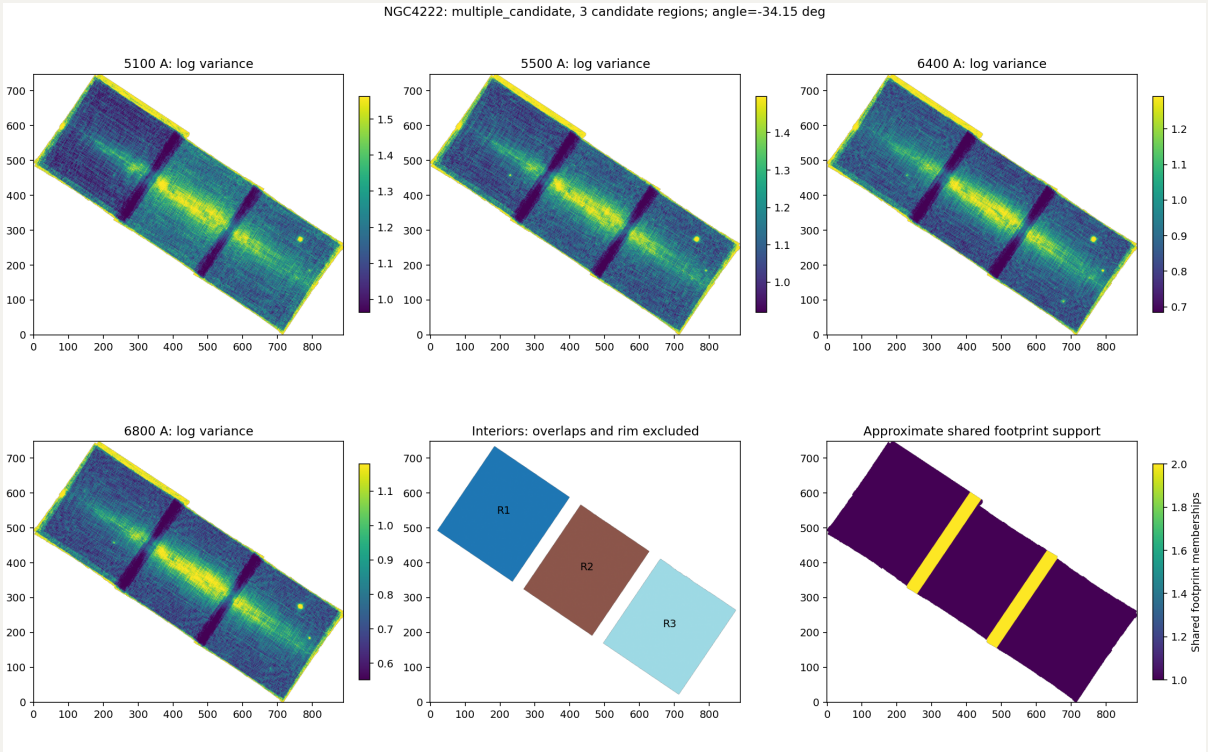
NGC4216 (needed)

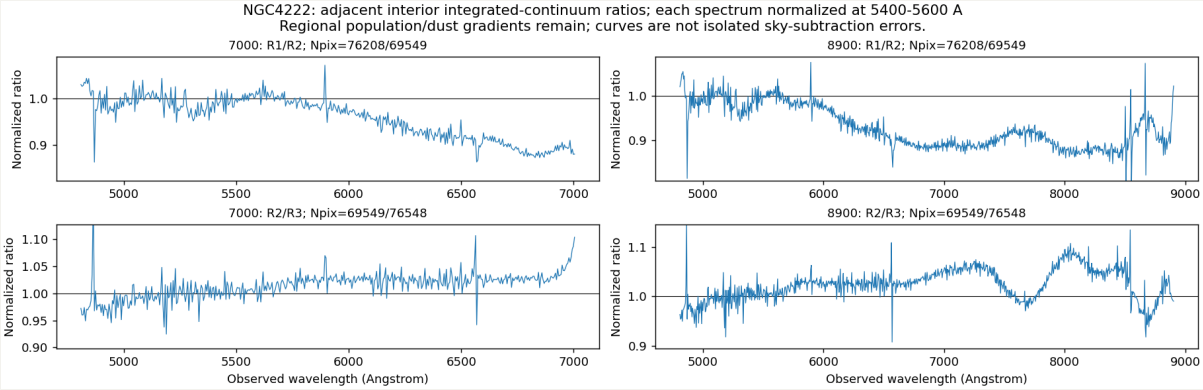

NGC4216: multiple_candidate, 7 candidate regions; angle=19.53 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R7
R6
R5
R4
R3
R2
R1
Approximate shared footprint support
Shared footprint memberships

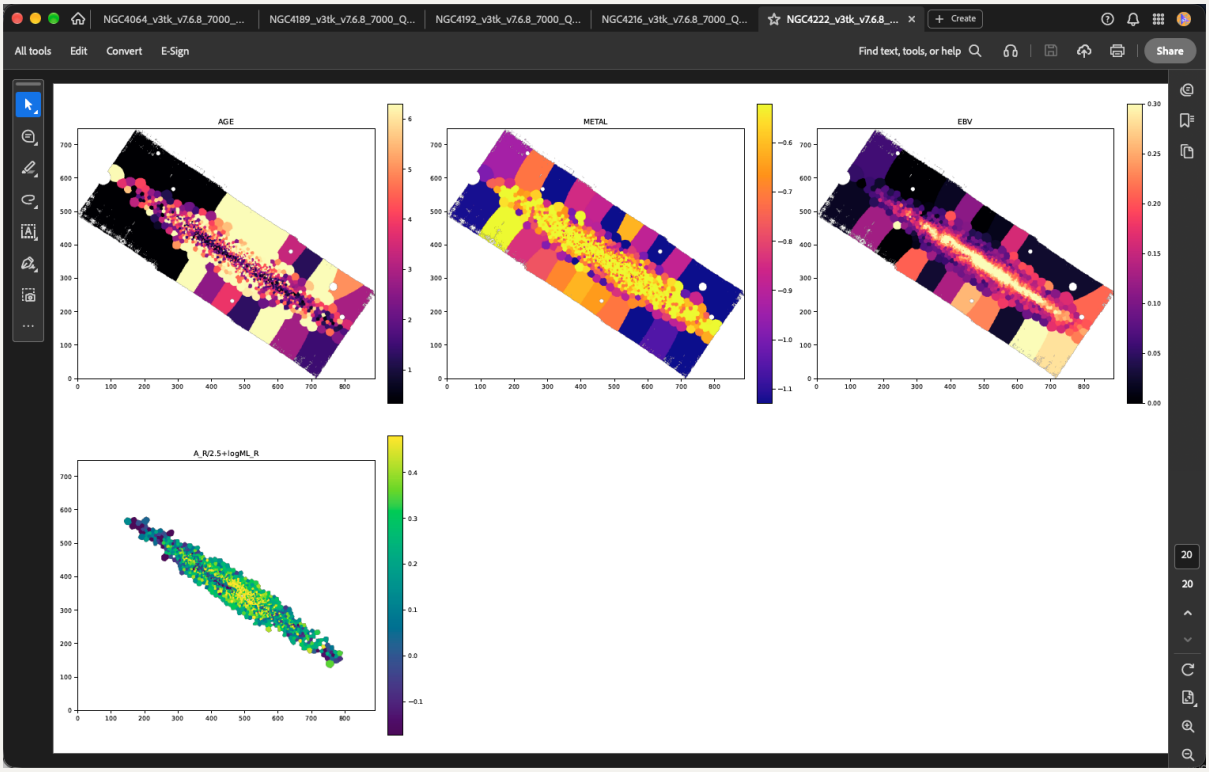

NGC4216: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=73822/65756
8900: R1/R2; Npix=73822/65756
7000: R2/R3; Npix=65756/66314
8900: R2/R3; Npix=65756/66314
7000: R3/R4; Npix=66314/66875
8900: R3/R4; Npix=66314/66875
7000: R4/R5; Npix=66875/66312
8900: R4/R5; Npix=66875/66312
7000: R5/R6; Npix=66312/66314
8900: R5/R6; Npix=66312/66314
7000: R6/R7; Npix=66314/73097
8900: R6/R7; Npix=66314/73097
Normalized ratio
Observed wavelength (Angstrom)



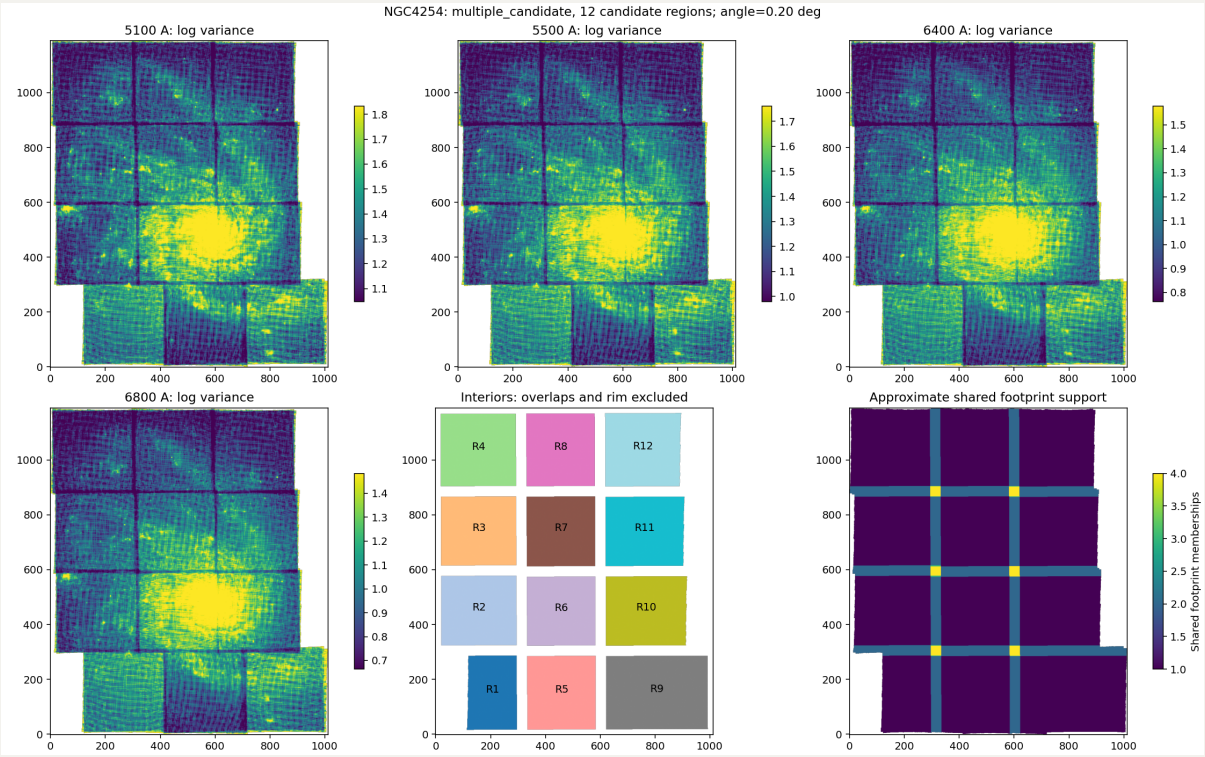
NGC4222 (fine)

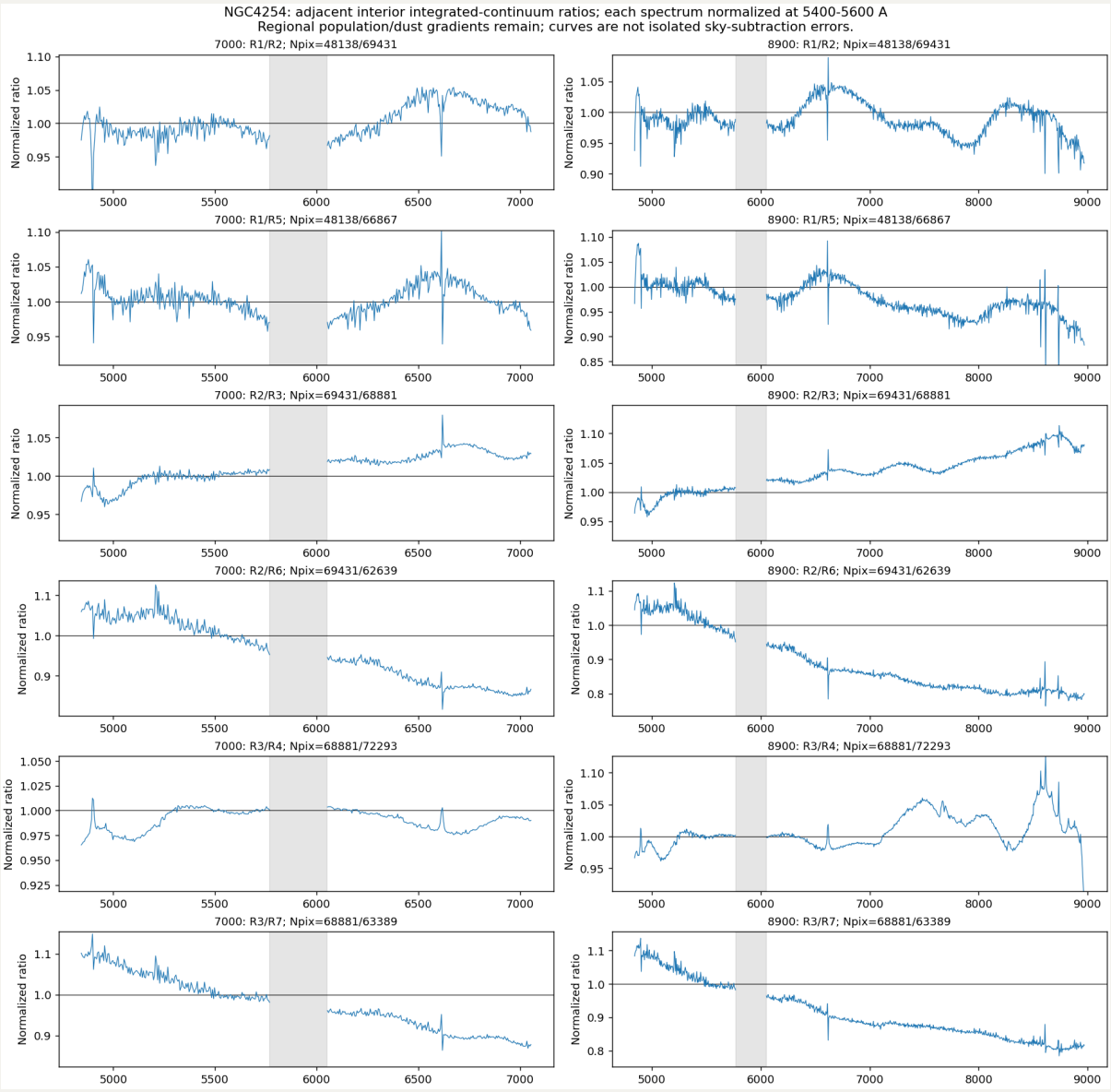


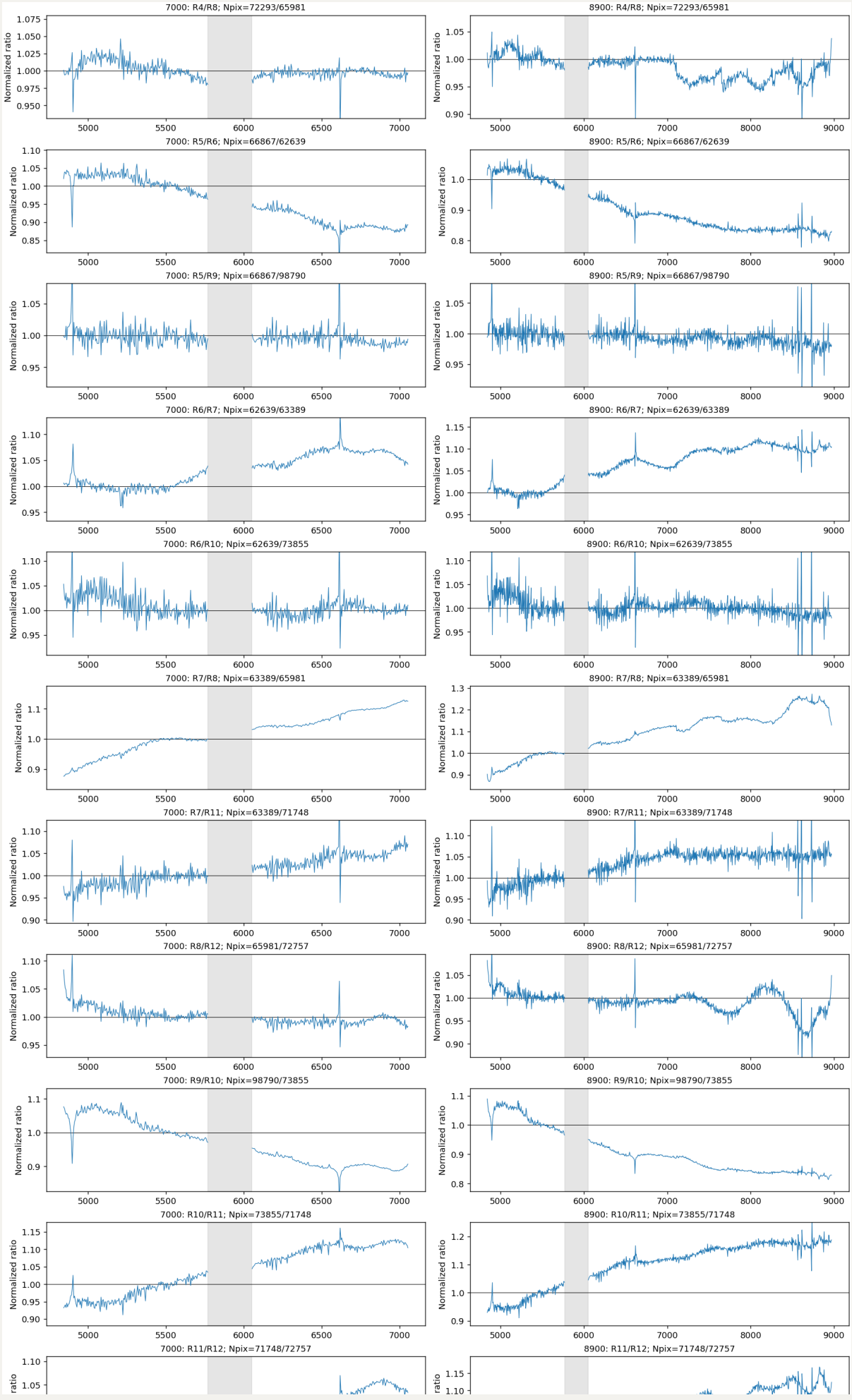


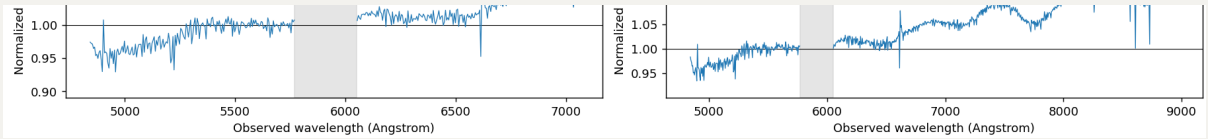


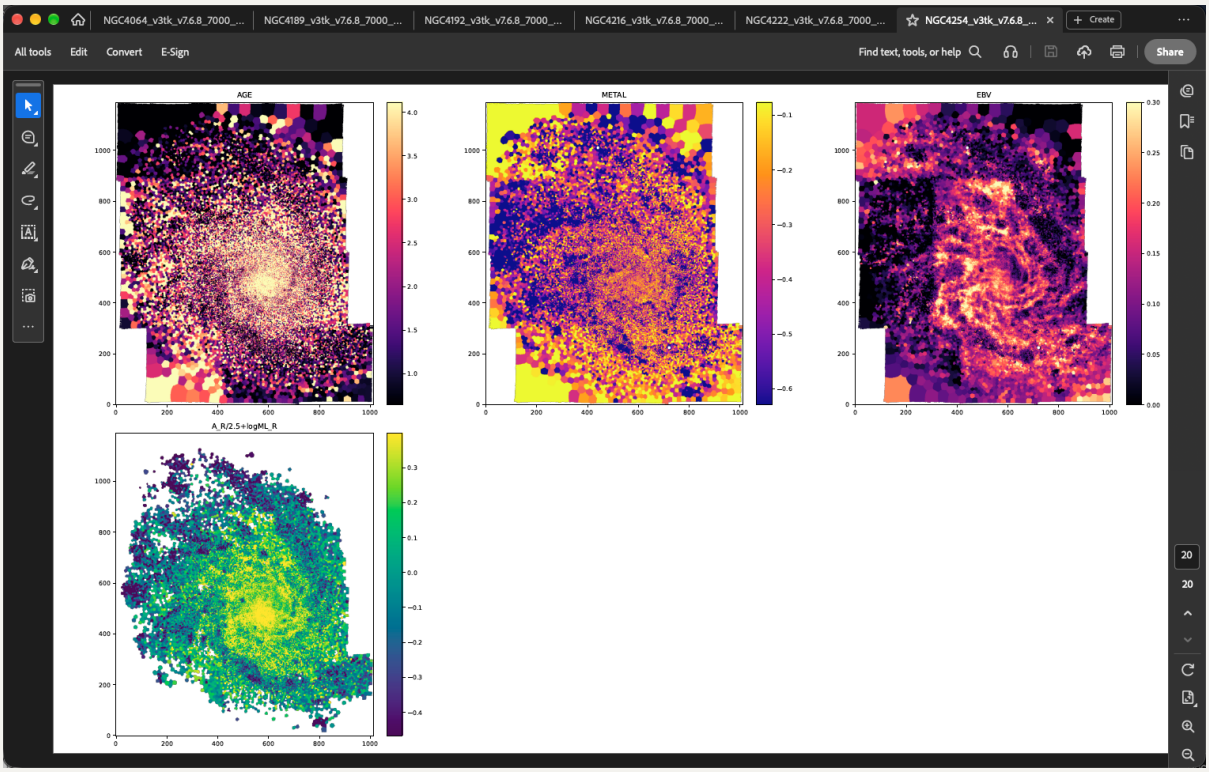
NGC4254 (maybe)


NGC4254: multiple_candidate, 12 candidate regions; angle=0.20 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
Approximate shared footprint support
R4
R8
R12
R3
R7
R11
R2
R6
R10
R1
R5
R9
Shared footprint memberships

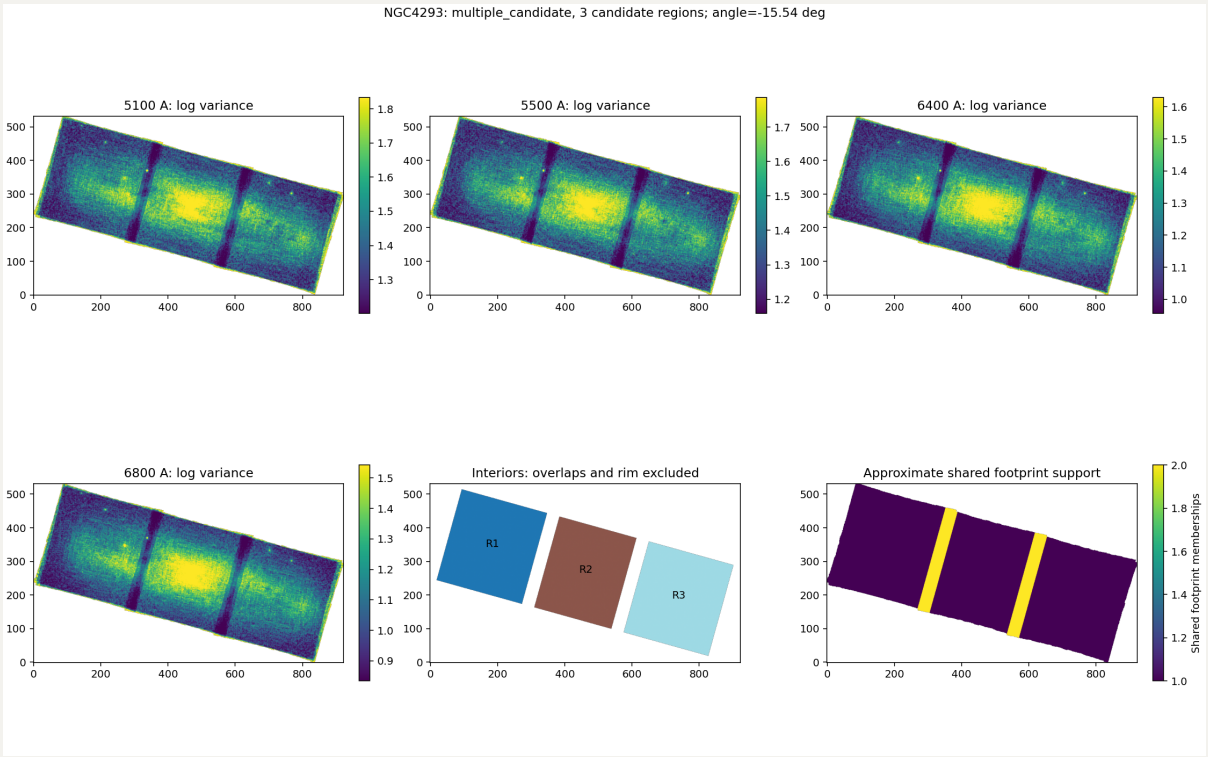

NGC4254: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=48138/69431
8900: R1/R2; Npix=48138/69431
7000: R1/R5; Npix=48138/66867
8900: R1/R5; Npix=48138/66867
7000: R2/R3; Npix=69431/68881
8900: R2/R3; Npix=69431/68881
7000: R2/R6; Npix=69431/62639
8900: R2/R6; Npix=69431/62639
7000: R3/R4; Npix=68881/72293
8900: R3/R4; Npix=68881/72293
7000: R3/R7; Npix=68881/63389
8900: R3/R7; Npix=68881/63389
Normalized ratio

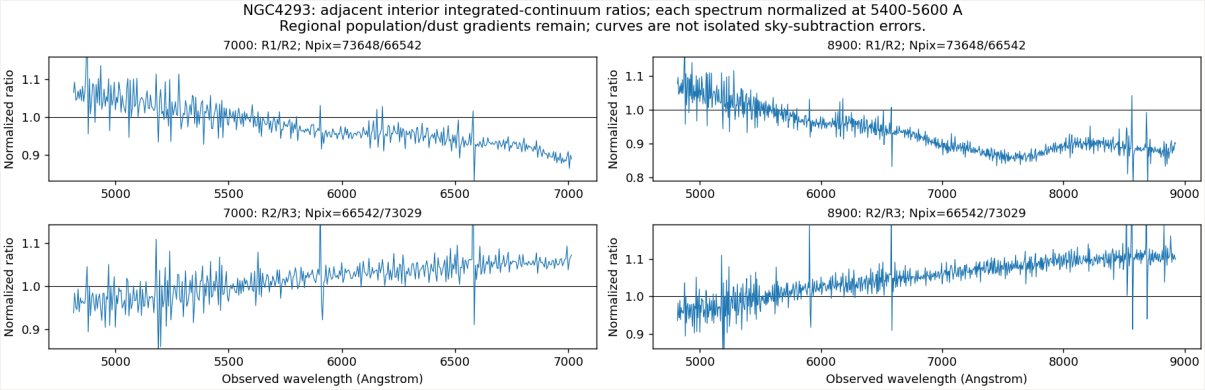

7000: R4/R8; Npix=72293/65981
8900: R4/R8; Npix=72293/65981
7000: R5/R6; Npix=66867/62639
8900: R5/R6; Npix=66867/62639
7000: R5/R9; Npix=66867/98790
8900: R5/R9; Npix=66867/98790
7000: R6/R7; Npix=62639/63389
8900: R6/R7; Npix=62639/63389
7000: R6/R10; Npix=62639/73855
8900: R6/R10; Npix=62639/73855
7000: R7/R8; Npix=63389/65981
8900: R7/R8; Npix=63389/65981
7000: R7/R11; Npix=63389/71748
8900: R7/R11; Npix=63389/71748
7000: R8/R12; Npix=65981/72757
8900: R8/R12; Npix=65981/72757
7000: R9/R10; Npix=98790/73855
8900: R9/R10; Npix=98790/73855
7000: R10/R11; Npix=73855/71748
8900: R10/R11; Npix=73855/71748
7000: R11/R12; Npix=71748/72757
8900: R11/R12; Npix=71748/72757
Normalized ratio

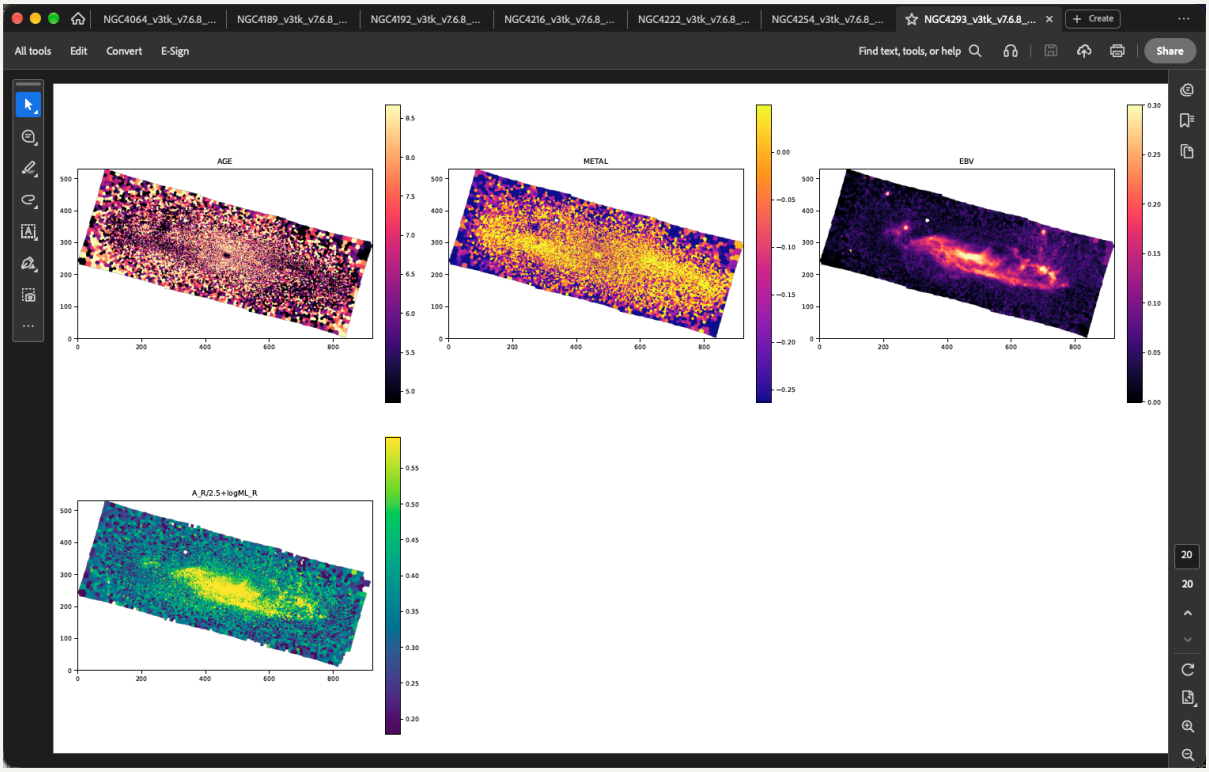




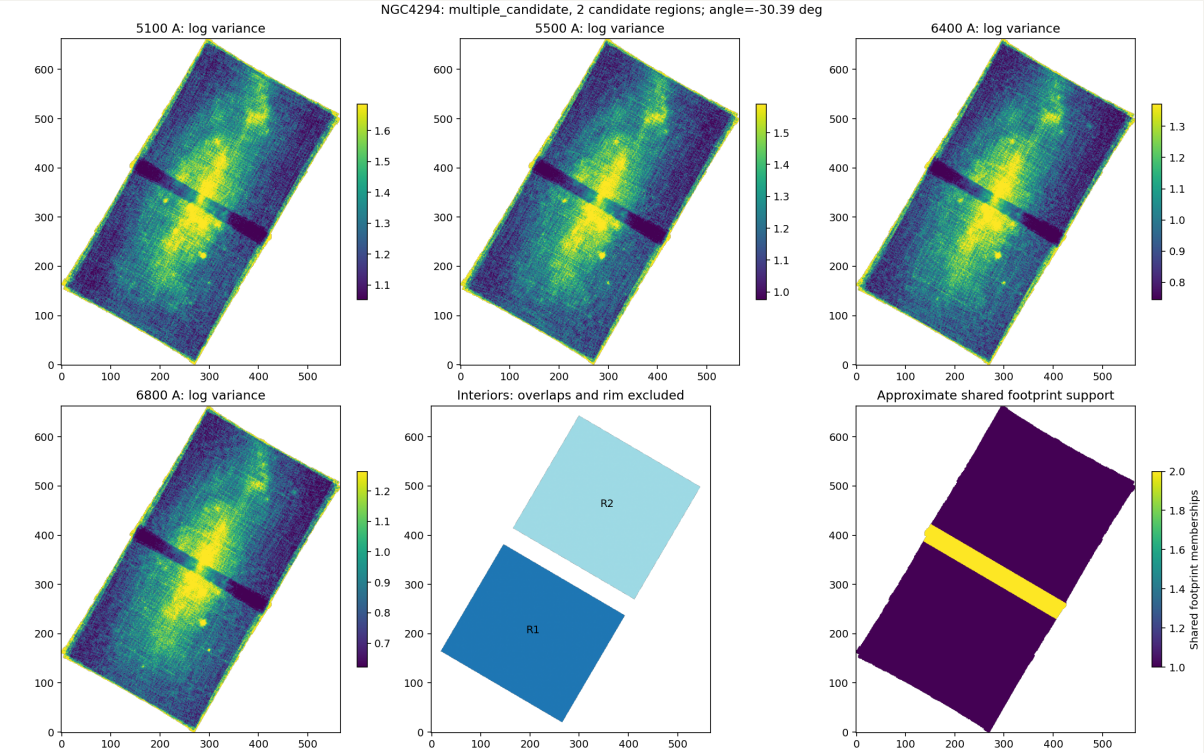
NGC4293 (fine)

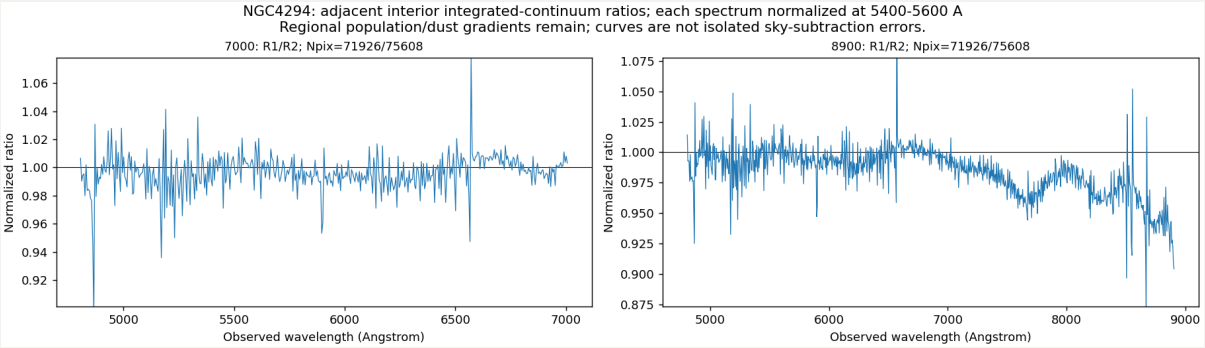


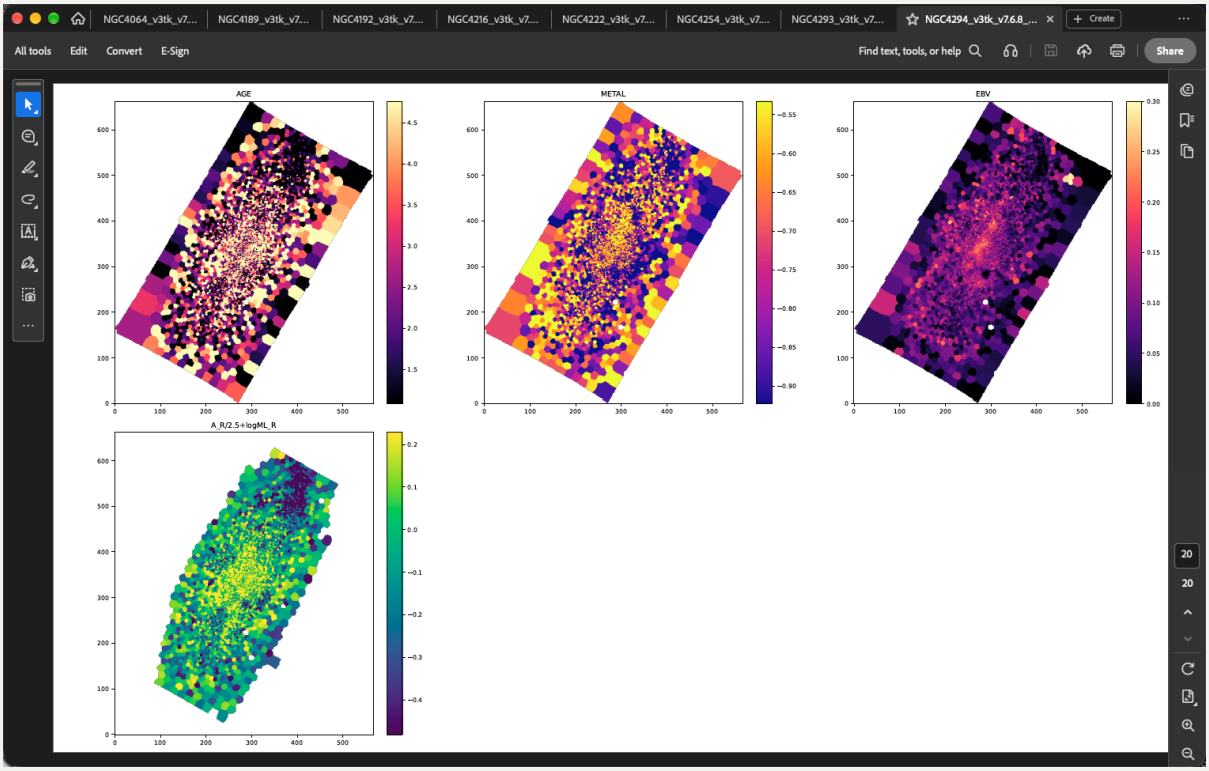




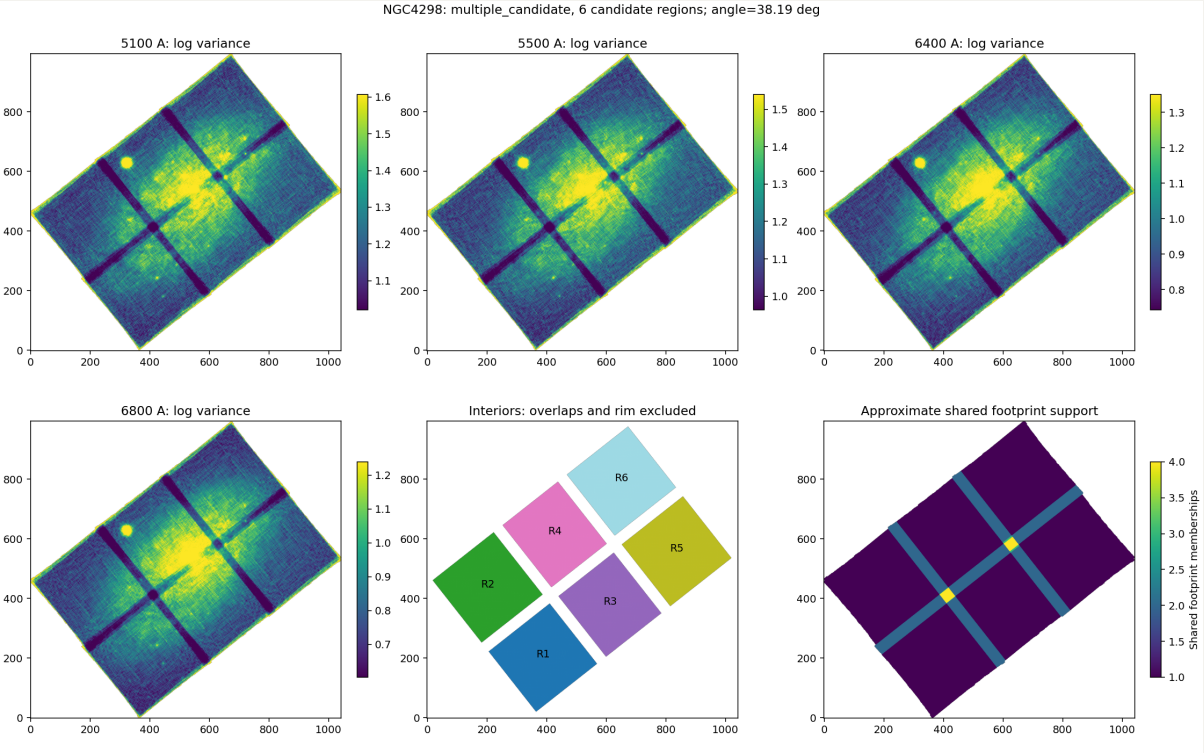
NGC4294 (fine)

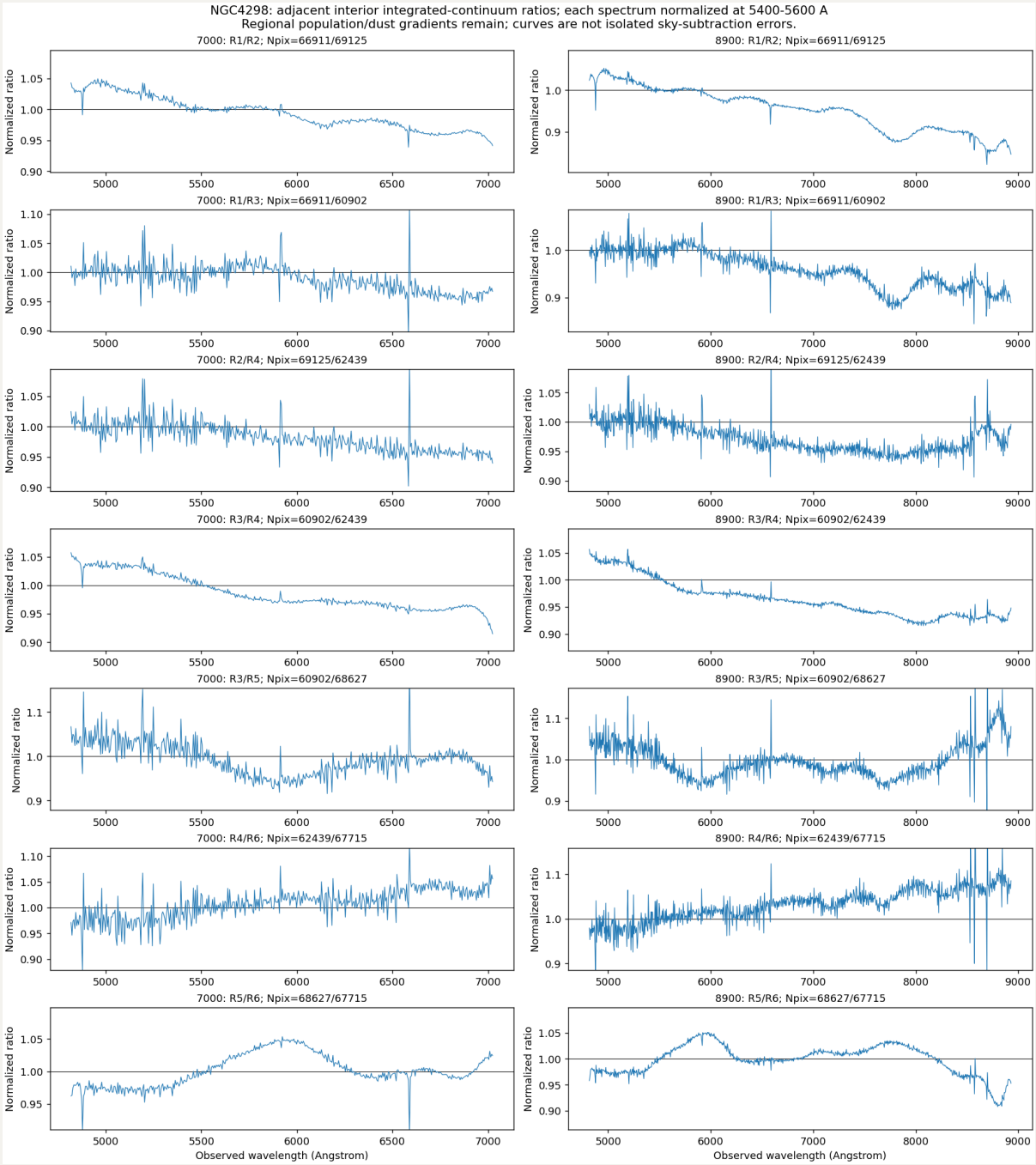

NGC4294: multiple_candidate, 2 candidate regions; angle=-30.39 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R2
R1
Approximate shared footprint support
Shared footprint memberships

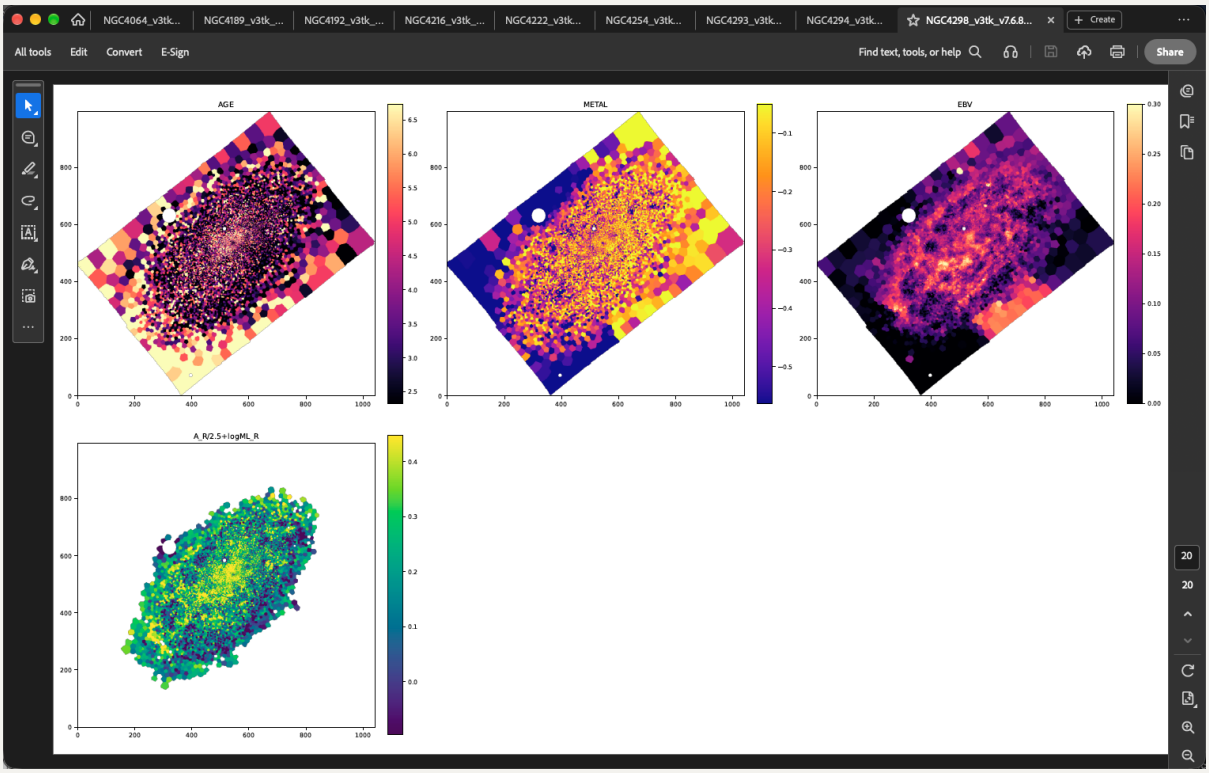

NGC4294: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=71926/75608
8900: R1/R2; Npix=71926/75608
Normalized ratio
Observed wavelength (Angstrom)


All tools
Edit
Convert
E-Sign
Share
AGE
METAL
EBV
A_R/2.5+logML_R

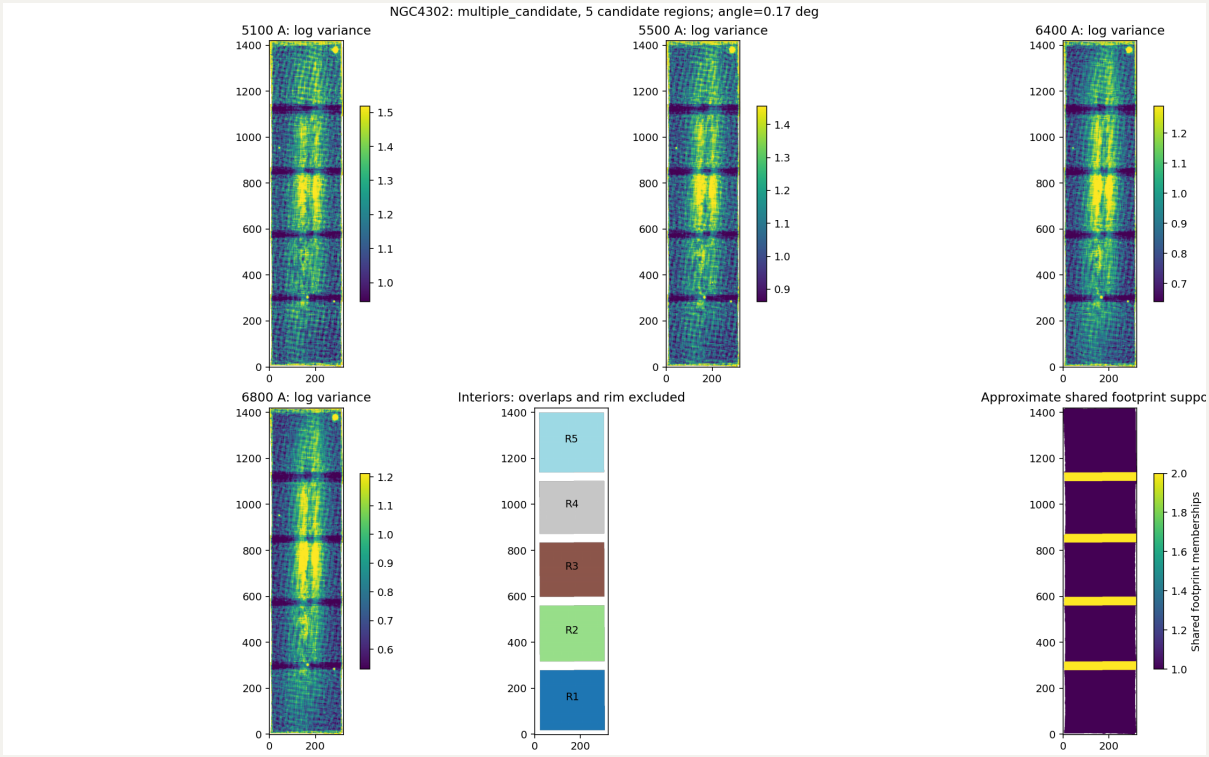
NGC4298 (maybe)

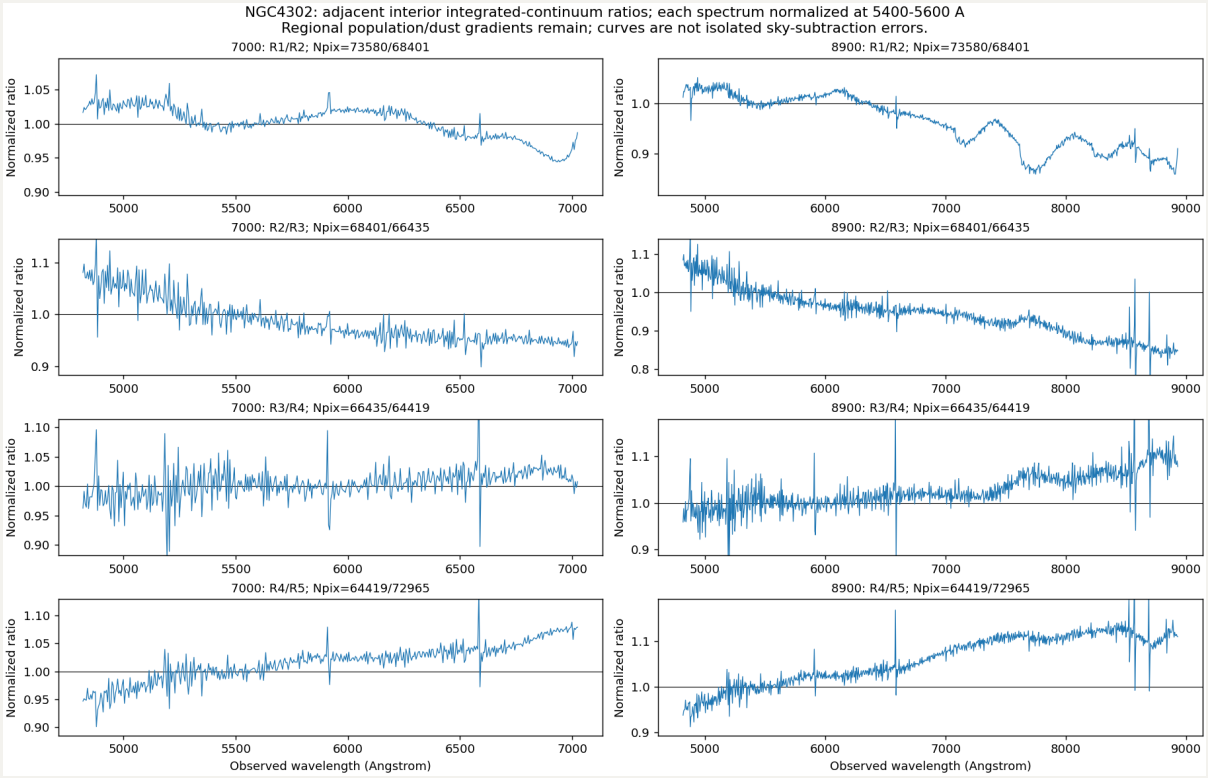


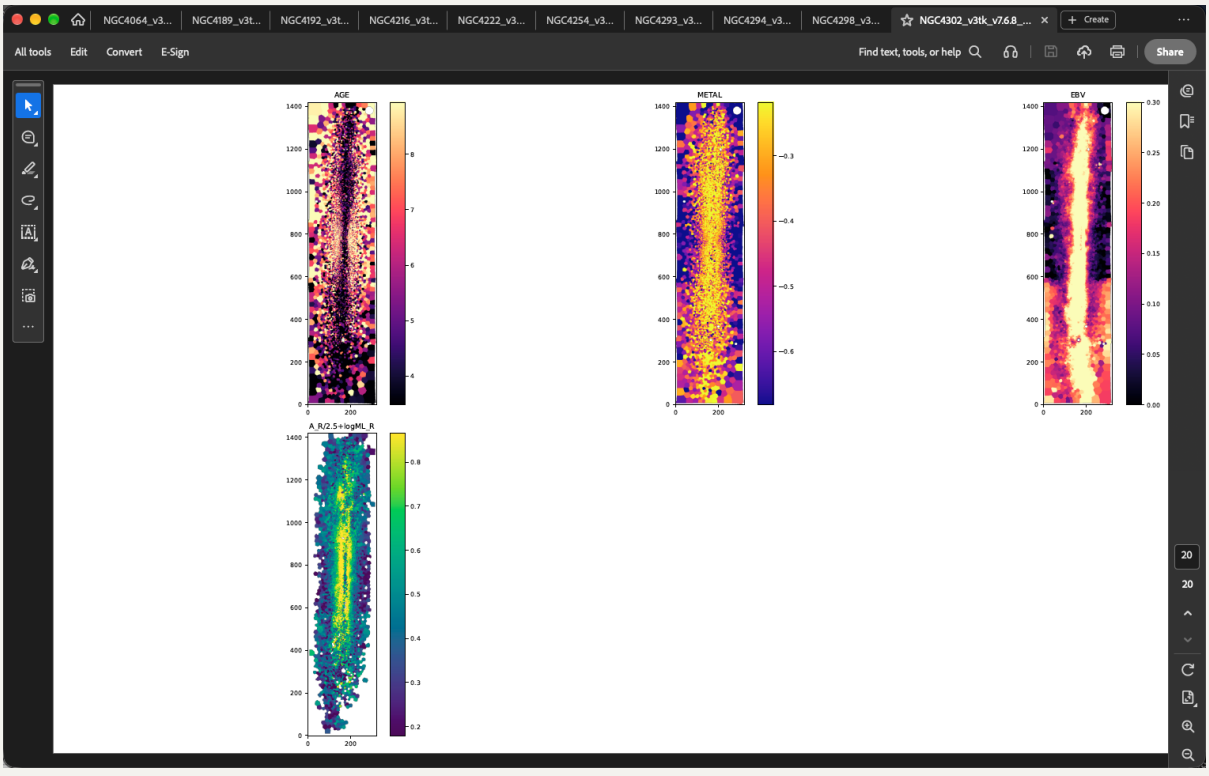

NGC4298: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=66911/69125
8900: R1/R2; Npix=66911/69125
7000: R1/R3; Npix=66911/60902
8900: R1/R3; Npix=66911/60902
7000: R2/R4; Npix=69125/62439
8900: R2/R4; Npix=69125/62439
7000: R3/R4; Npix=60902/62439
8900: R3/R4; Npix=60902/62439
7000: R3/R5; Npix=60902/68627
8900: R3/R5; Npix=60902/68627
7000: R4/R6; Npix=62439/67715
8900: R4/R6; Npix=62439/67715
7000: R5/R6; Npix=68627/67715
8900: R5/R6; Npix=68627/67715
Normalized ratio
Observed wavelength (Angstrom)



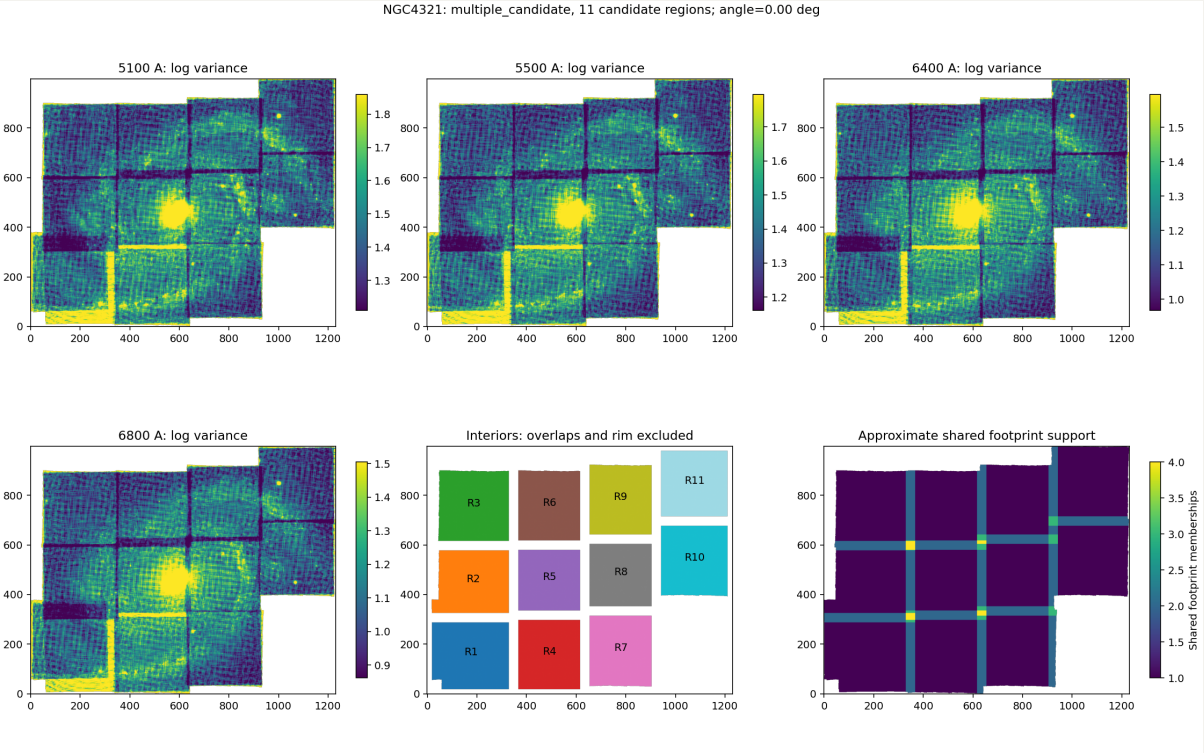
NGC4302 (fine)

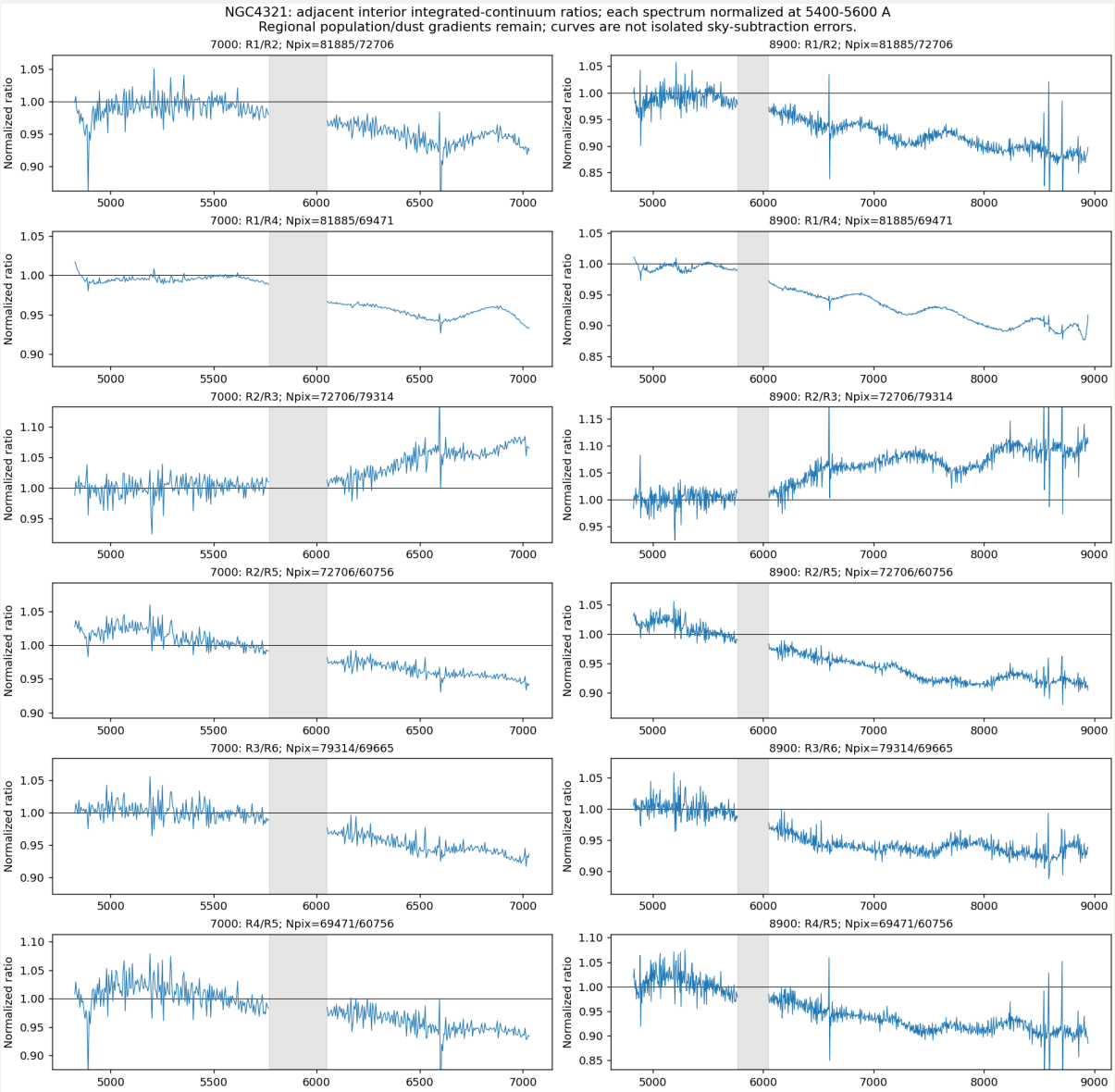


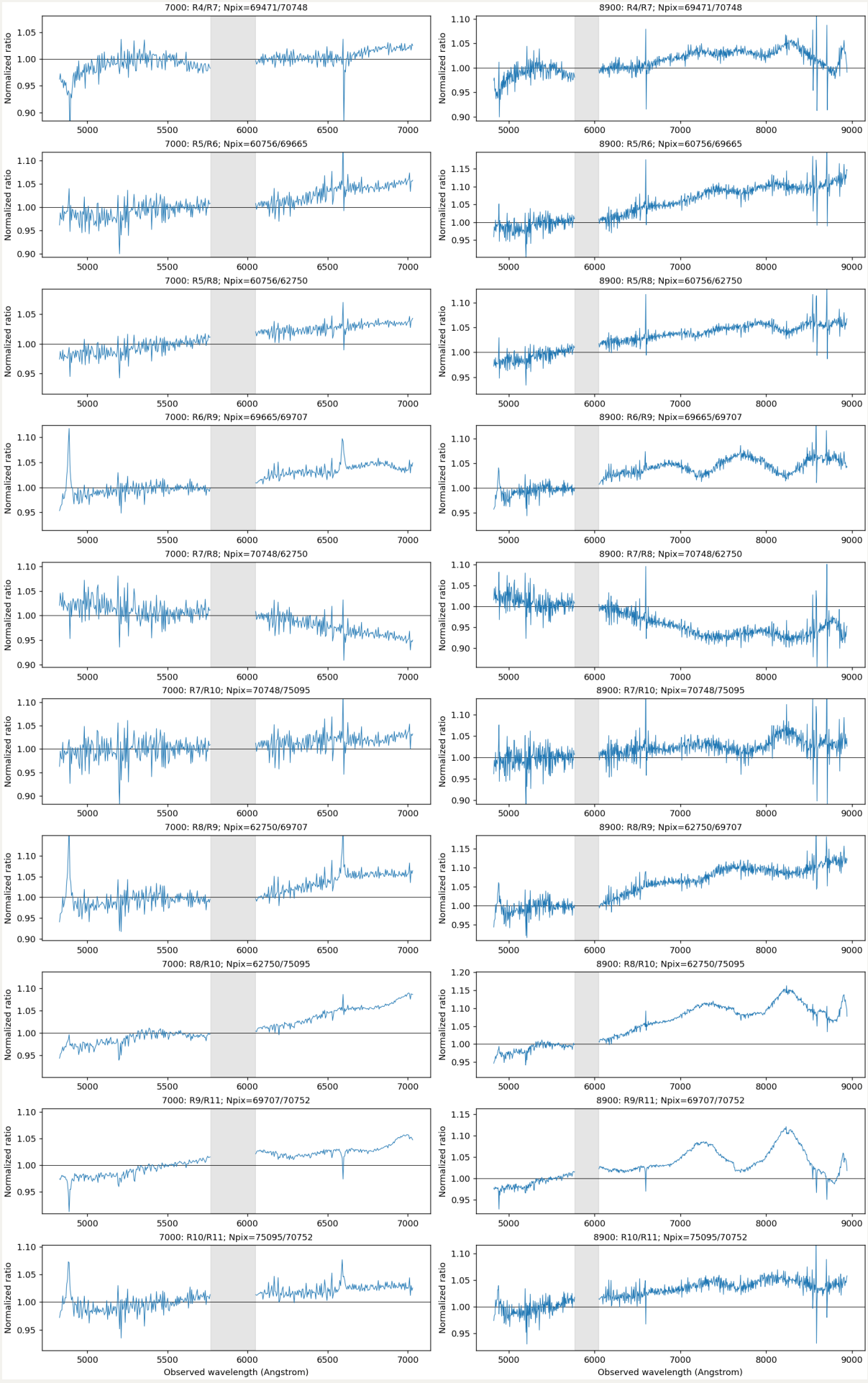


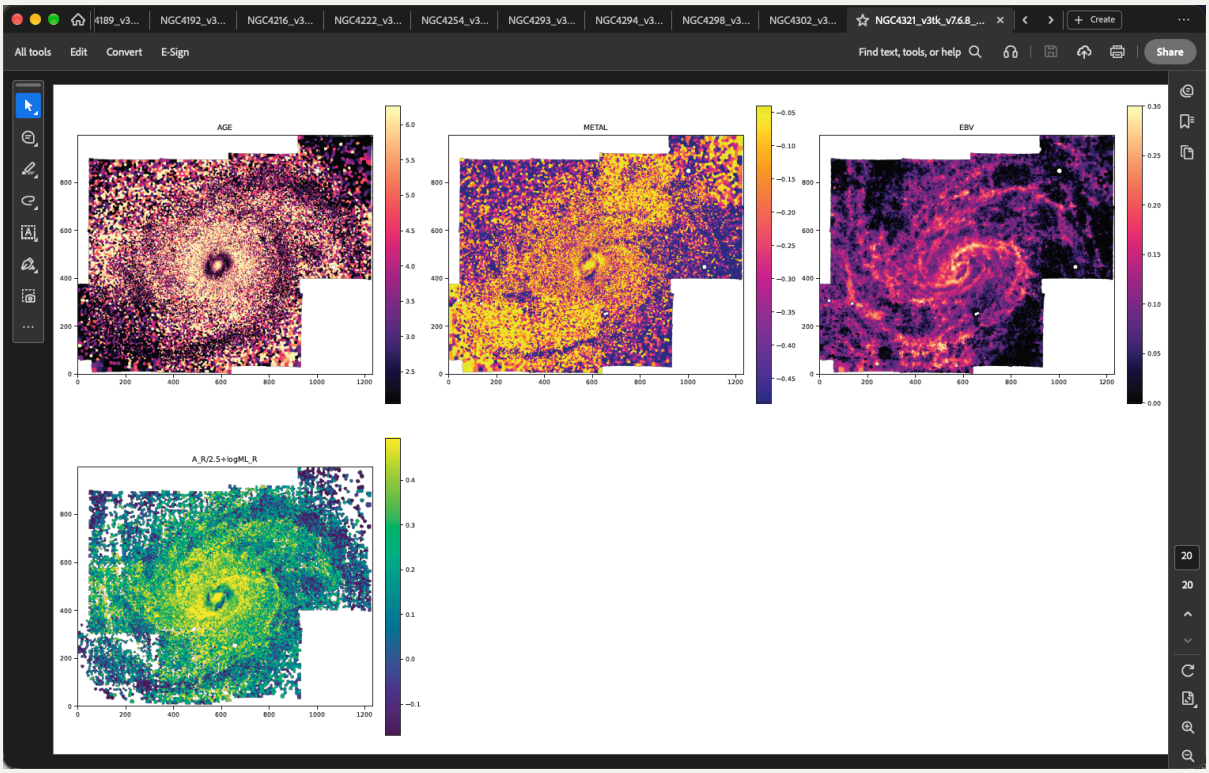


NGC4321 (fine)

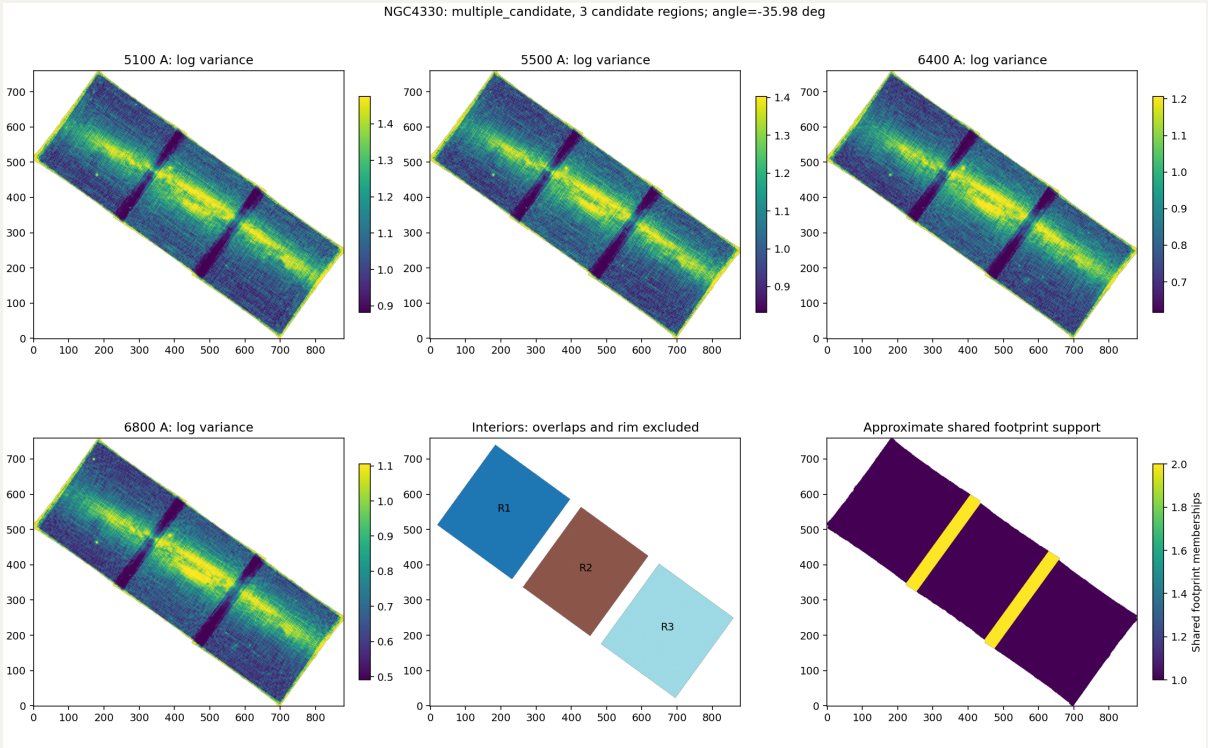

NGC4321: multiple_candidate, 11 candidate regions; angle=0.00 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
Approximate shared footprint support
Shared footprint memberships

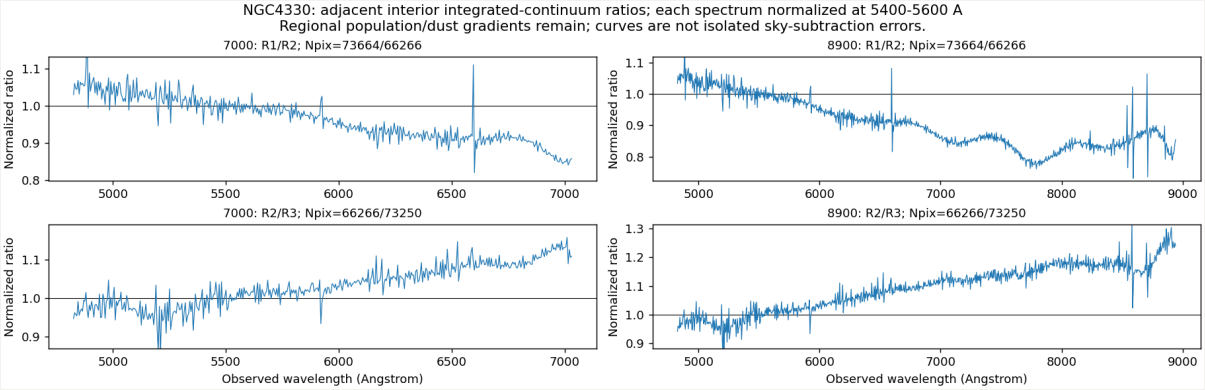

NGC4321: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=81885/72706
8900: R1/R2; Npix=81885/72706
7000: R1/R4; Npix=81885/69471
8900: R1/R4; Npix=81885/69471
7000: R2/R3; Npix=72706/79314
8900: R2/R3; Npix=72706/79314
7000: R2/R5; Npix=72706/60756
8900: R2/R5; Npix=72706/60756
7000: R3/R6; Npix=79314/69665
8900: R3/R6; Npix=79314/69665
7000: R4/R5; Npix=69471/60756
8900: R4/R5; Npix=69471/60756
Normalized ratio

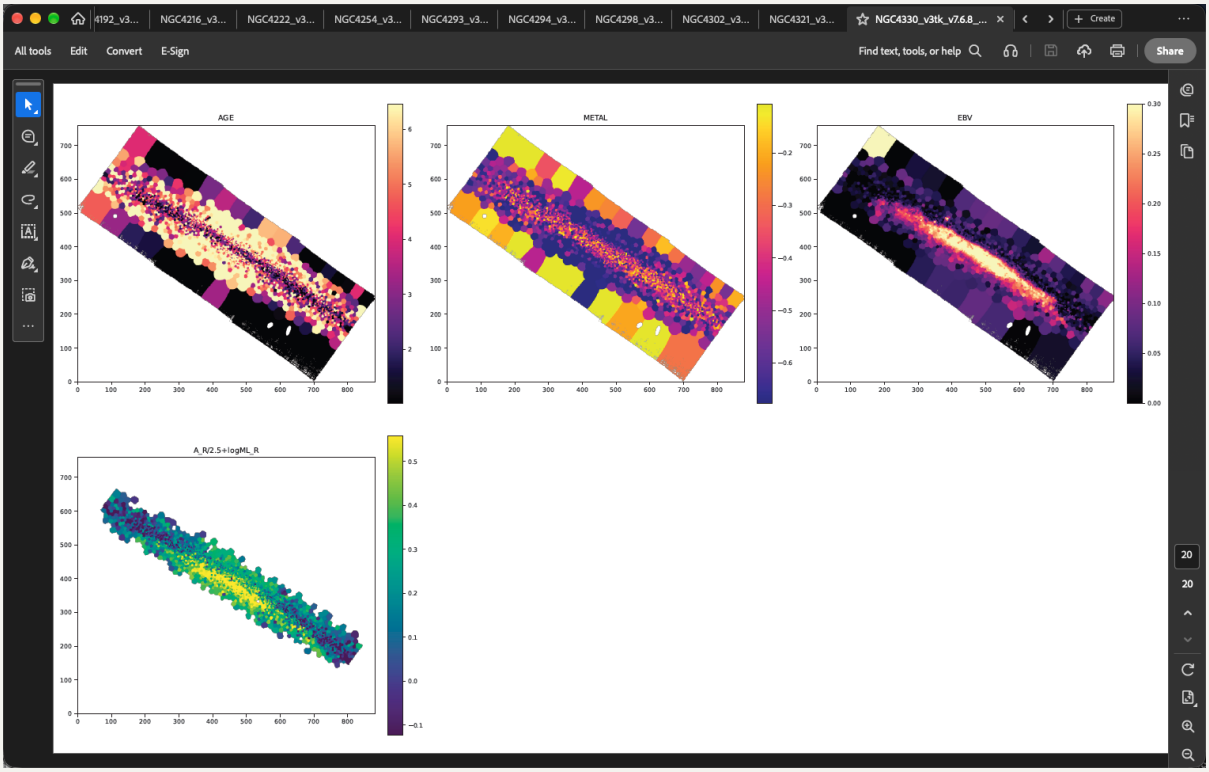

7000: R4/R7; Npix=69471/70748
8900: R4/R7; Npix=69471/70748
7000: R5/R6; Npix=60756/69665
8900: R5/R6; Npix=60756/69665
7000: R5/R8; Npix=60756/62750
8900: R5/R8; Npix=60756/62750
7000: R6/R9; Npix=69665/69707
8900: R6/R9; Npix=69665/69707
7000: R7/R8; Npix=70748/62750
8900: R7/R8; Npix=70748/62750
7000: R7/R10; Npix=70748/75095
8900: R7/R10; Npix=70748/75095
7000: R8/R9; Npix=62750/69707
8900: R8/R9; Npix=62750/69707
7000: R8/R10; Npix=62750/75095
8900: R8/R10; Npix=62750/75095
7000: R9/R11; Npix=69707/70752
8900: R9/R11; Npix=69707/70752
7000: R10/R11; Npix=75095/70752
8900: R10/R11; Npix=75095/70752
Normalized ratio
Observed wavelength (Angstrom)



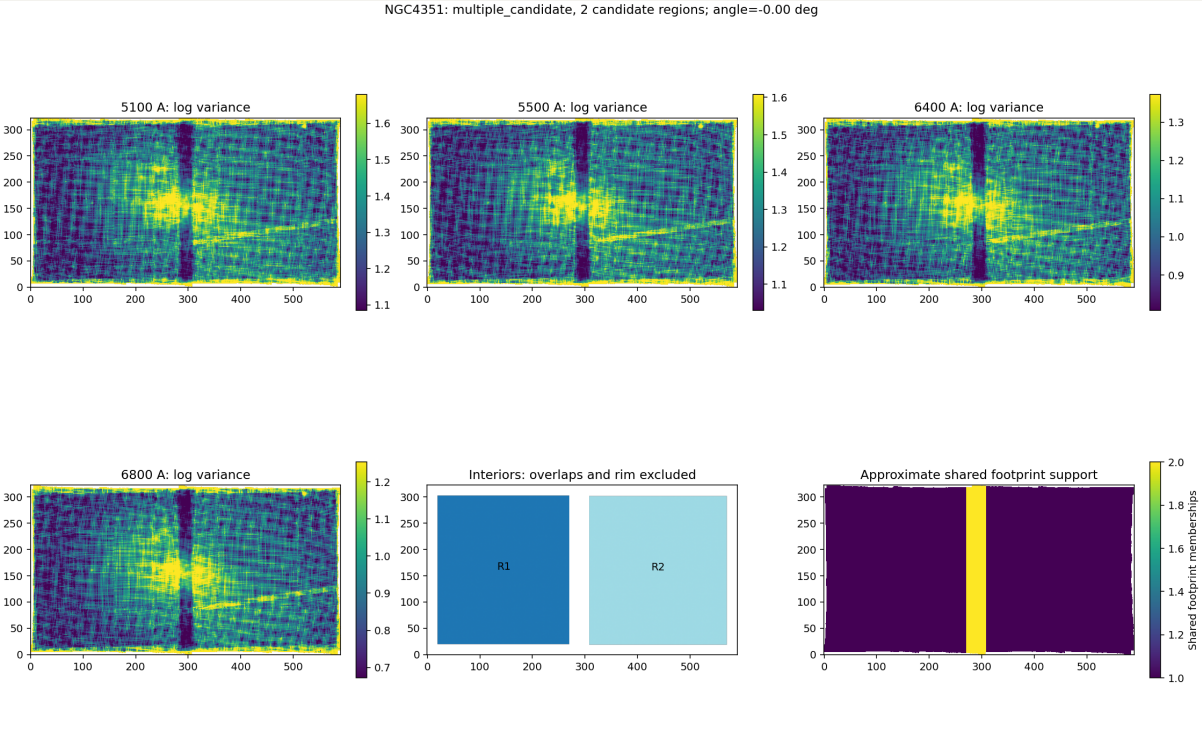
NGC4330 (fine)

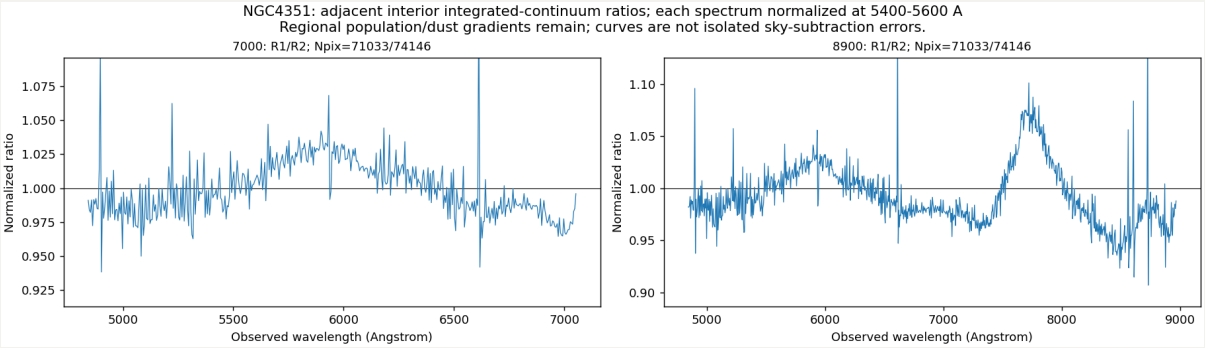


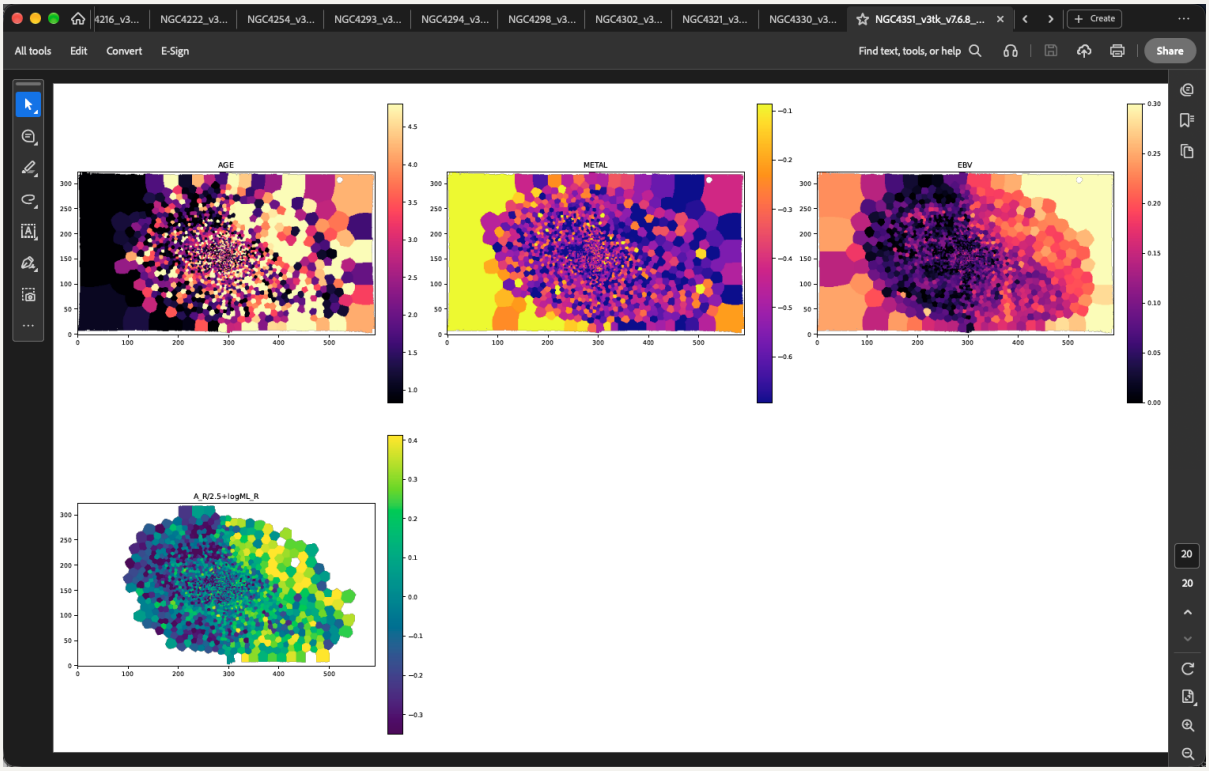




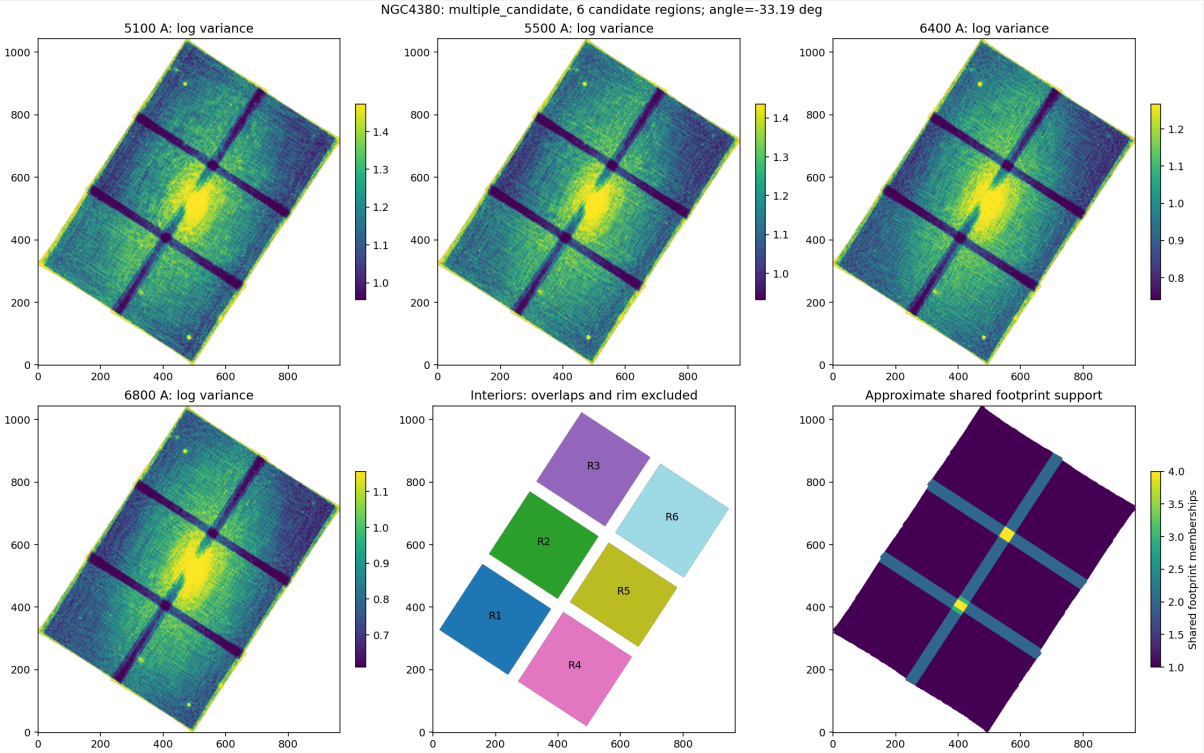
NGC4351 (needed)

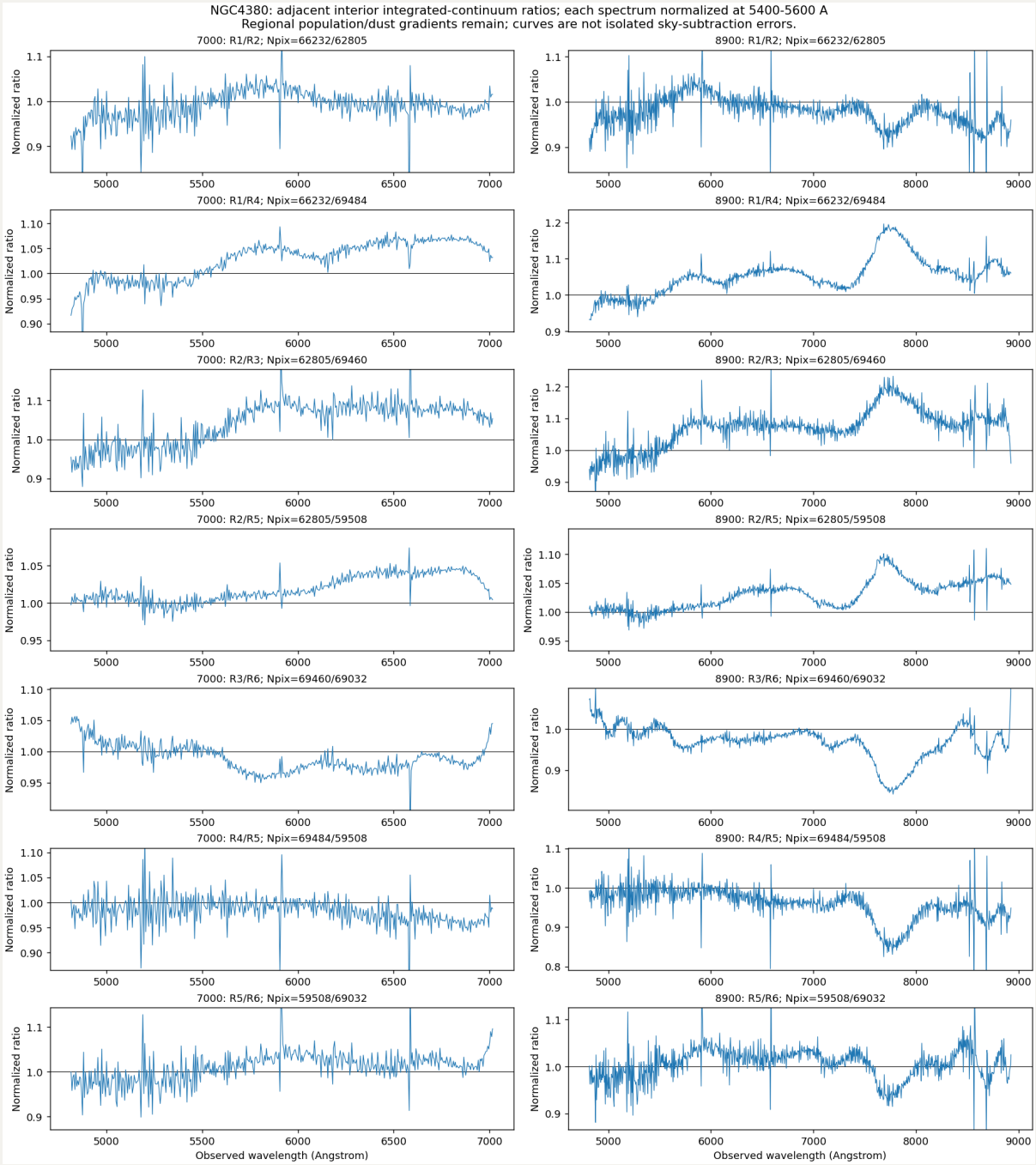

NGC4351: multiple_candidate, 2 candidate regions; angle=-0.00 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R1
R2
Approximate shared footprint support
Shared footprint memberships

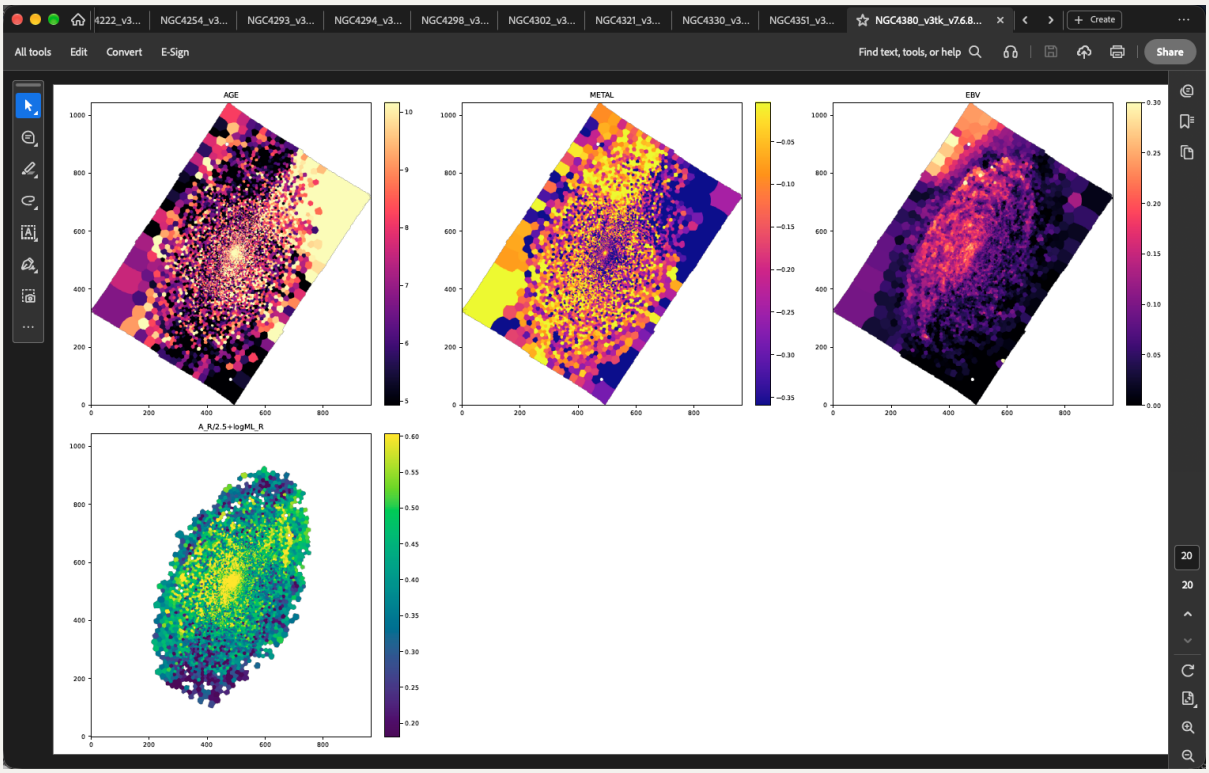

NGC4351: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=71033/74146
8900: R1/R2; Npix=71033/74146
Normalized ratio
Observed wavelength (Angstrom)


All tools
Edit
Convert
E-Sign
Find text, tools, or help
Share
AGE
METAL
EBV
A_R/2.5+logML_R

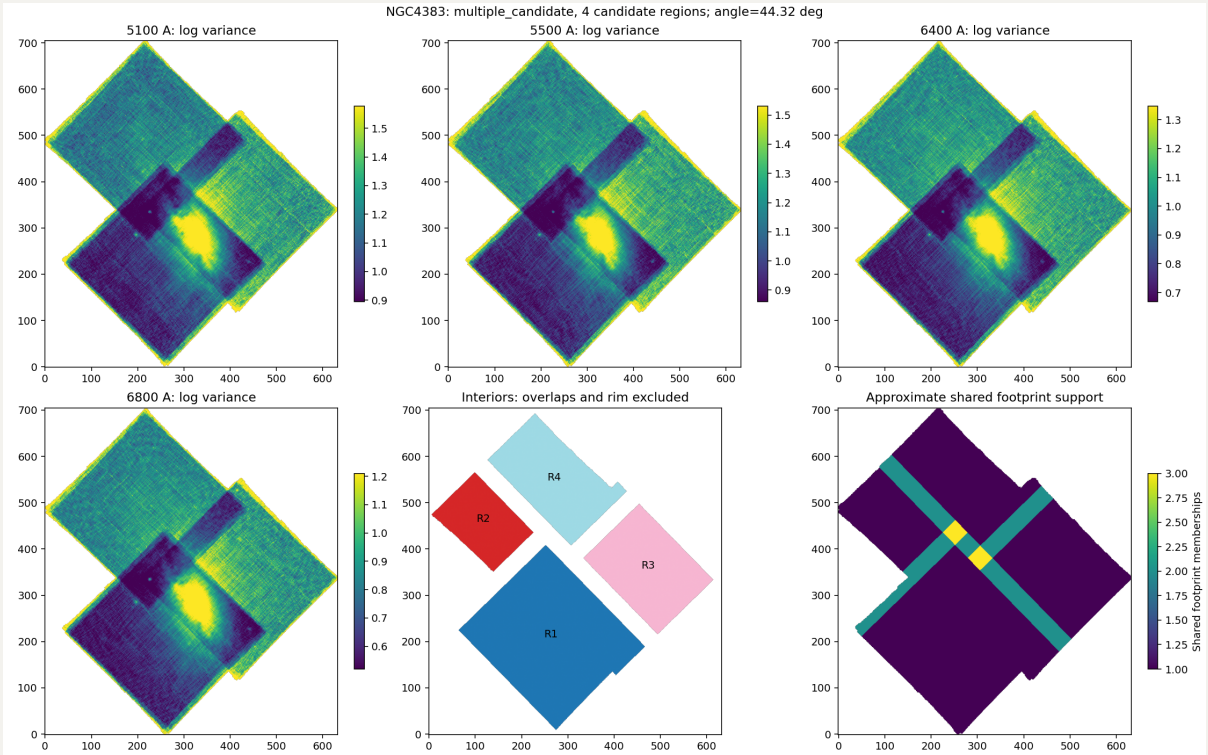
NGC4380 (needed)

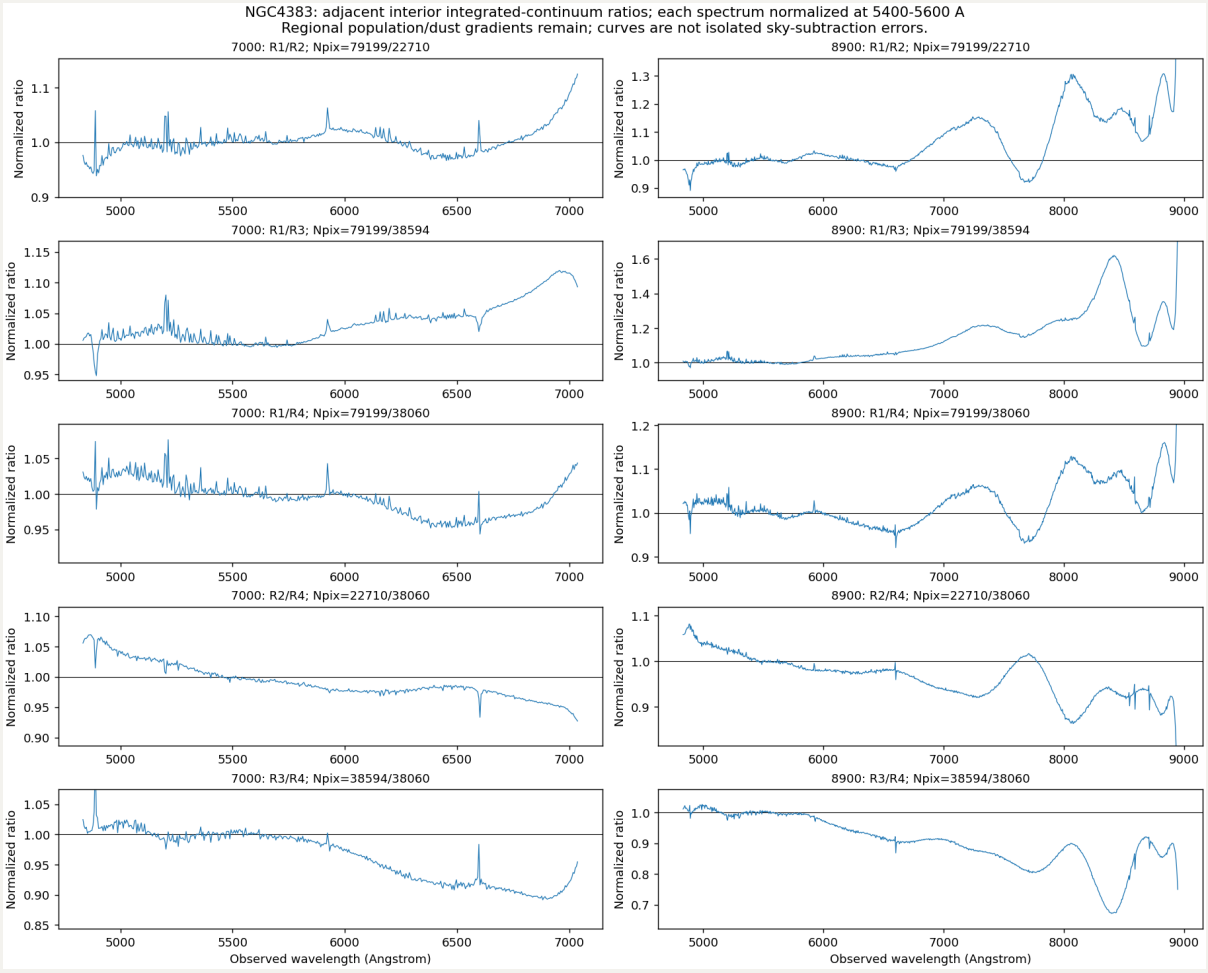


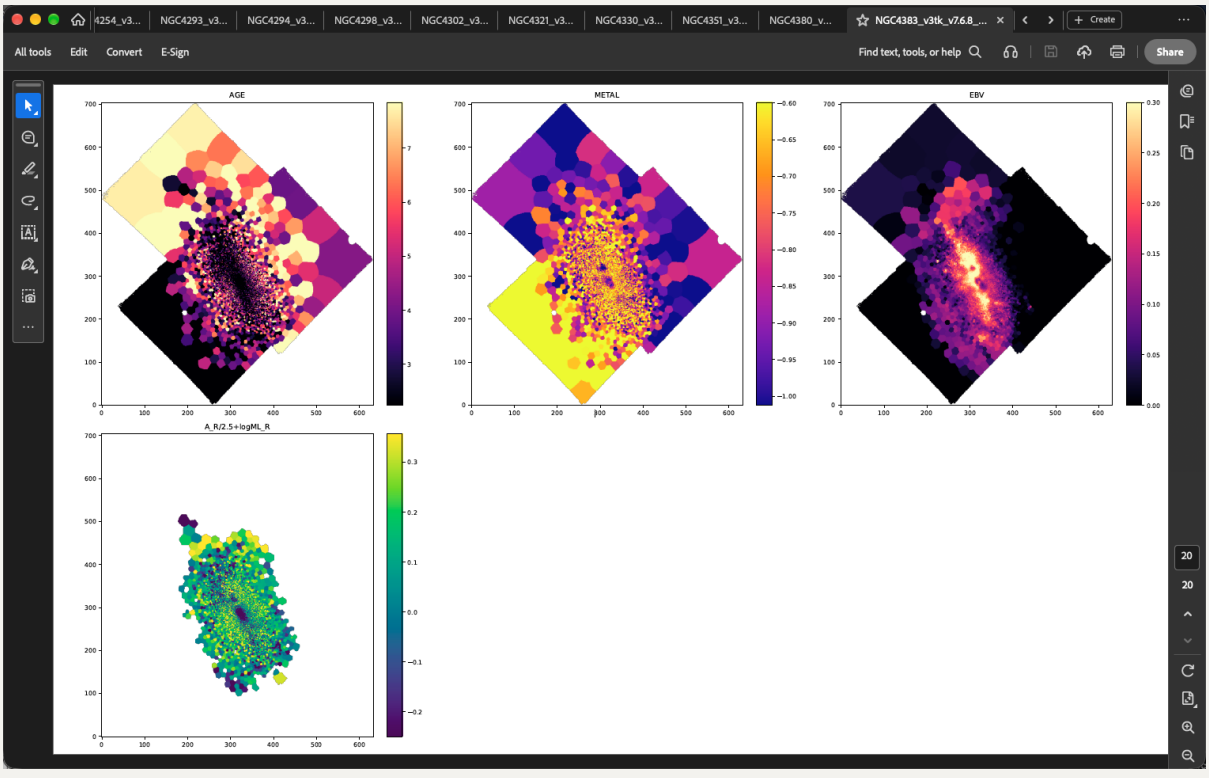

NGC4380: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=66232/62805
8900: R1/R2; Npix=66232/62805
7000: R1/R4; Npix=66232/69484
8900: R1/R4; Npix=66232/69484
7000: R2/R3; Npix=62805/69460
8900: R2/R3; Npix=62805/69460
7000: R2/R5; Npix=62805/59508
8900: R2/R5; Npix=62805/59508
7000: R3/R6; Npix=69460/69032
8900: R3/R6; Npix=69460/69032
7000: R4/R5; Npix=69484/59508
8900: R4/R5; Npix=69484/59508
7000: R5/R6; Npix=59508/69032
8900: R5/R6; Npix=59508/69032
Normalized ratio
Observed wavelength (Angstrom)



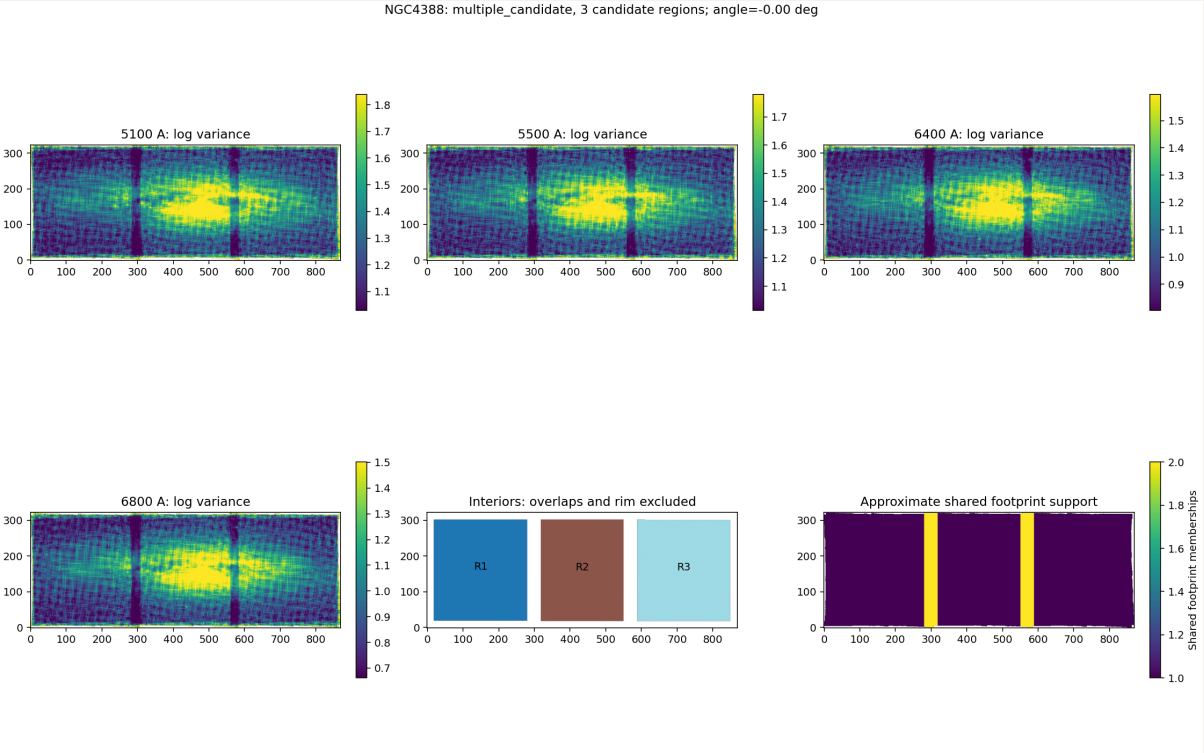
NGC4383 (maybe)

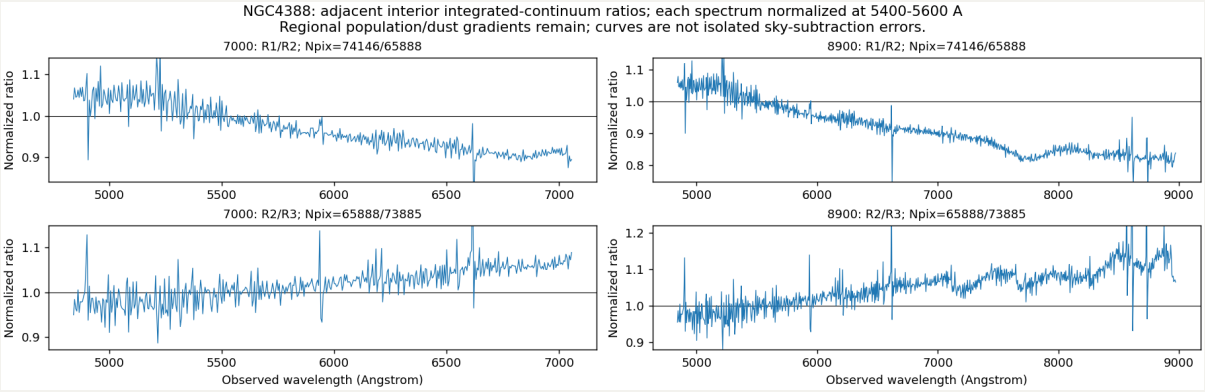


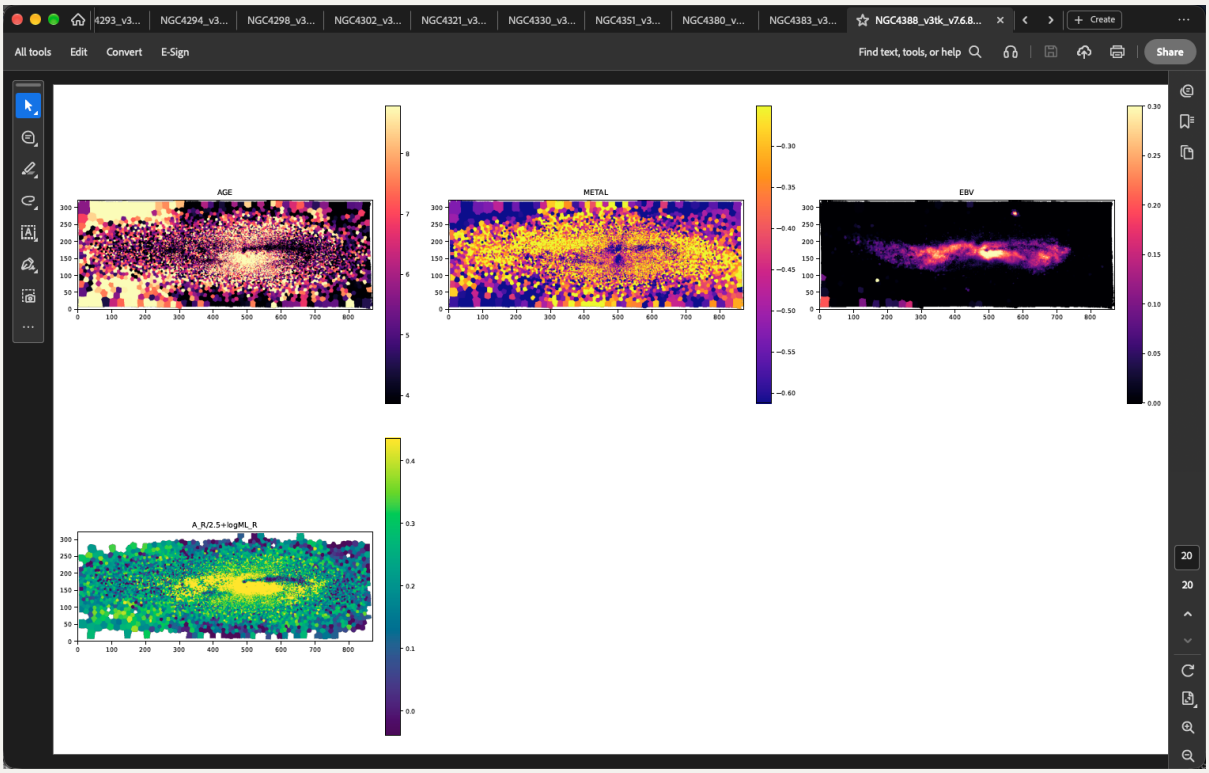




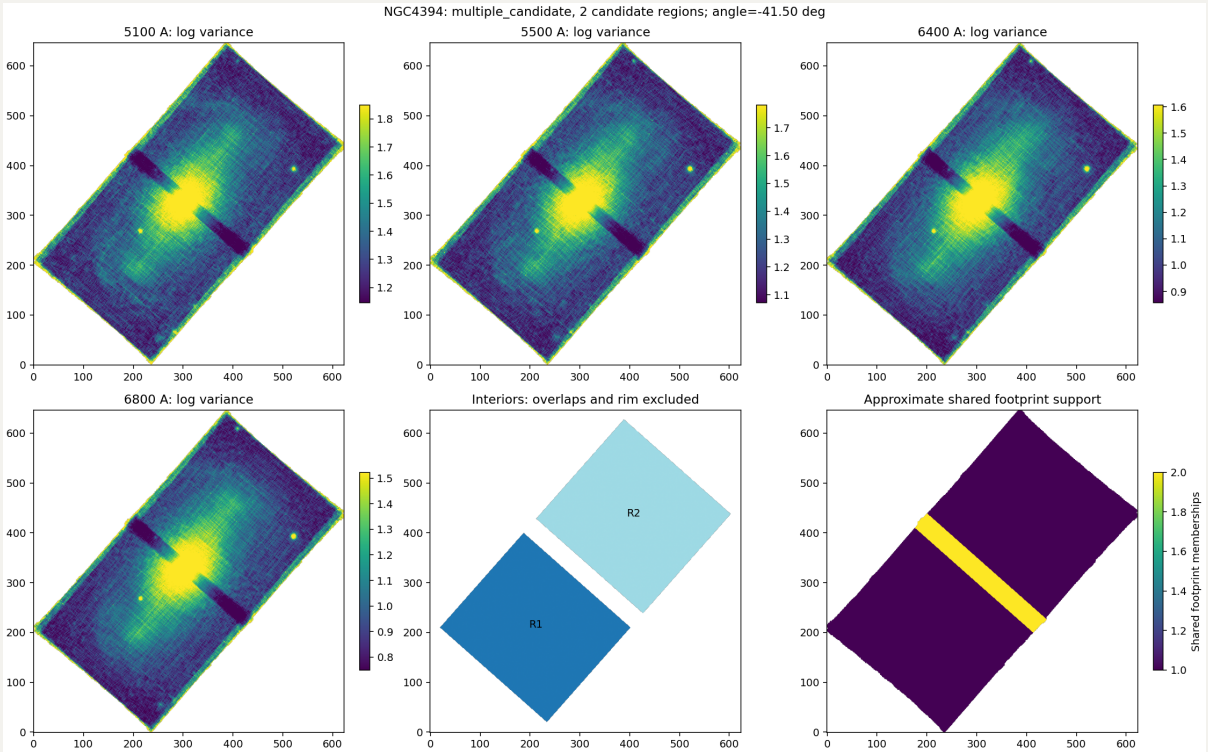
NGC4388 (fine)

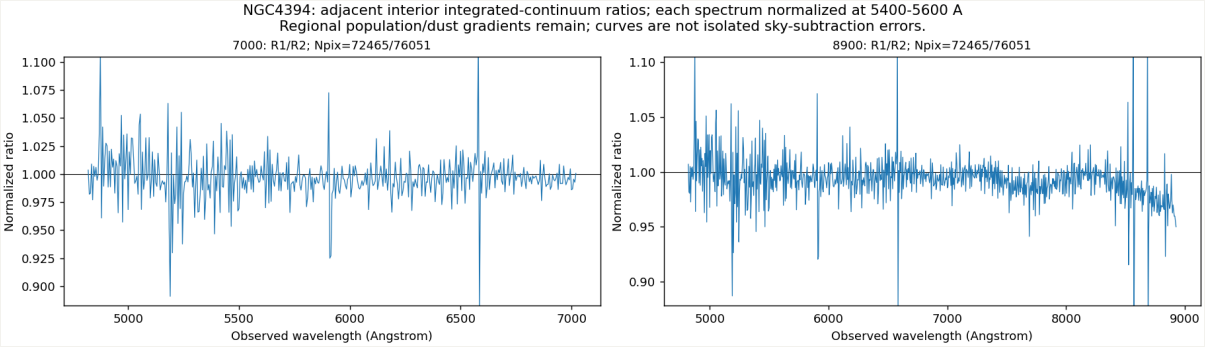

NGC4388: multiple_candidate, 3 candidate regions; angle=-0.00 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R1
R2
R3
Approximate shared footprint support
Shared footprint memberships

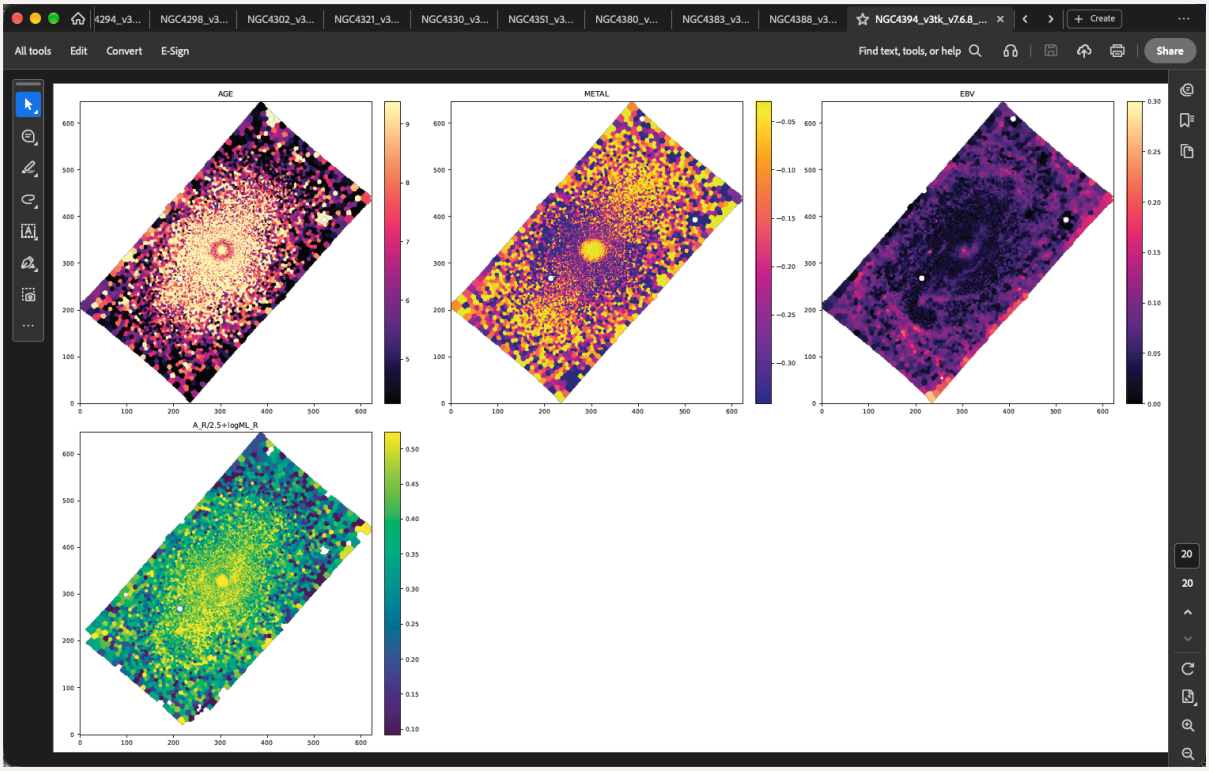

NGC4388: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=74146/65888
8900: R1/R2; Npix=74146/65888
7000: R2/R3; Npix=65888/73885
8900: R2/R3; Npix=65888/73885
Normalized ratio
Observed wavelength (Angstrom)



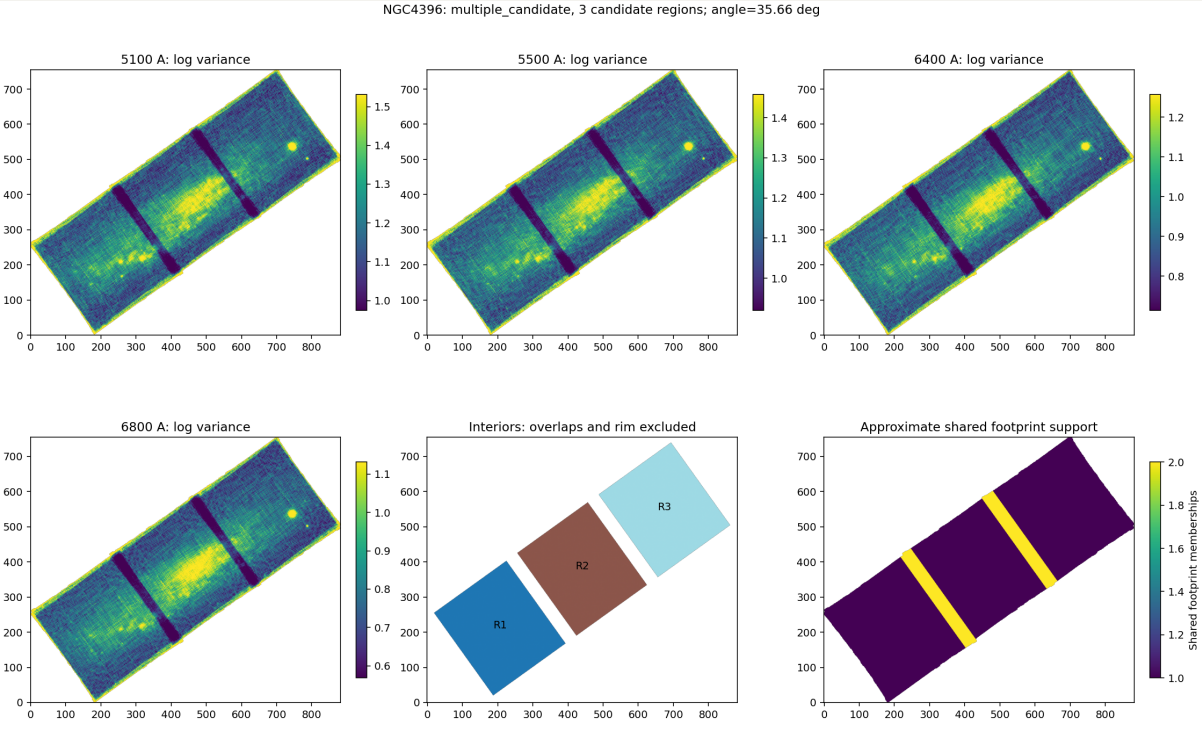
NGC4394 (fine)

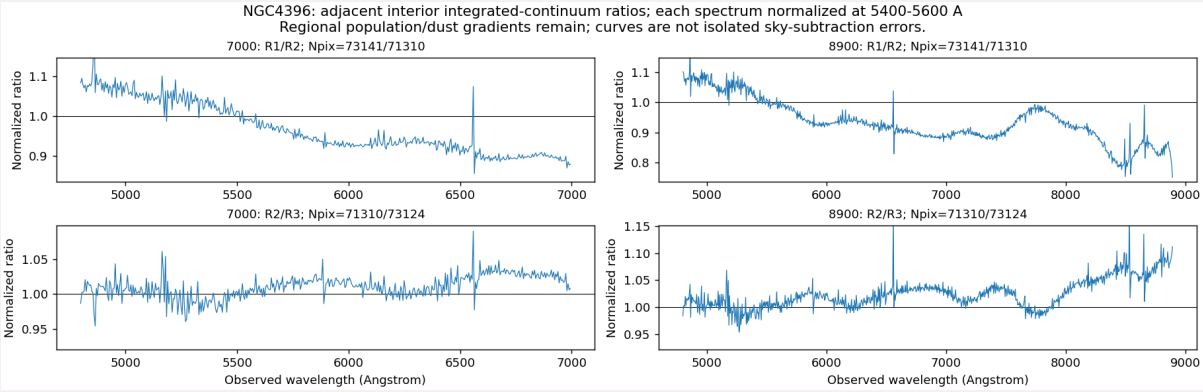


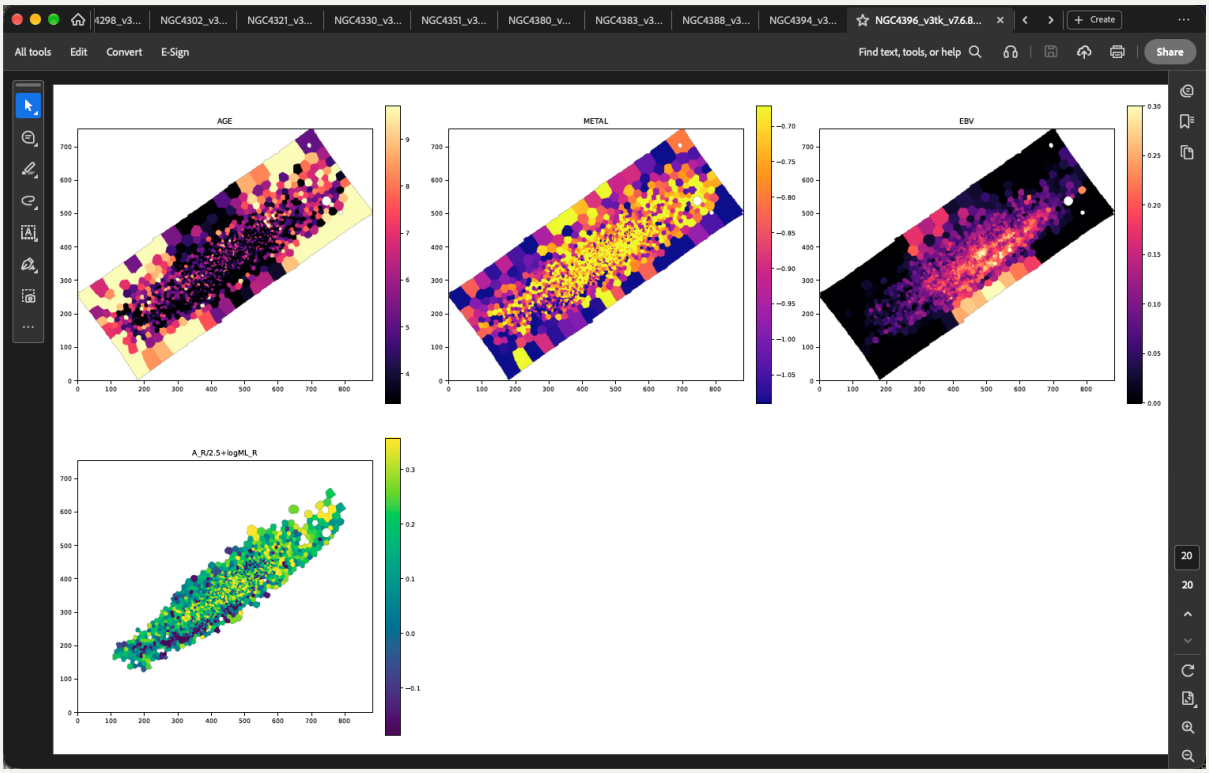




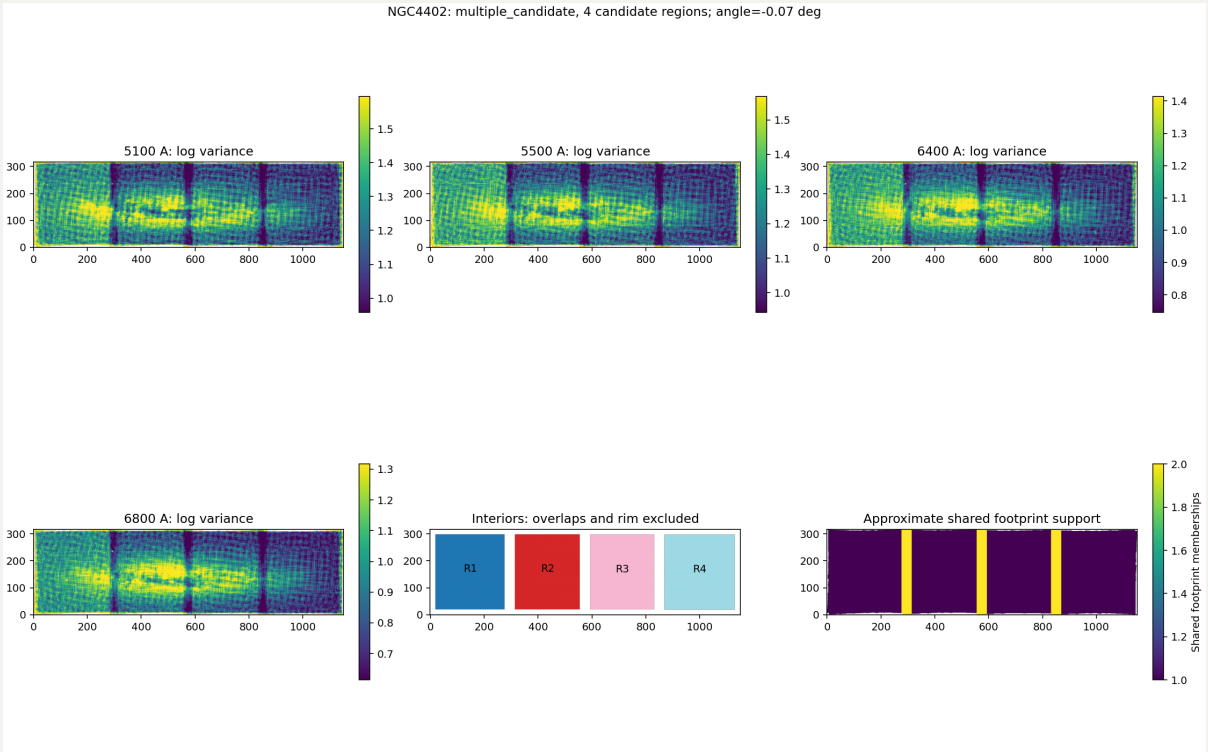
NGC4396 (fine)

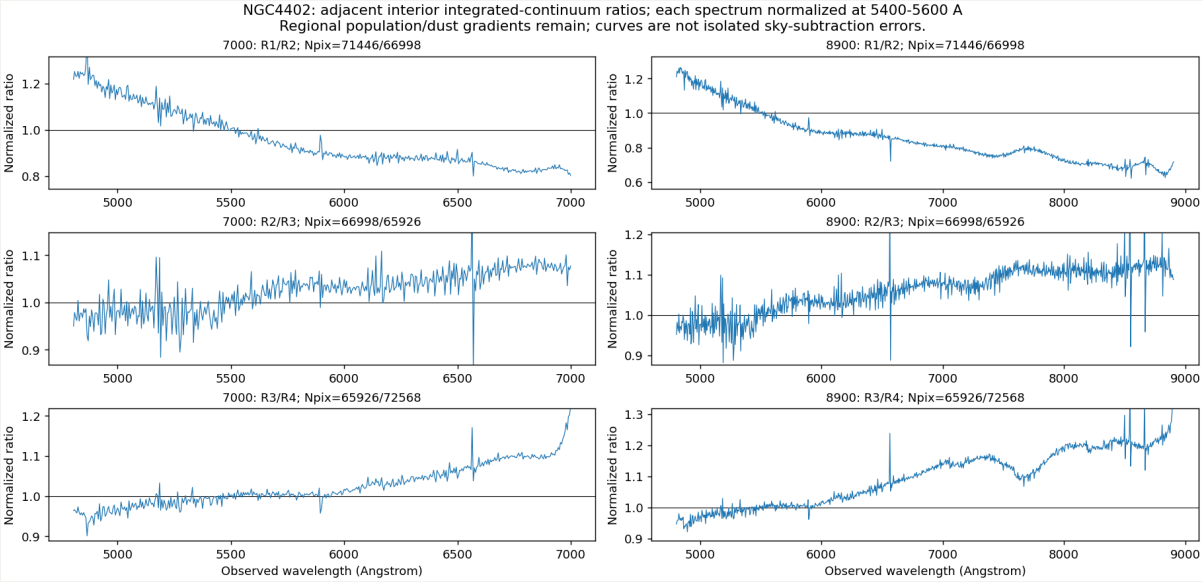

NGC4396: multiple_candidate, 3 candidate regions; angle=35.66 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R1
R2
R3
Approximate shared footprint support
Shared footprint memberships

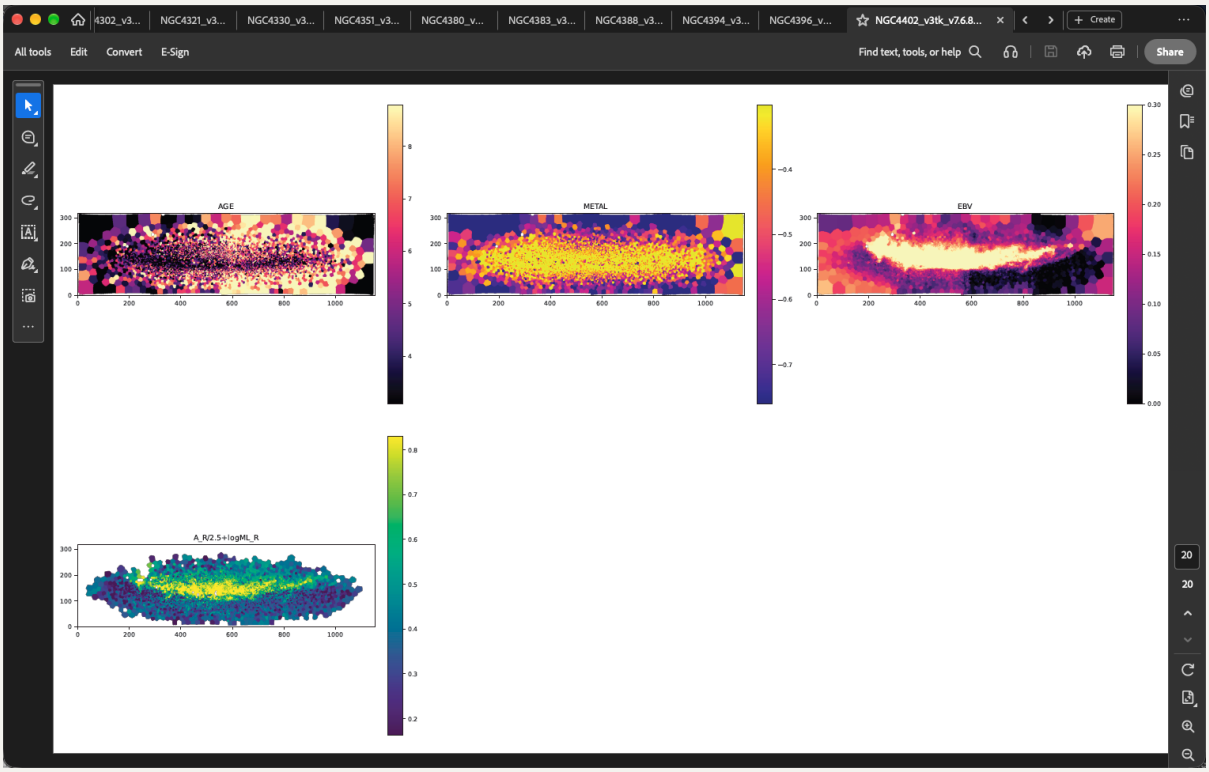

NGC4396: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=73141/71310
8900: R1/R2; Npix=73141/71310
7000: R2/R3; Npix=71310/73124
8900: R2/R3; Npix=71310/73124
Normalized ratio
Observed wavelength (Angstrom)



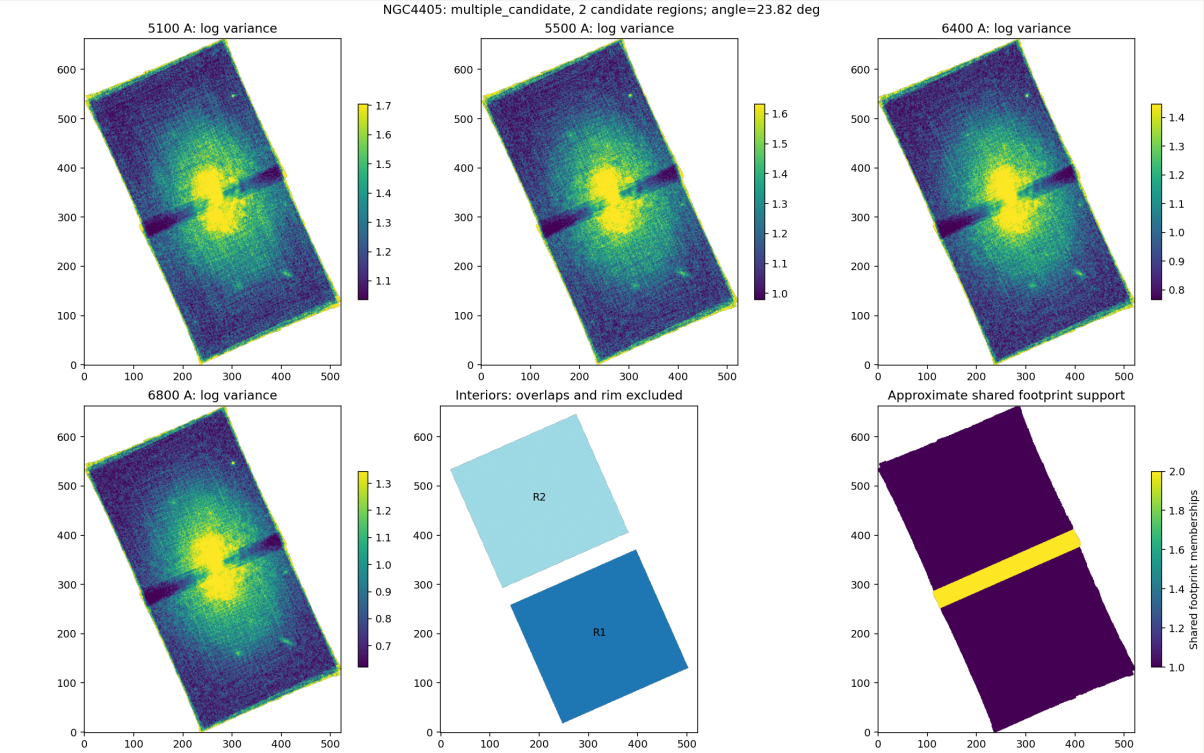
NGC4402 (fine)

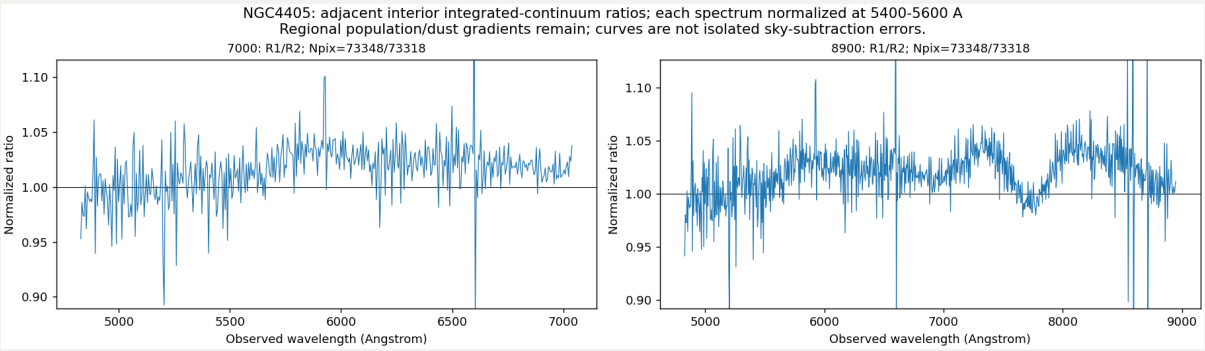


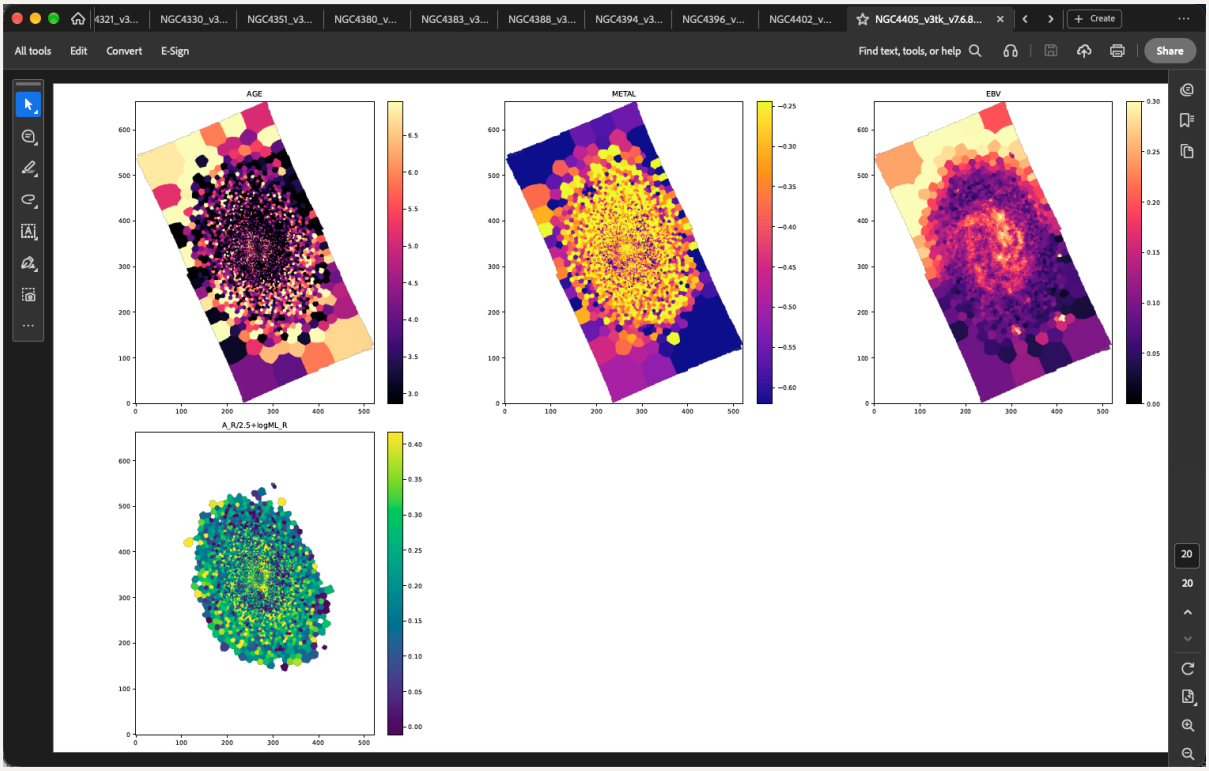




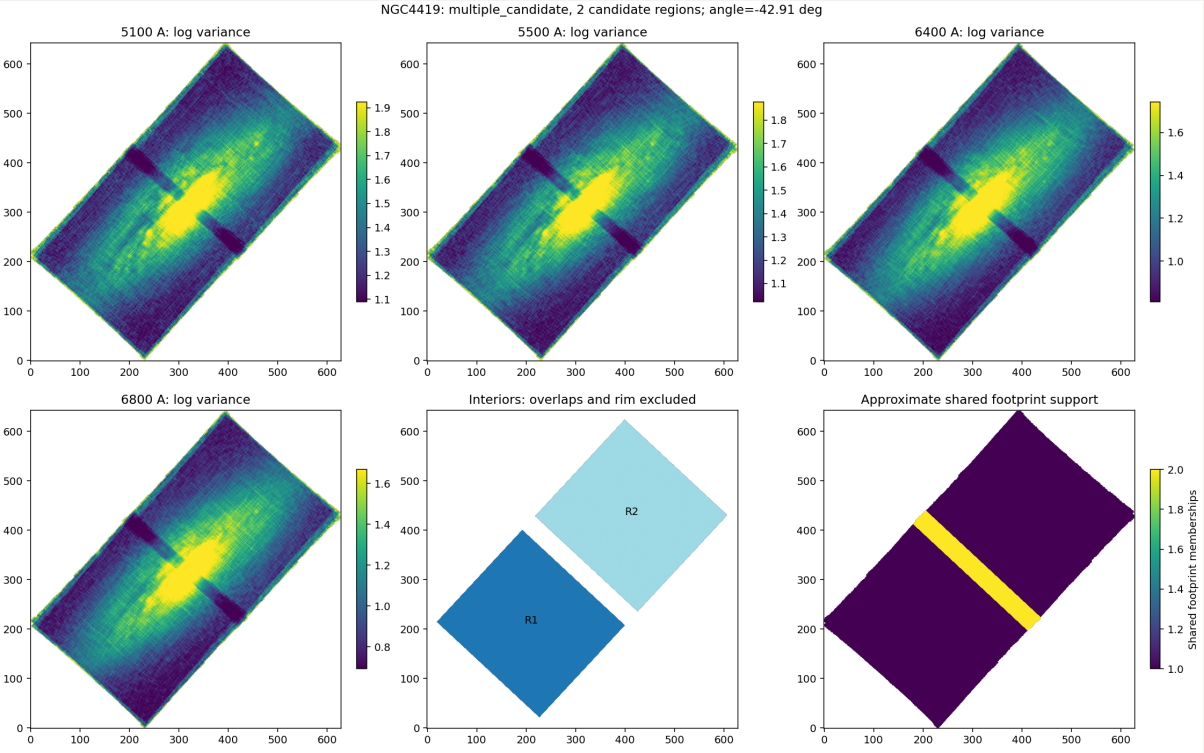
NGC4405 (fine)

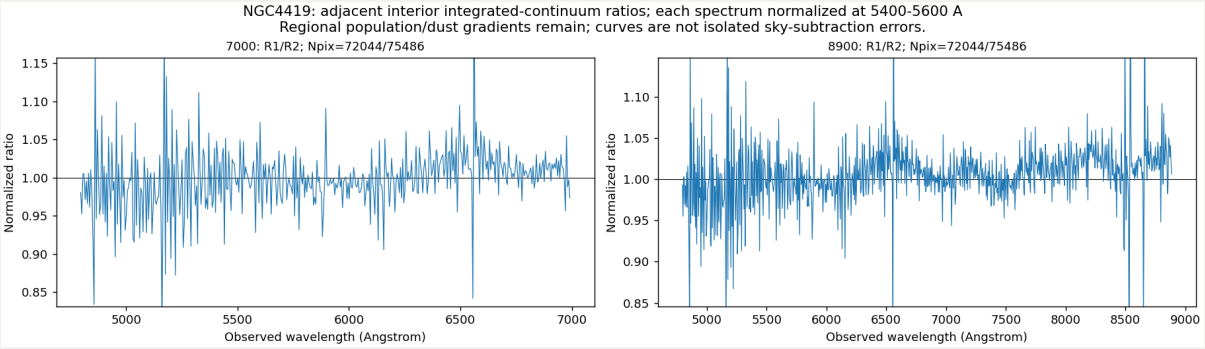

NGC4405: multiple_candidate, 2 candidate regions; angle=23.82 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R2
R1
Approximate shared footprint support
Shared footprint memberships

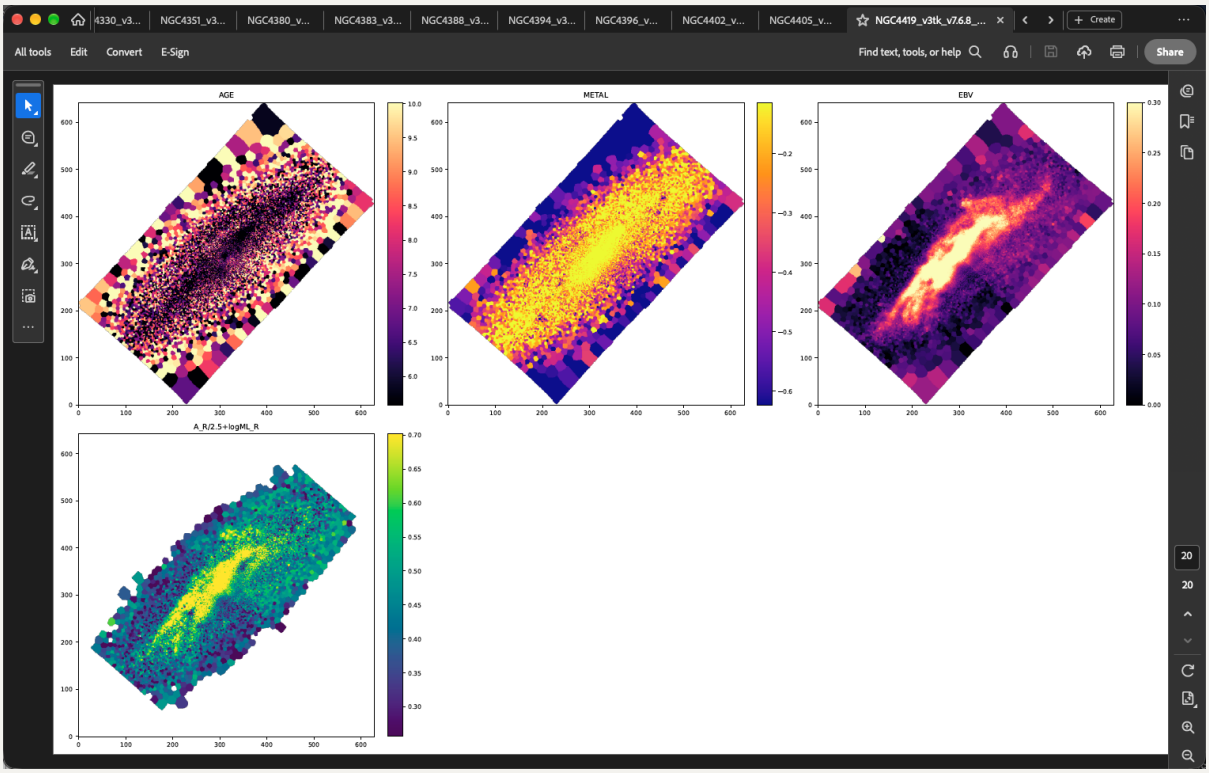

NGC4405: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=73348/73318
8900: R1/R2; Npix=73348/73318
Normalized ratio
Observed wavelength (Angstrom)


All tools
Edit
Convert
E-Sign
Share
AGE
METAL
EBV
A_R/2.5+logML_R

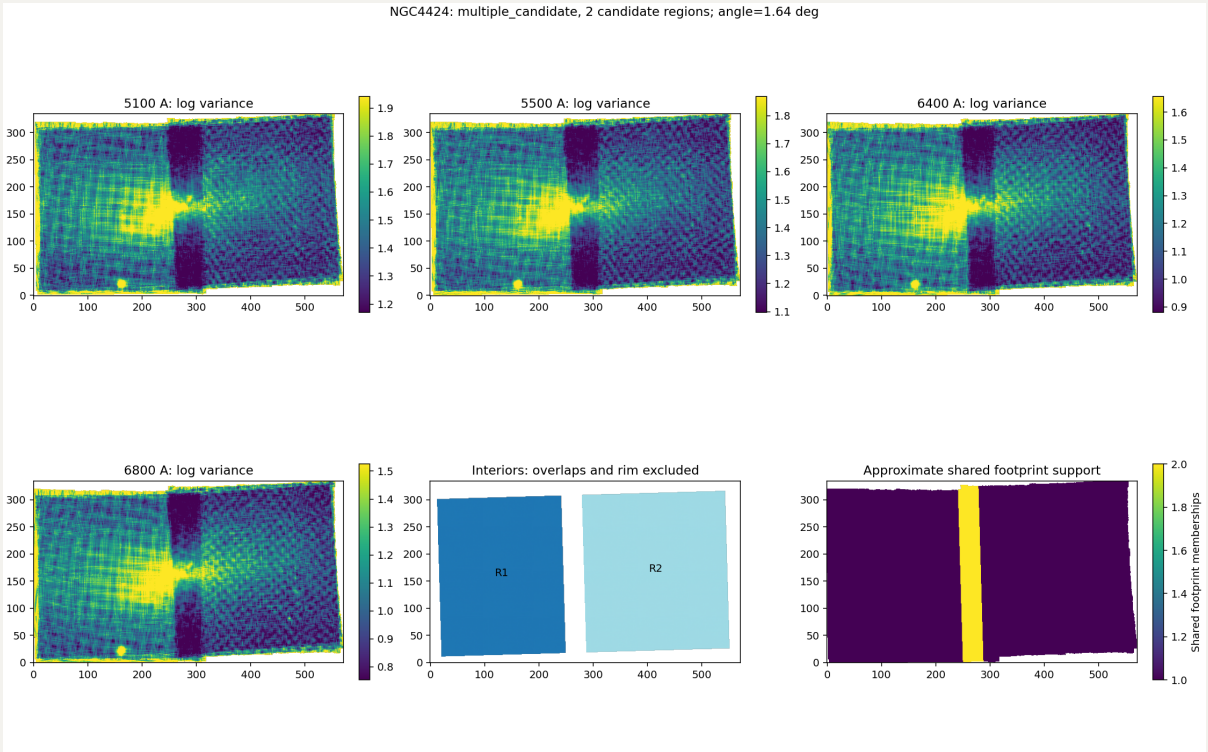
NGC4419 (fine)

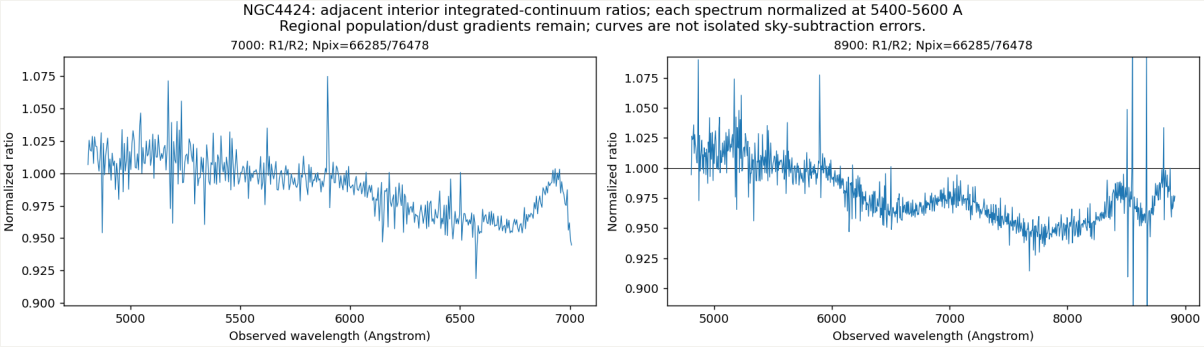


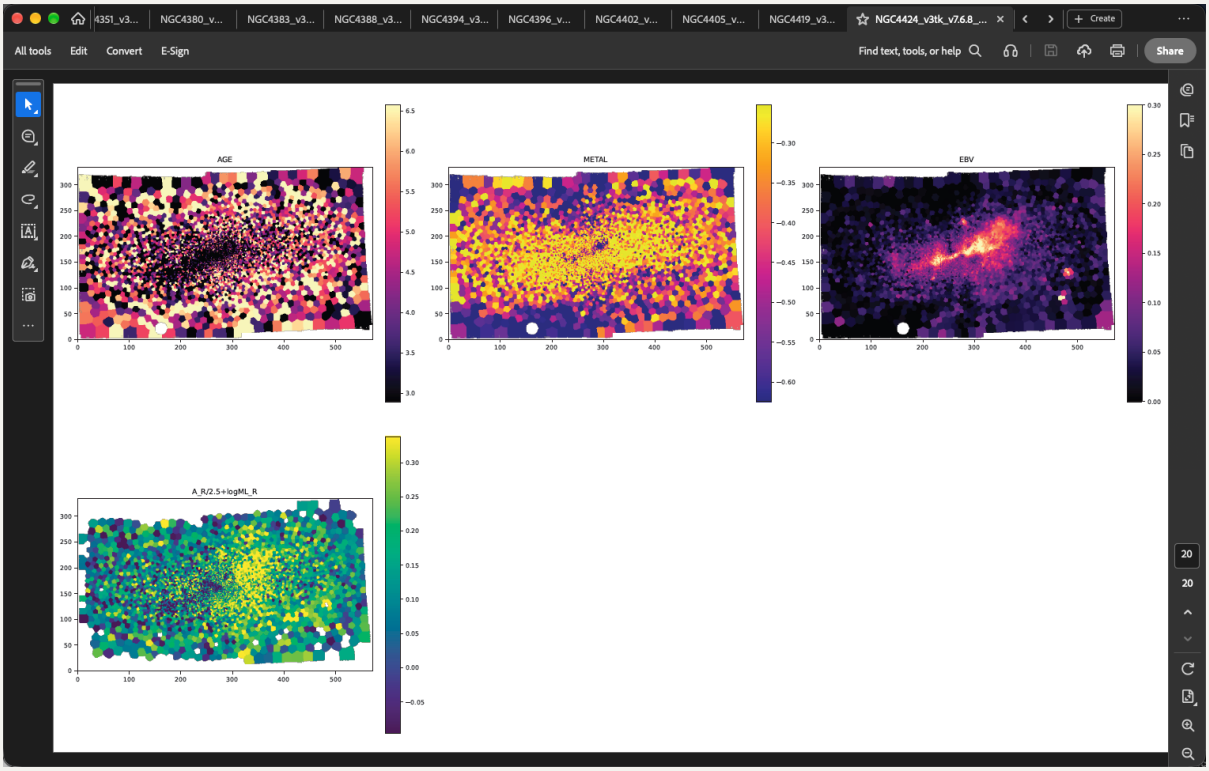




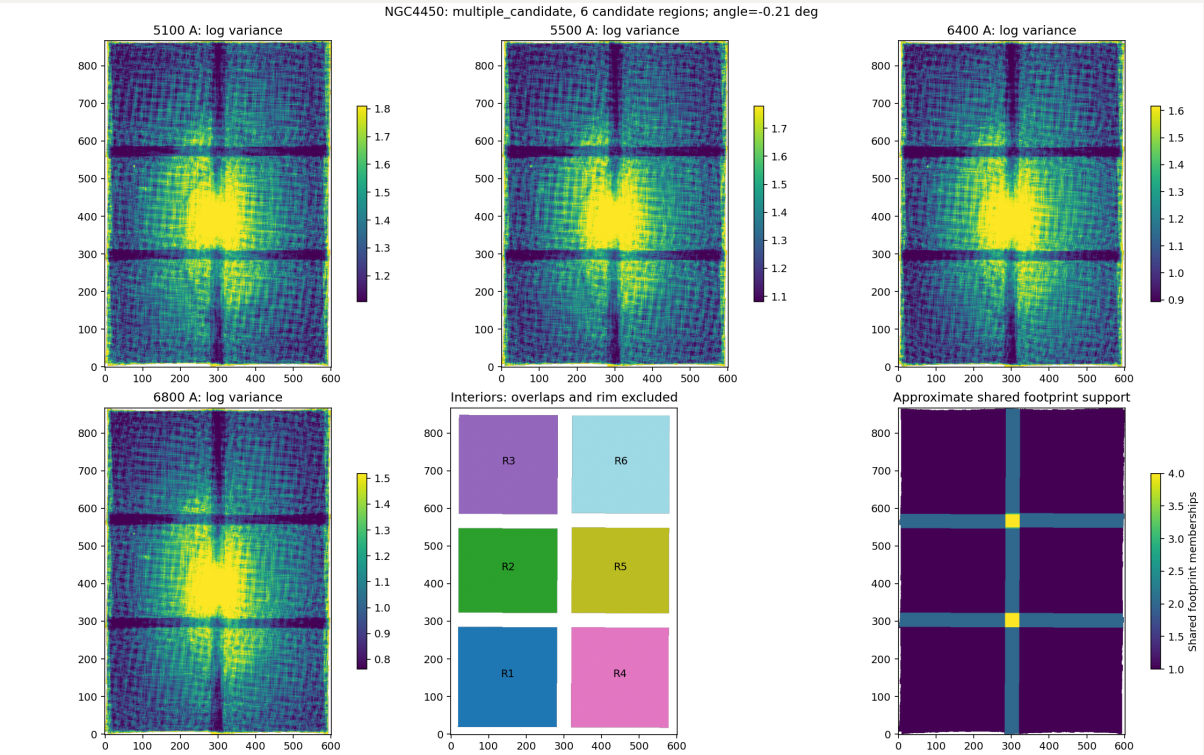
NGC4424 (fine)

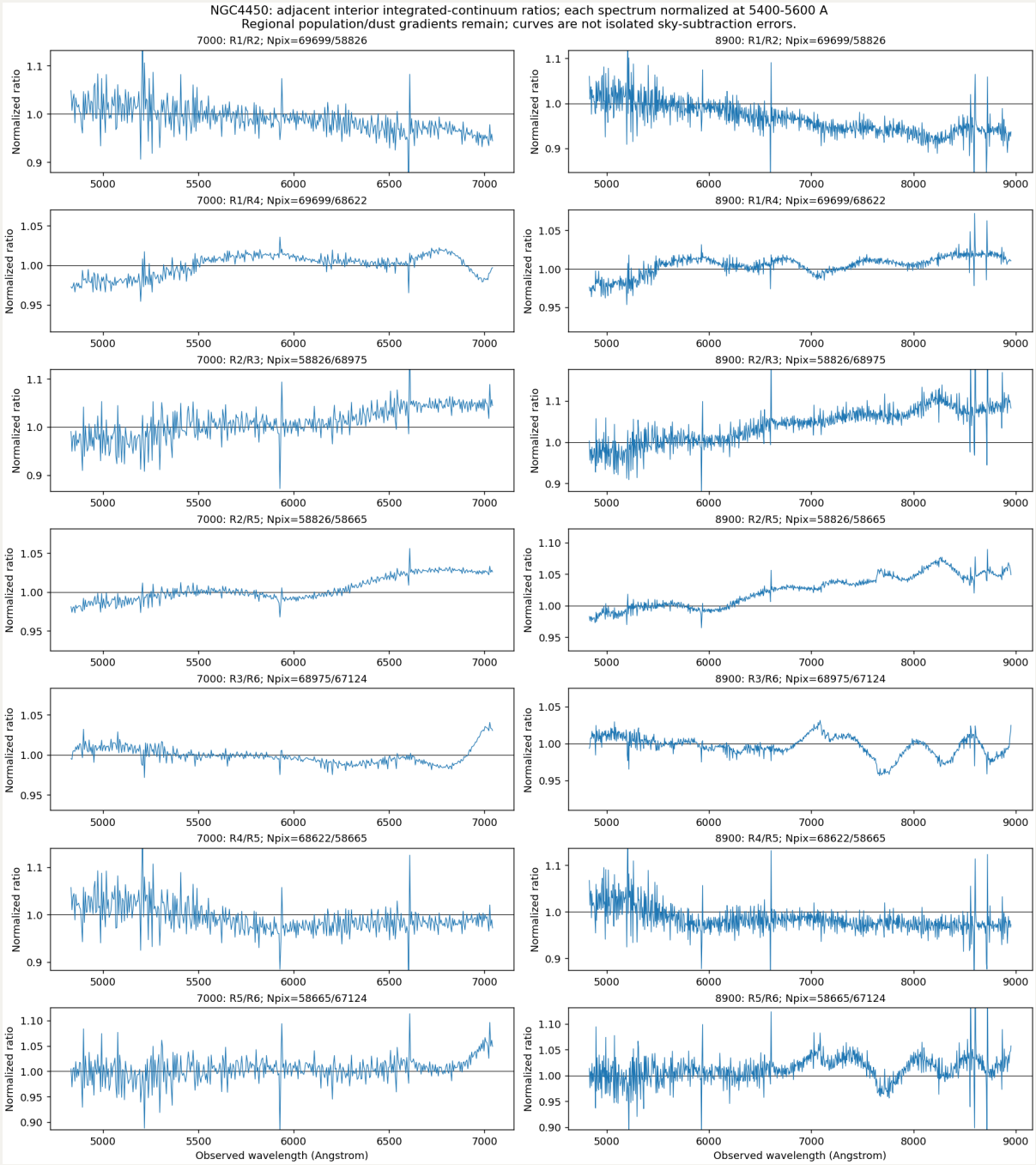


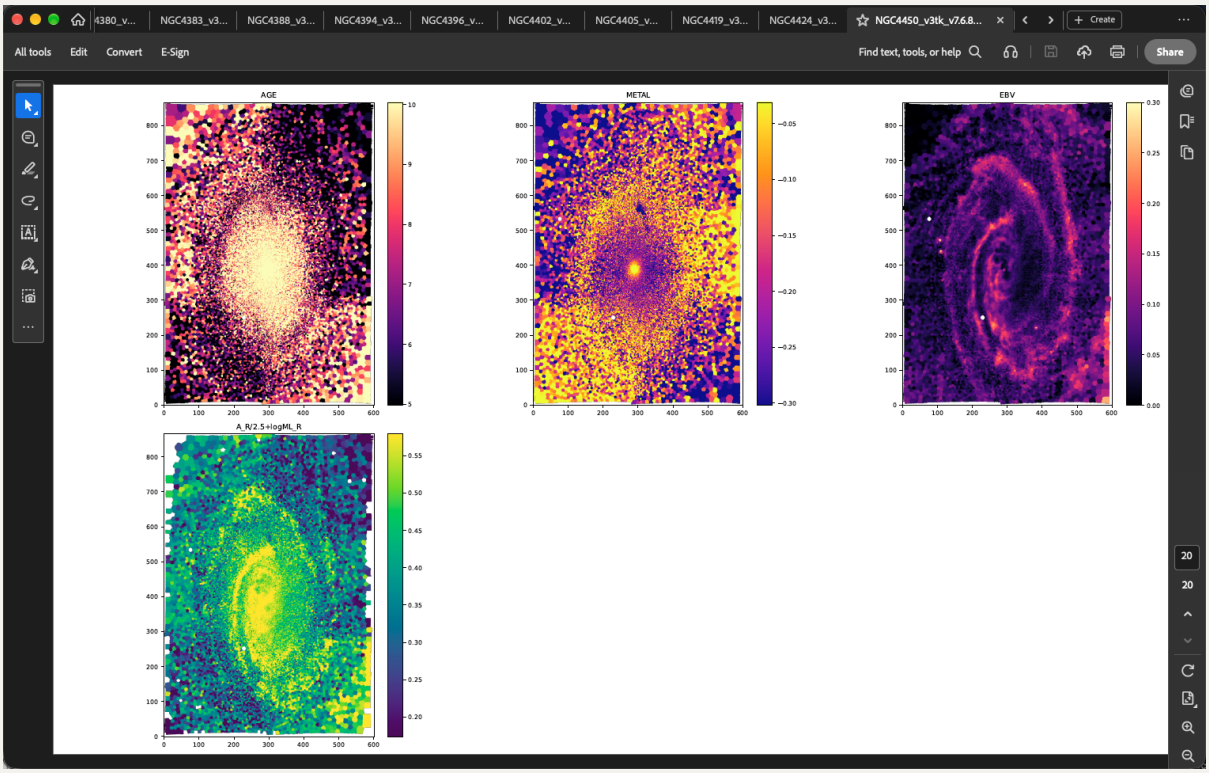




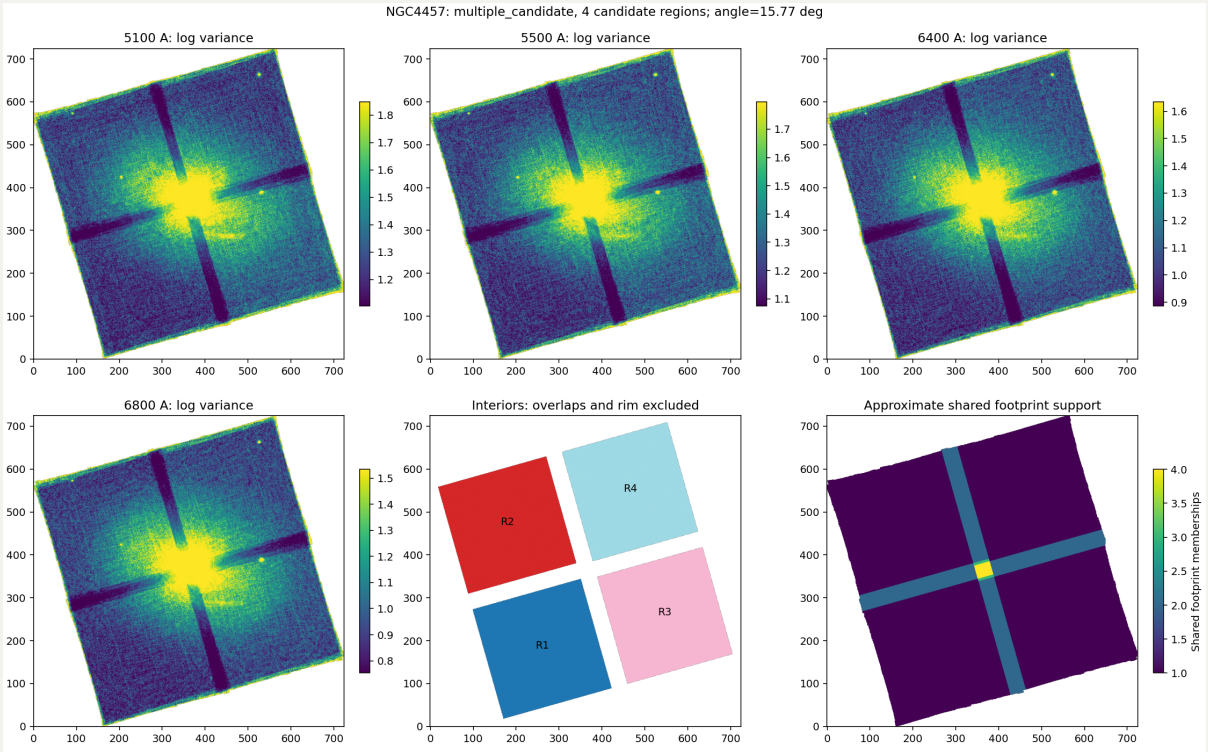
NGC4450 (needed)

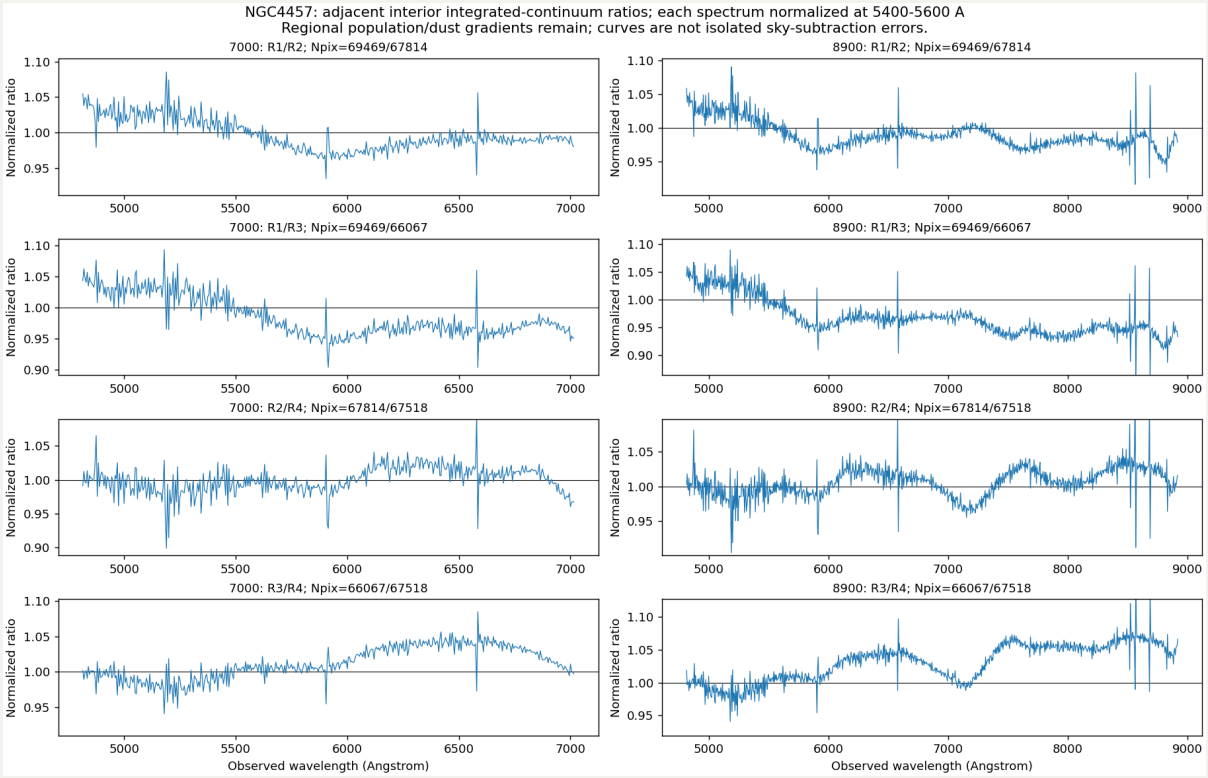

NGC4450: multiple_candidate, 6 candidate regions; angle=-0.21 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
Approximate shared footprint support
R1
R2
R3
R4
R5
R6
Shared footprint memberships

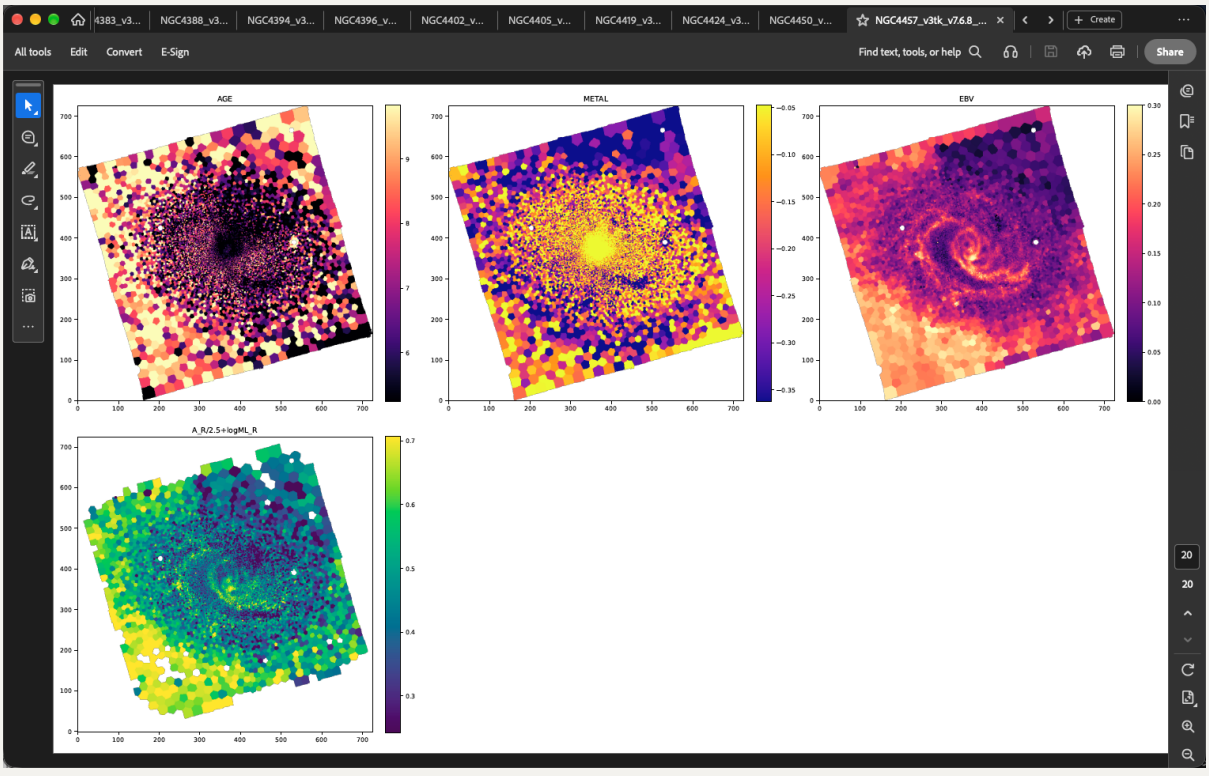

NGC4450: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=69699/58826
8900: R1/R2; Npix=69699/58826
7000: R1/R4; Npix=69699/68622
8900: R1/R4; Npix=69699/68622
7000: R2/R3; Npix=58826/68975
8900: R2/R3; Npix=58826/68975
7000: R2/R5; Npix=58826/58665
8900: R2/R5; Npix=58826/58665
7000: R3/R6; Npix=68975/67124
8900: R3/R6; Npix=68975/67124
7000: R4/R5; Npix=68622/58665
8900: R4/R5; Npix=68622/58665
7000: R5/R6; Npix=58665/67124
8900: R5/R6; Npix=58665/67124
Normalized ratio
Observed wavelength (Angstrom)



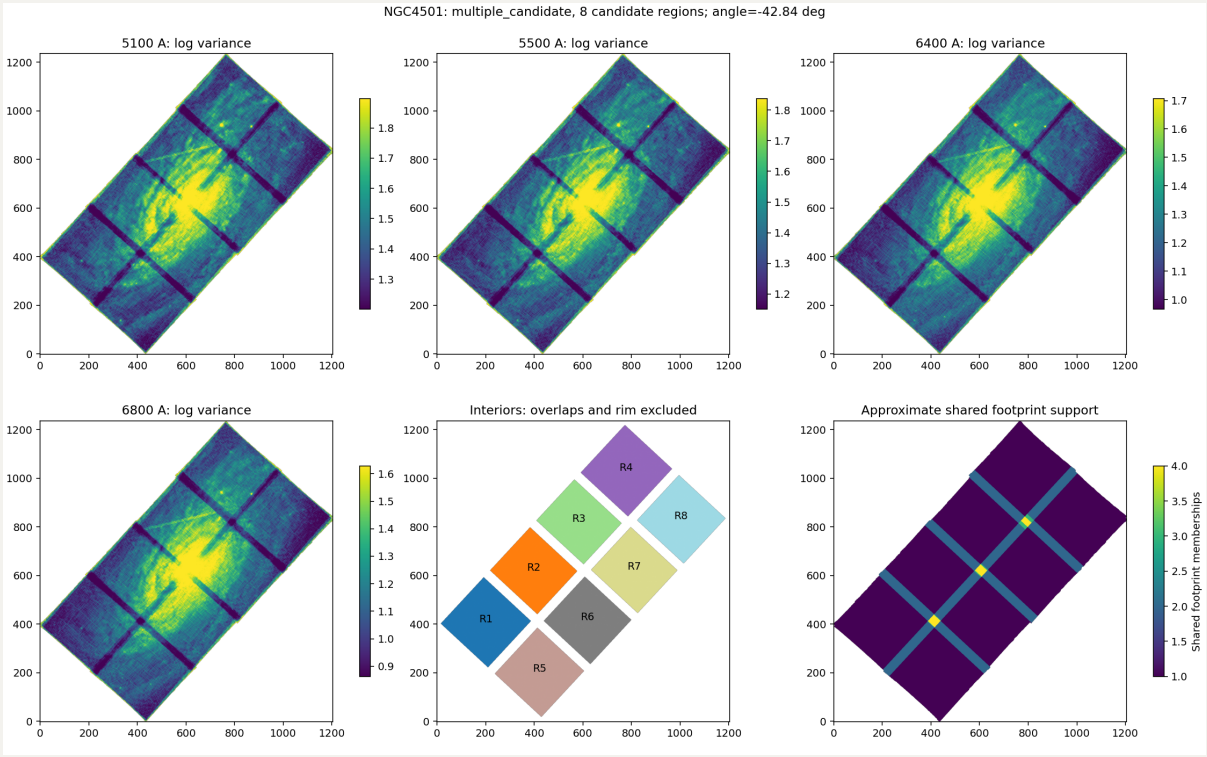
NGC4457 (needed)

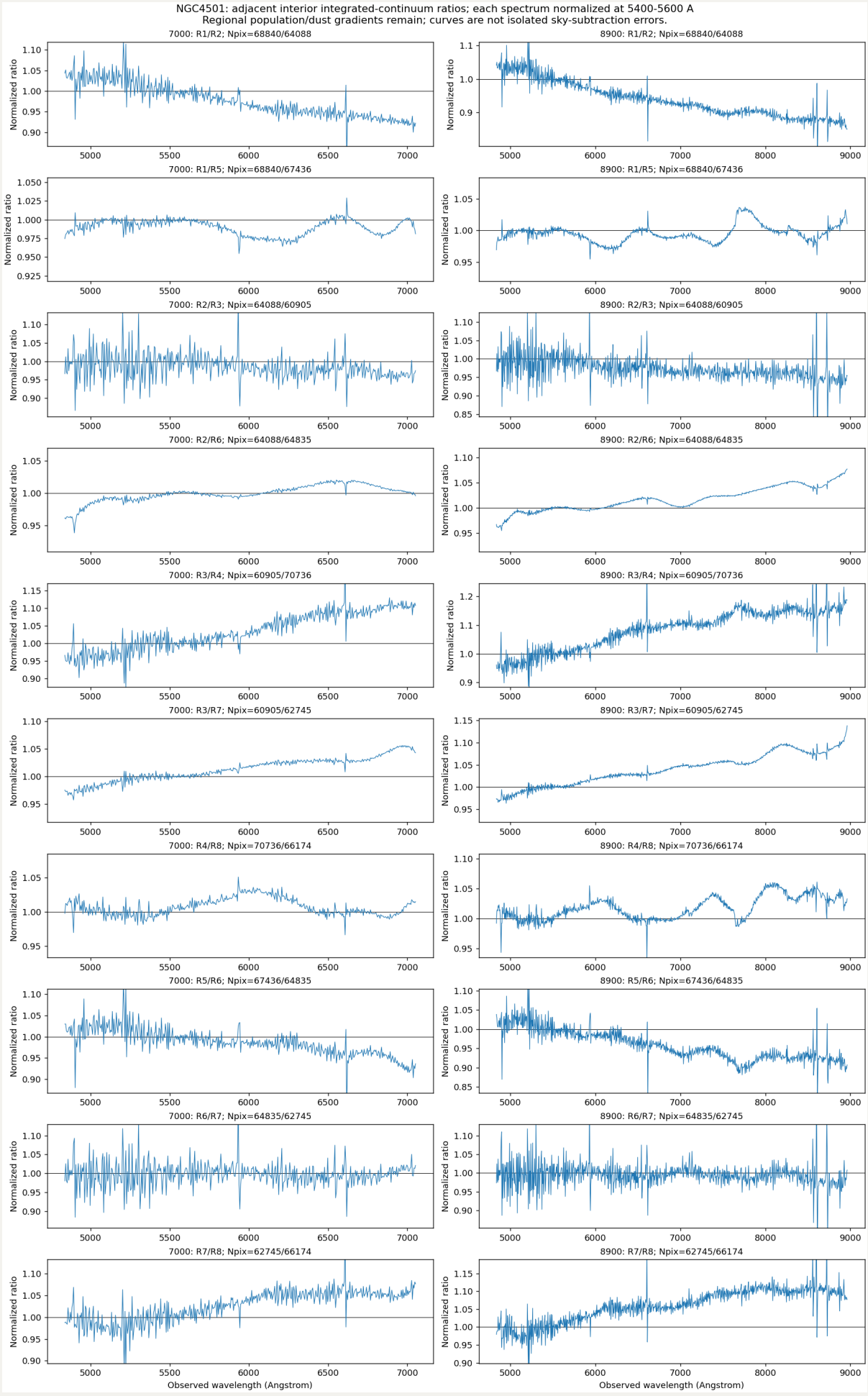


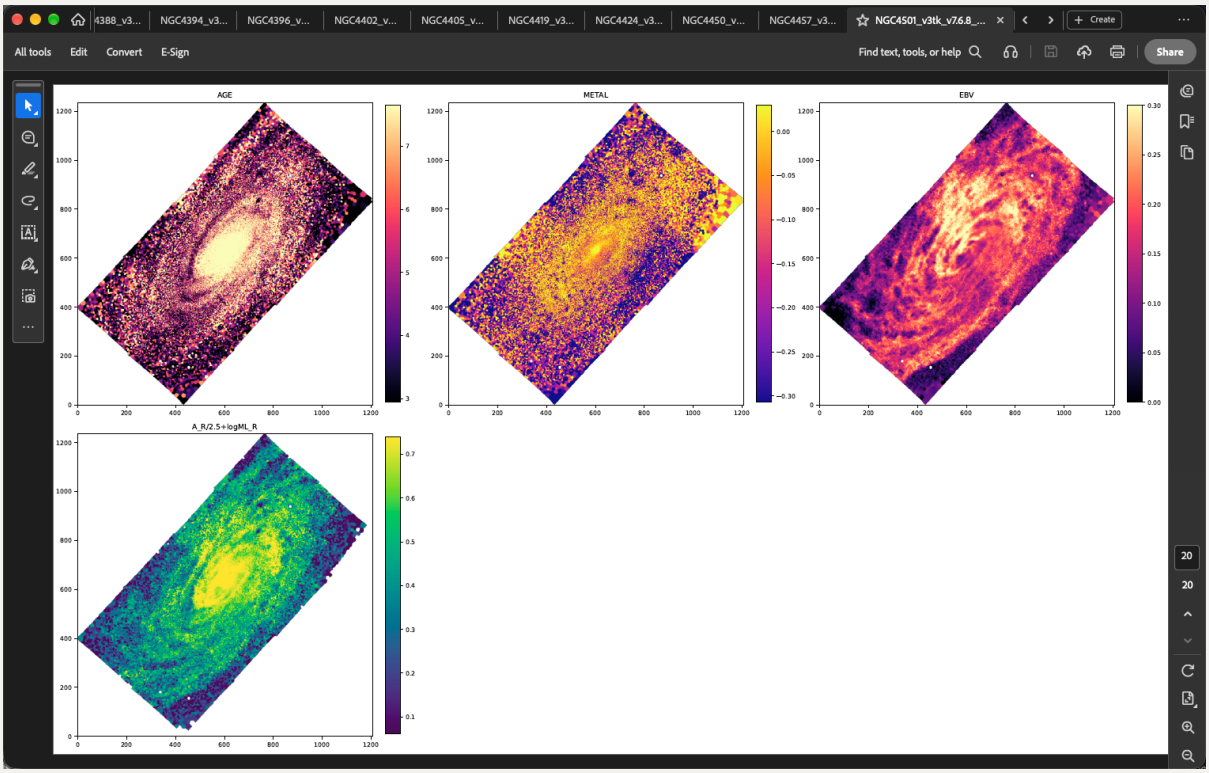




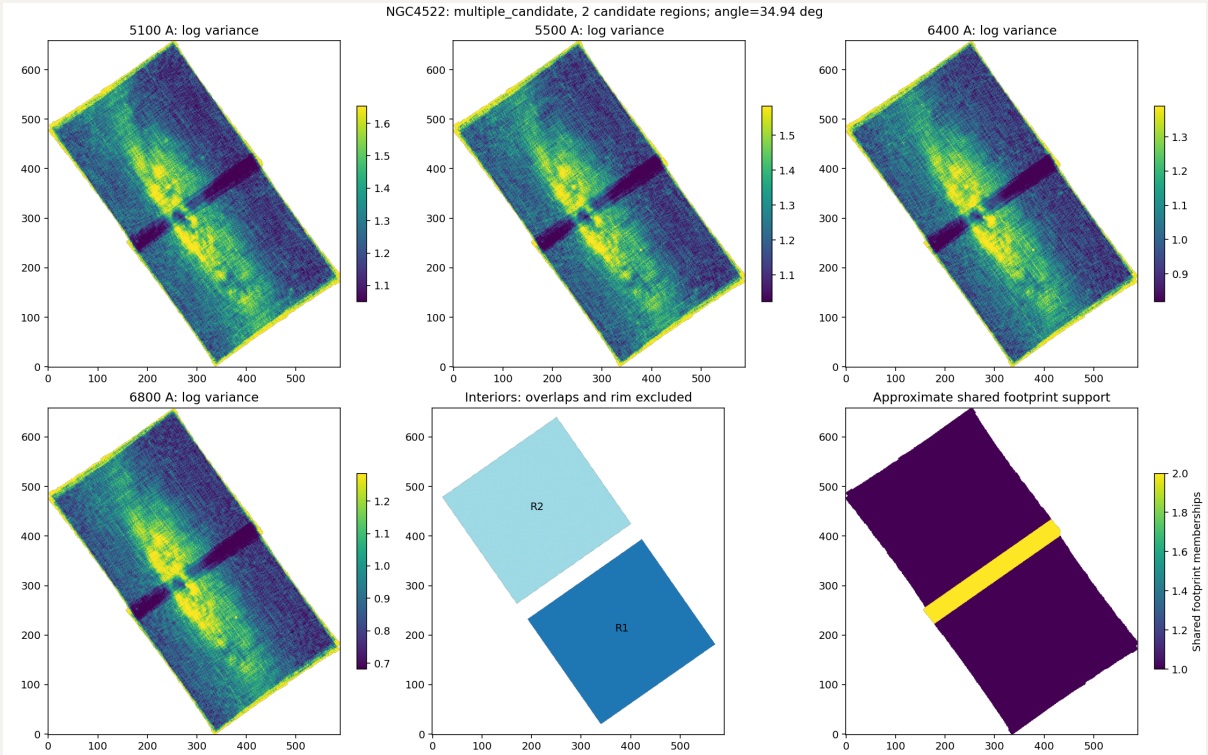
NGC4501 (needed)

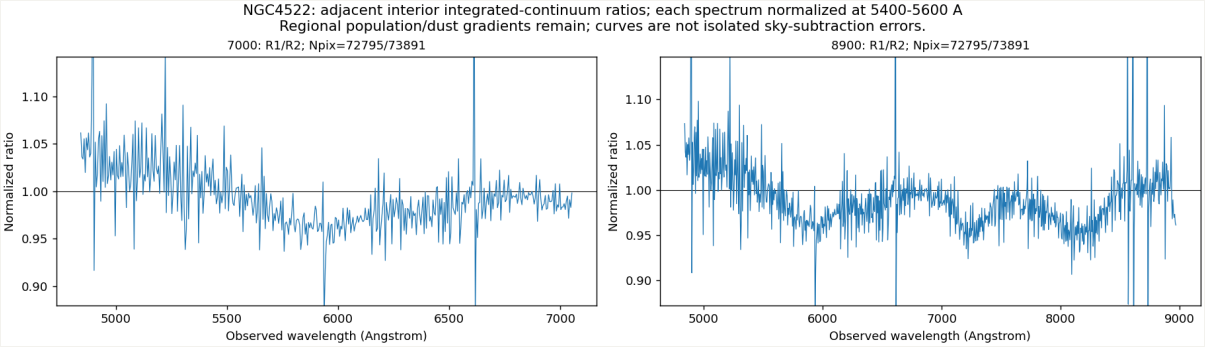

NGC4501: multiple_candidate, 8 candidate regions; angle=-42.84 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R1
R2
R3
R4
R5
R6
R7
R8
Approximate shared footprint support
Shared footprint memberships

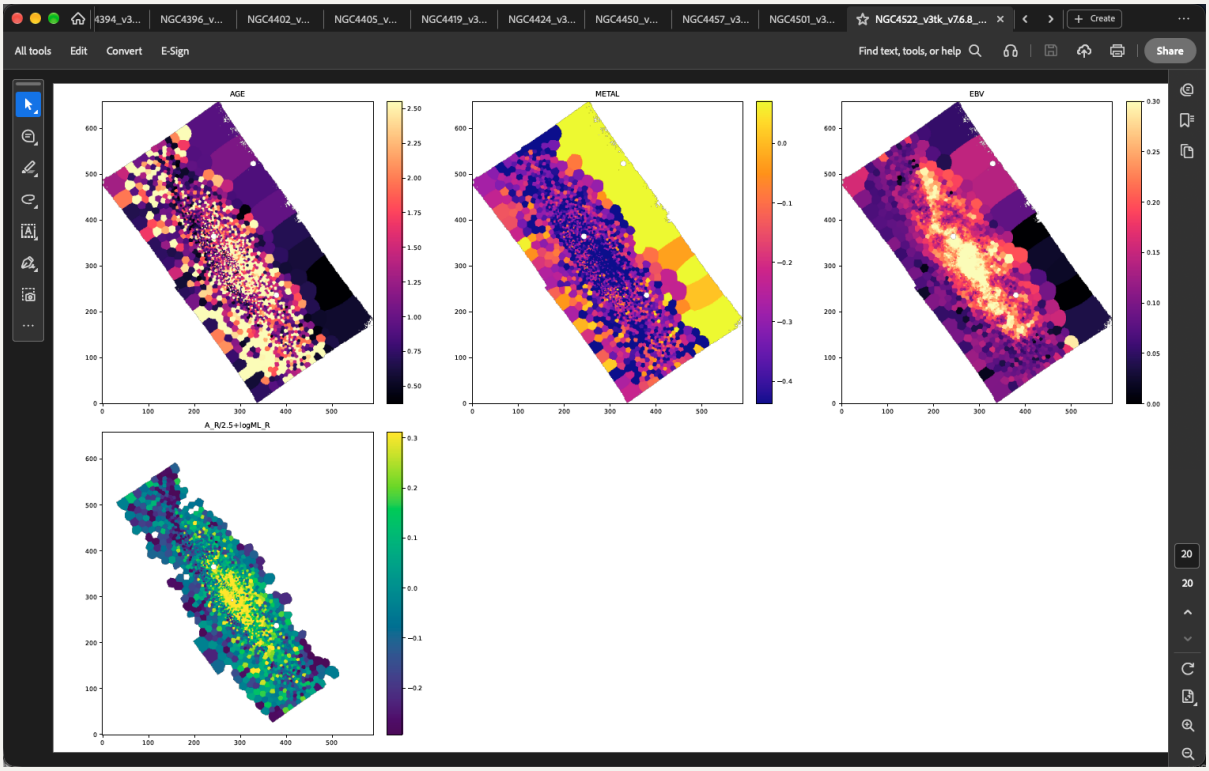

NGC4501: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=68840/64088
8900: R1/R2; Npix=68840/64088
7000: R1/R5; Npix=68840/67436
8900: R1/R5; Npix=68840/67436
7000: R2/R3; Npix=64088/60905
8900: R2/R3; Npix=64088/60905
7000: R2/R6; Npix=64088/64835
8900: R2/R6; Npix=64088/64835
7000: R3/R4; Npix=60905/70736
8900: R3/R4; Npix=60905/70736
7000: R3/R7; Npix=60905/62745
8900: R3/R7; Npix=60905/62745
7000: R4/R8; Npix=70736/66174
8900: R4/R8; Npix=70736/66174
7000: R5/R6; Npix=67436/64835
8900: R5/R6; Npix=67436/64835
7000: R6/R7; Npix=64835/62745
8900: R6/R7; Npix=64835/62745
7000: R7/R8; Npix=62745/66174
8900: R7/R8; Npix=62745/66174
Normalized ratio
Observed wavelength (Angstrom)



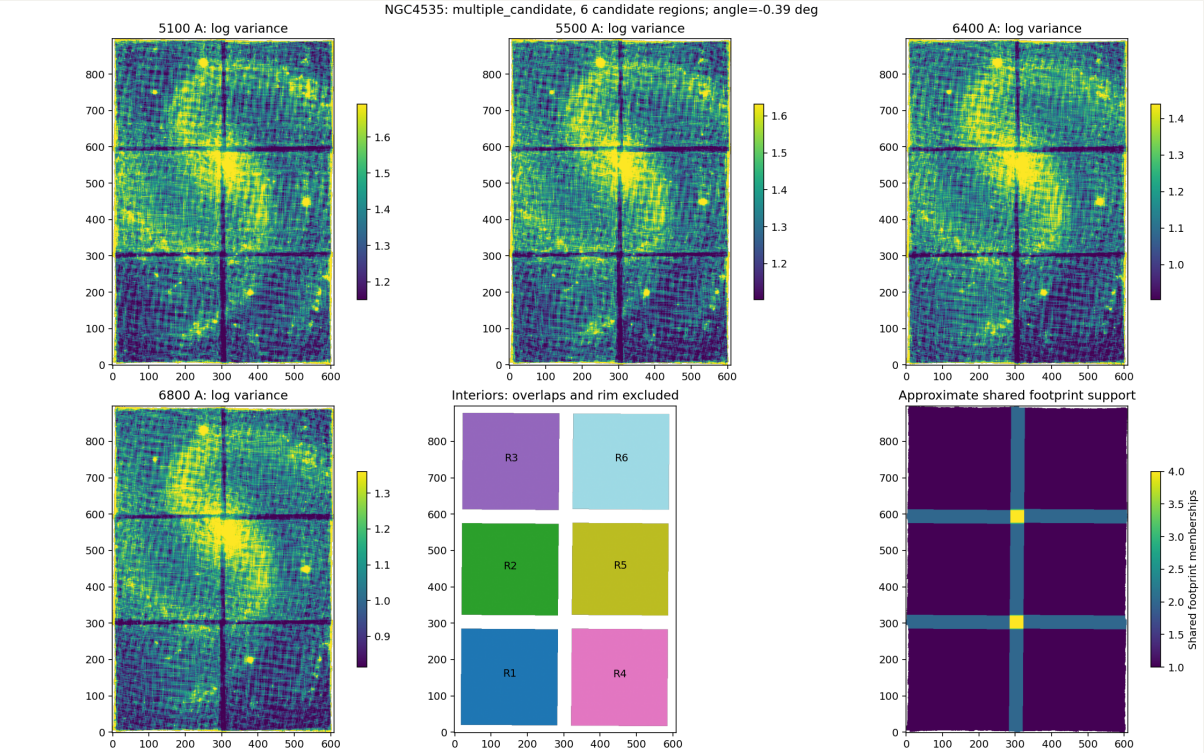
NGC4522 (fine)

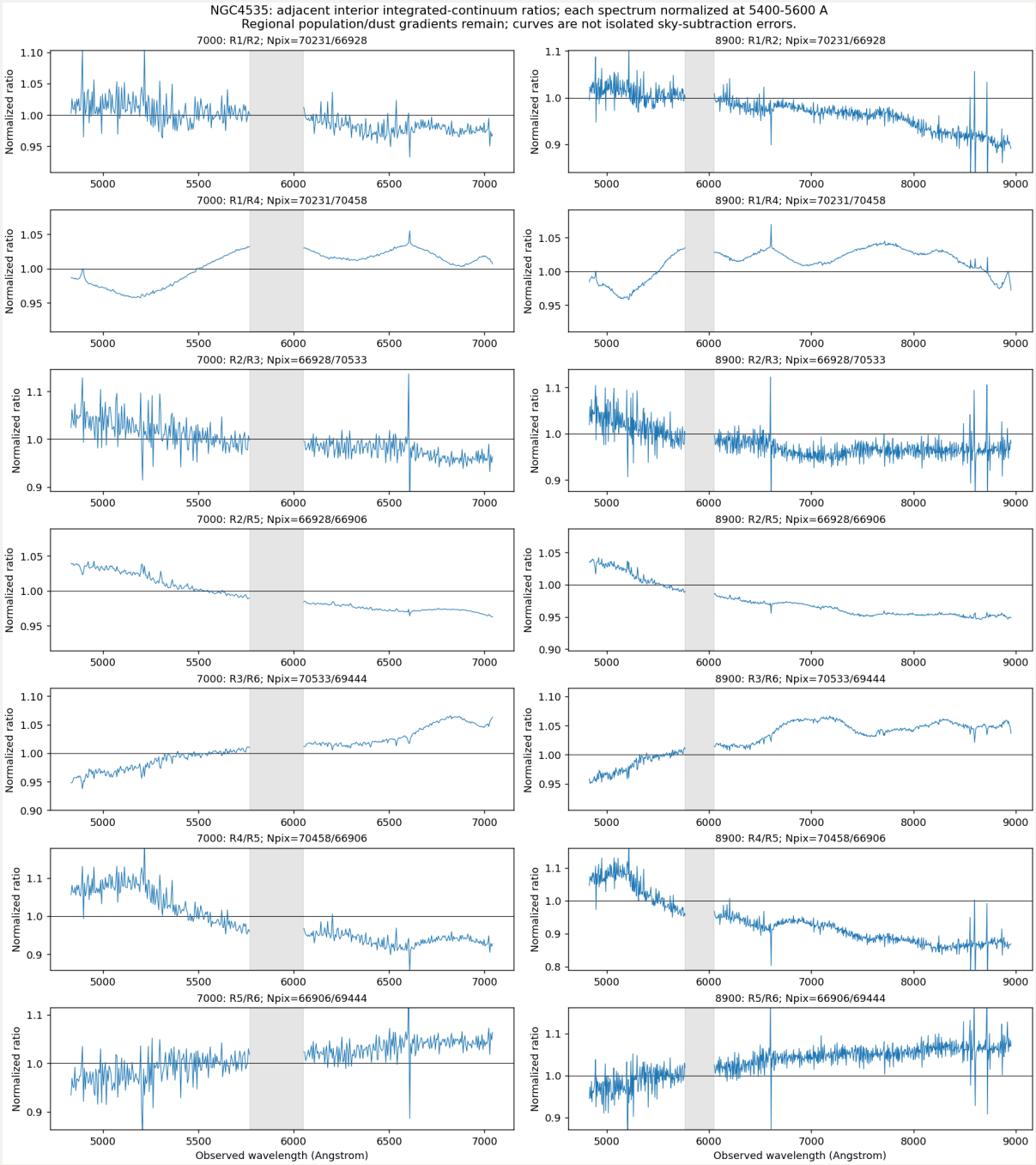


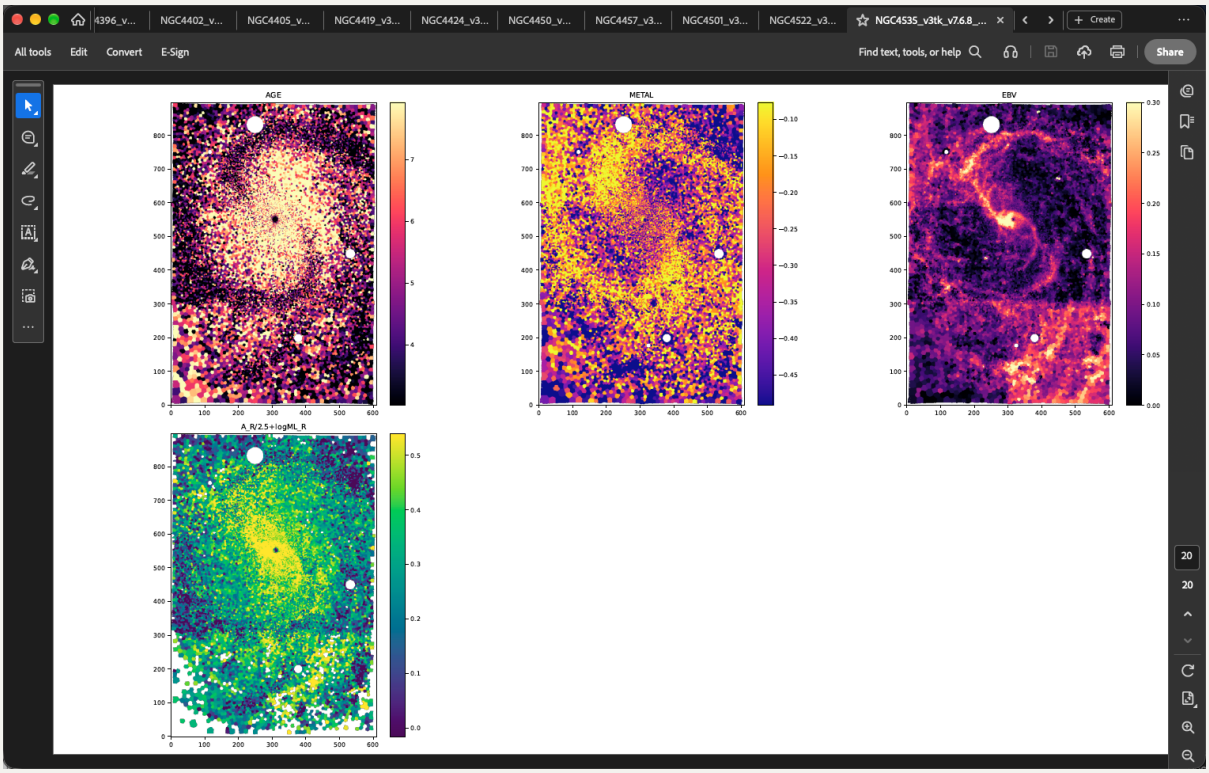




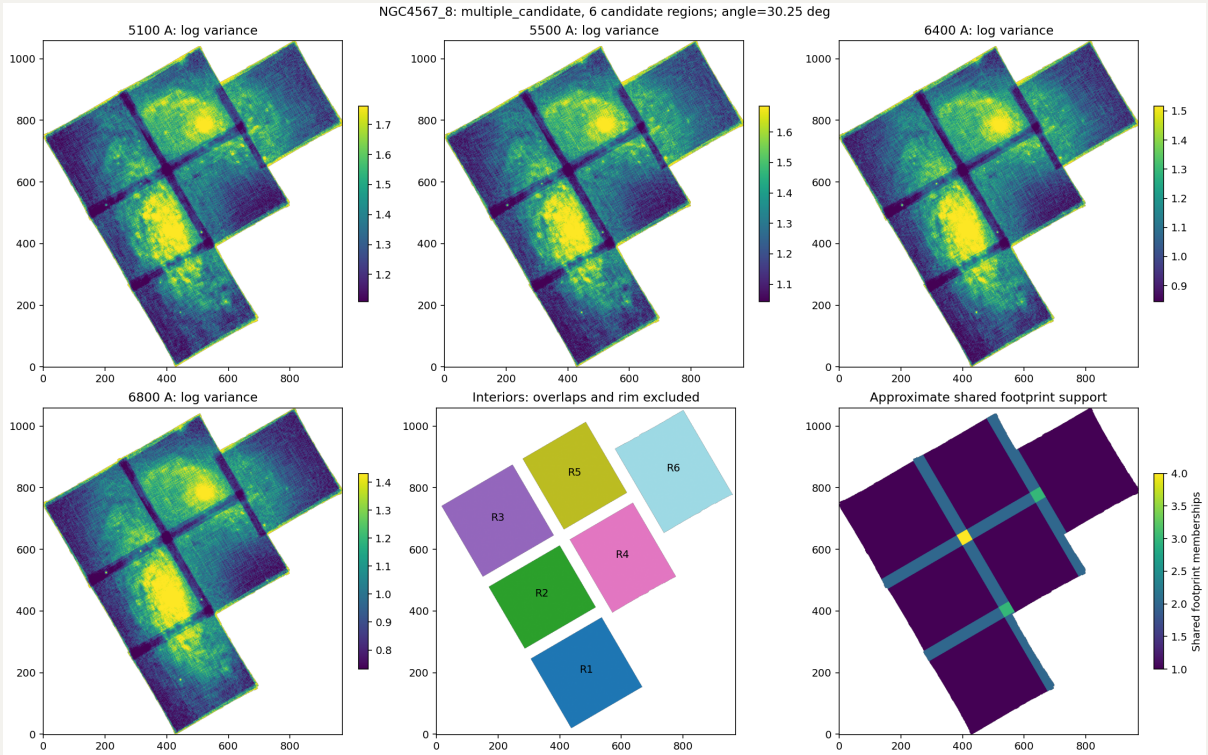
NGC4535 (needed)

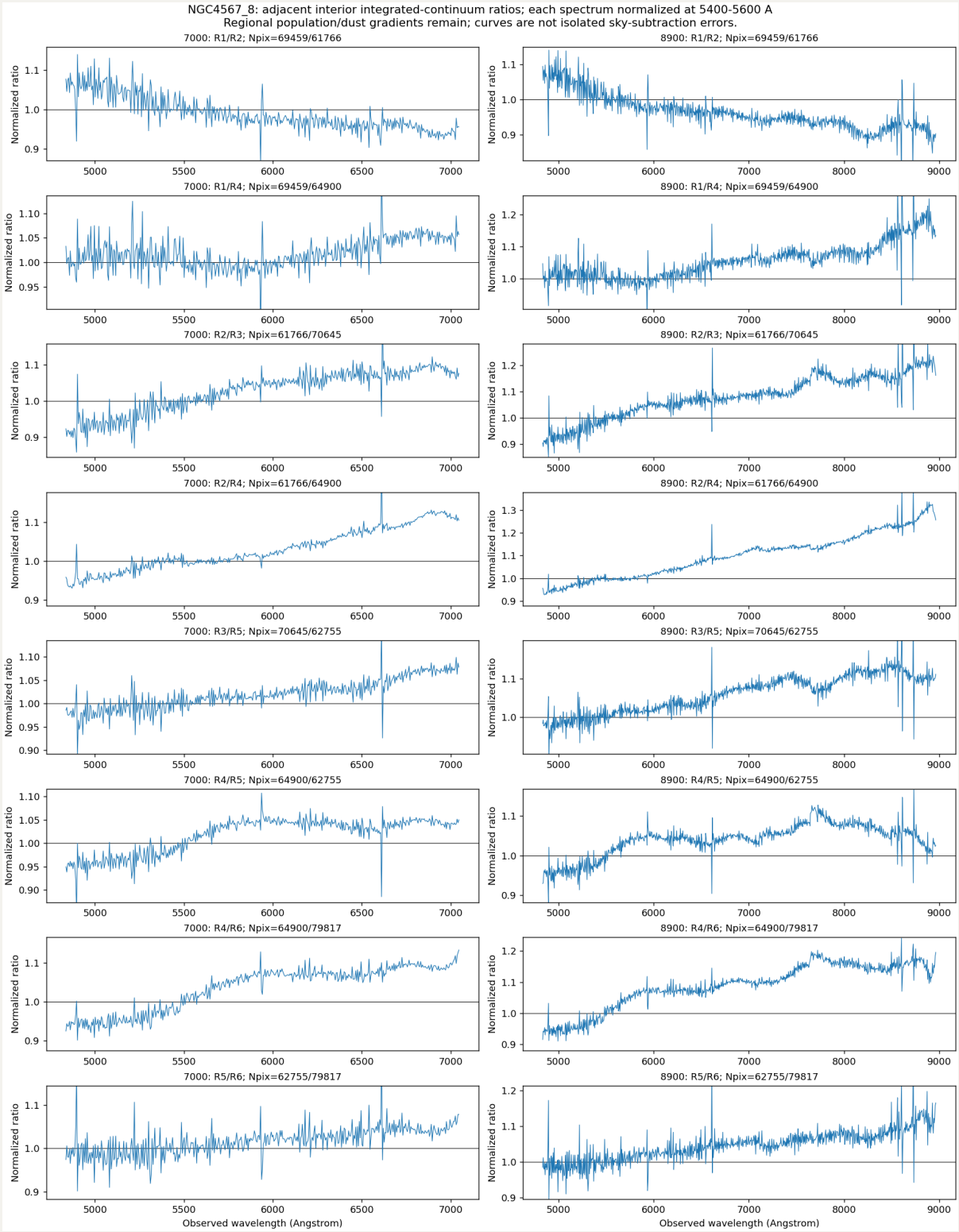

NGC4535: multiple_candidate, 6 candidate regions; angle=-0.39 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
Approximate shared footprint support
R1
R2
R3
R4
R5
R6
Shared footprint memberships

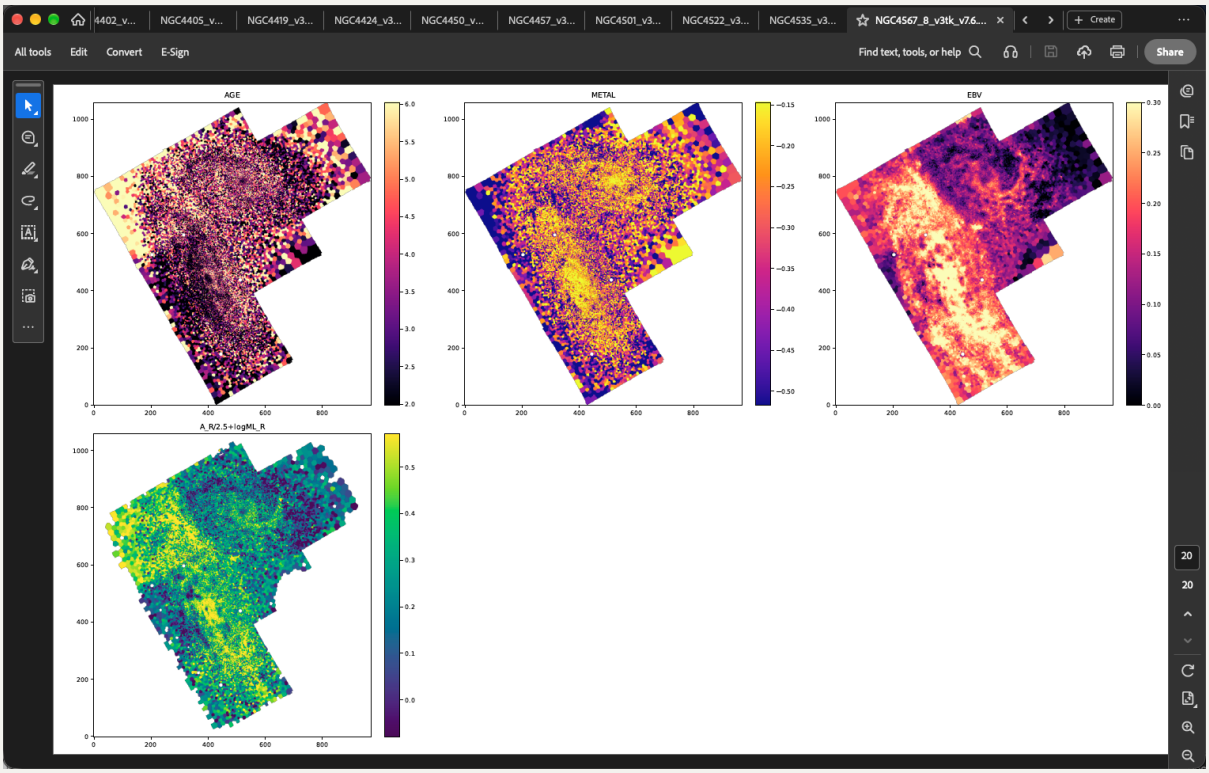

NGC4535: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=70231/66928
8900: R1/R2; Npix=70231/66928
7000: R1/R4; Npix=70231/70458
8900: R1/R4; Npix=70231/70458
7000: R2/R3; Npix=66928/70533
8900: R2/R3; Npix=66928/70533
7000: R2/R5; Npix=66928/66906
8900: R2/R5; Npix=66928/66906
7000: R3/R6; Npix=70533/69444
8900: R3/R6; Npix=70533/69444
7000: R4/R5; Npix=70458/66906
8900: R4/R5; Npix=70458/66906
7000: R5/R6; Npix=66906/69444
8900: R5/R6; Npix=66906/69444
Normalized ratio
Observed wavelength (Angstrom)



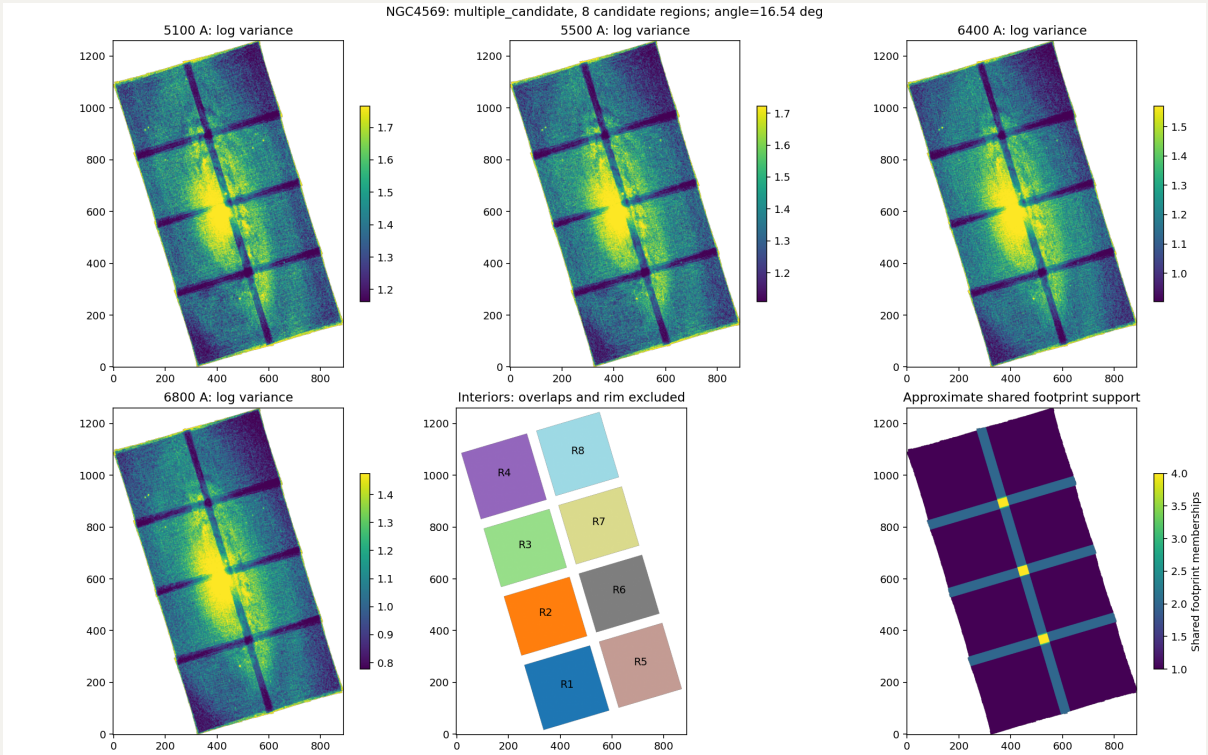
NGC4567_8 (maybe)

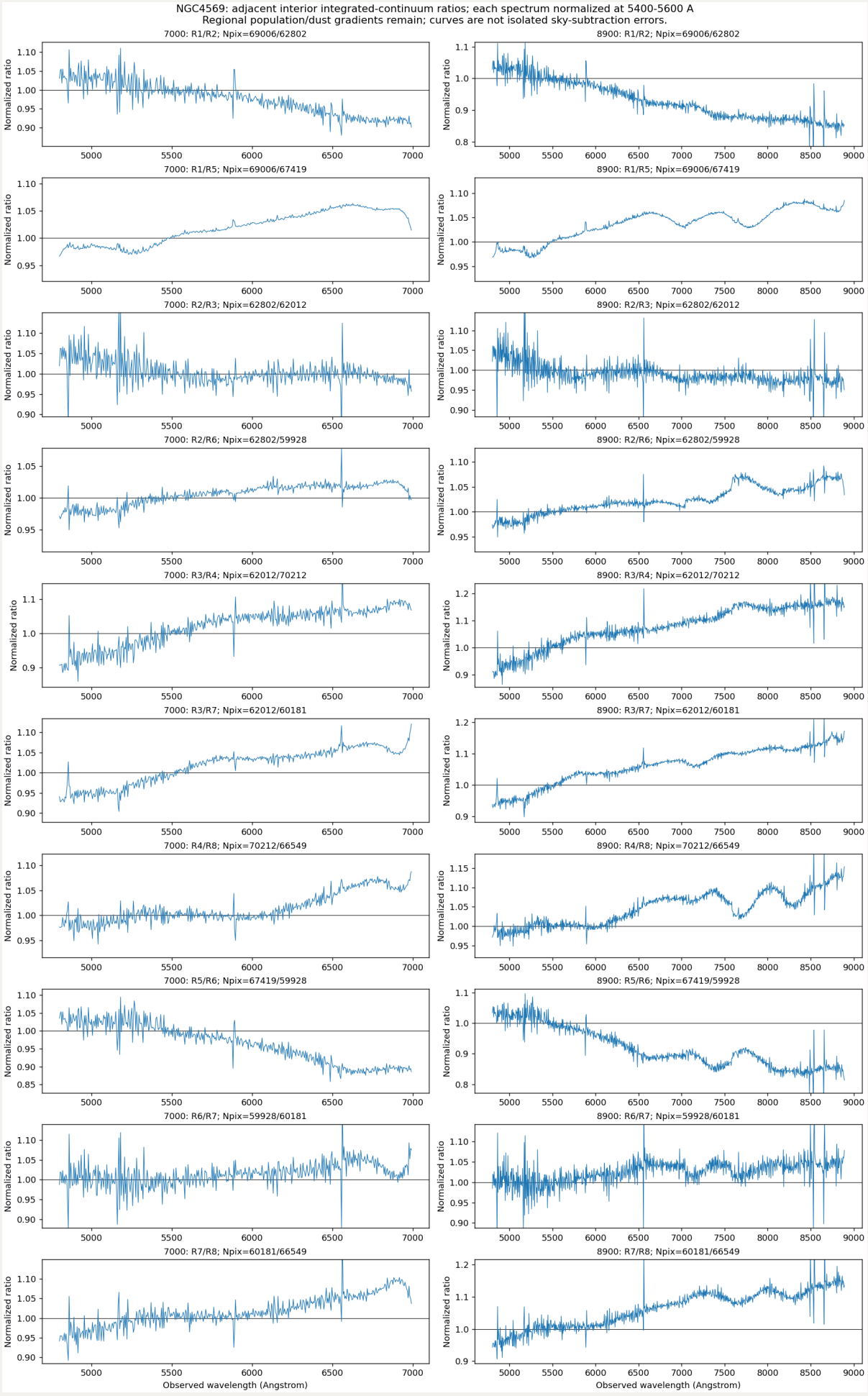


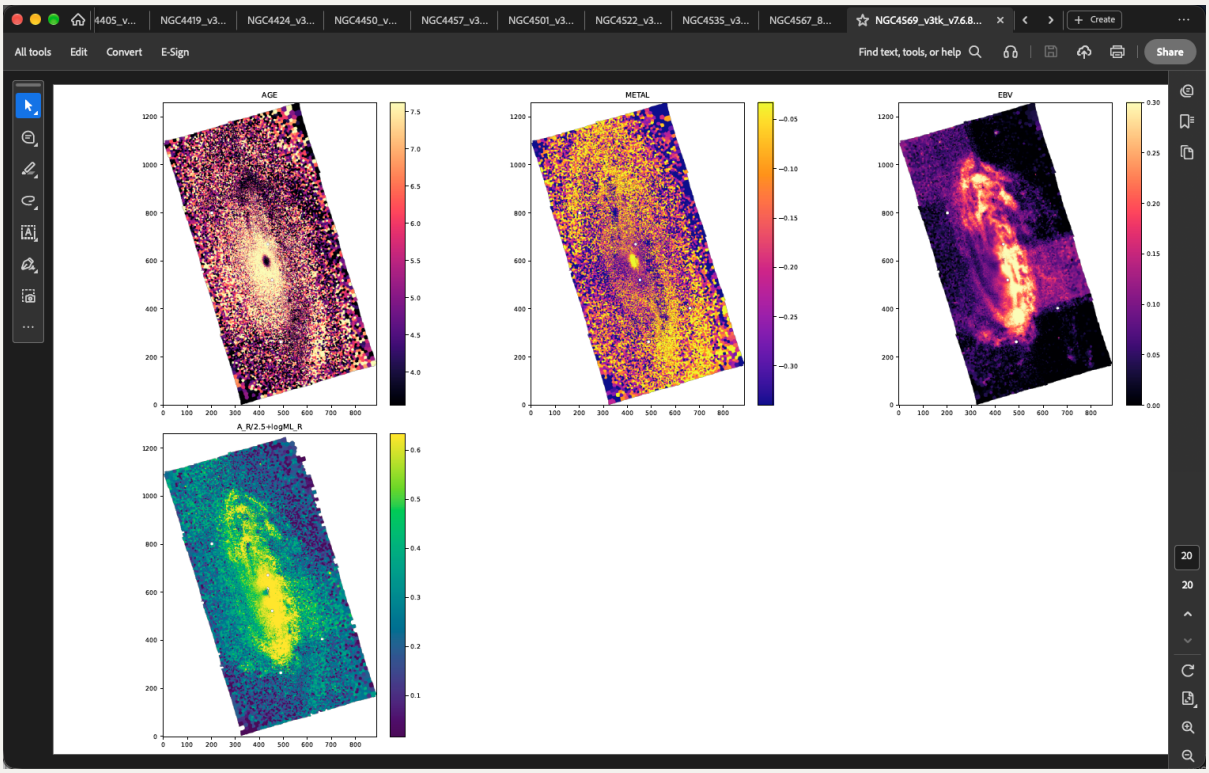

NGC4567_8: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=69459/61766
8900: R1/R2; Npix=69459/61766
7000: R1/R4; Npix=69459/64900
8900: R1/R4; Npix=69459/64900
7000: R2/R3; Npix=61766/70645
8900: R2/R3; Npix=61766/70645
7000: R2/R4; Npix=61766/64900
8900: R2/R4; Npix=61766/64900
7000: R3/R5; Npix=70645/62755
8900: R3/R5; Npix=70645/62755
7000: R4/R5; Npix=64900/62755
8900: R4/R5; Npix=64900/62755
7000: R4/R6; Npix=64900/79817
8900: R4/R6; Npix=64900/79817
7000: R5/R6; Npix=62755/79817
8900: R5/R6; Npix=62755/79817
Normalized ratio
Observed wavelength (Angstrom)



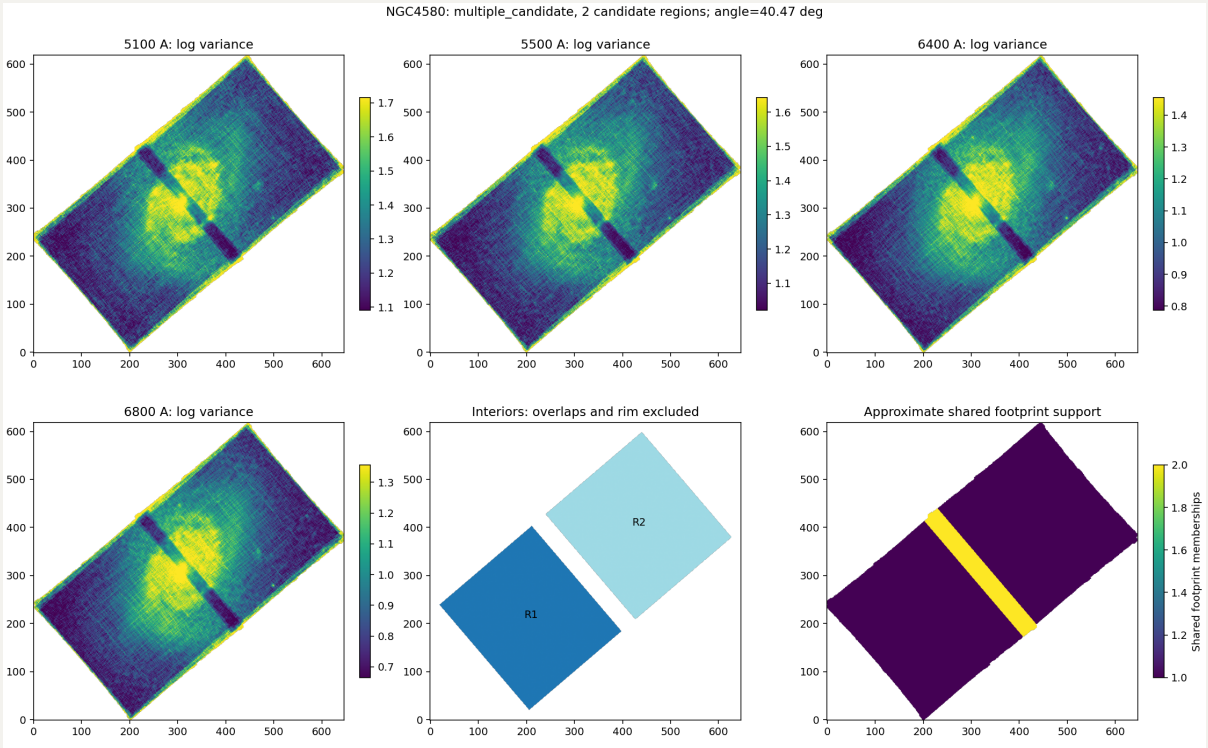
NGC4569 (maybe)

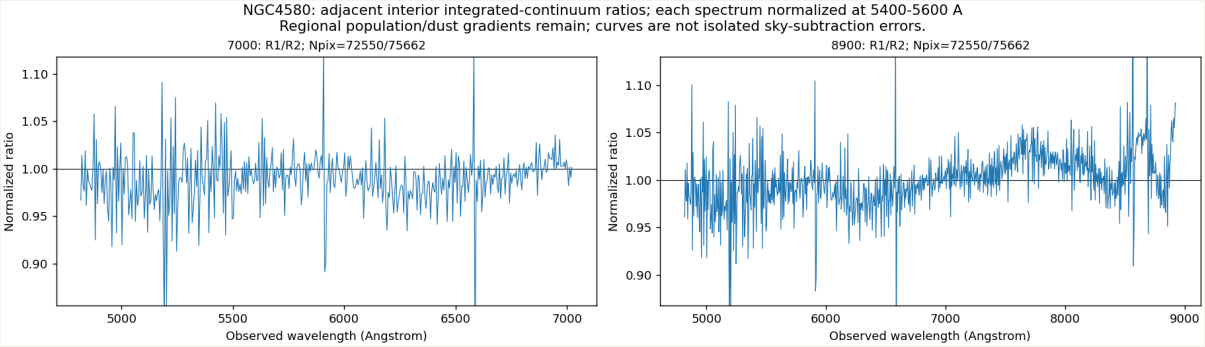


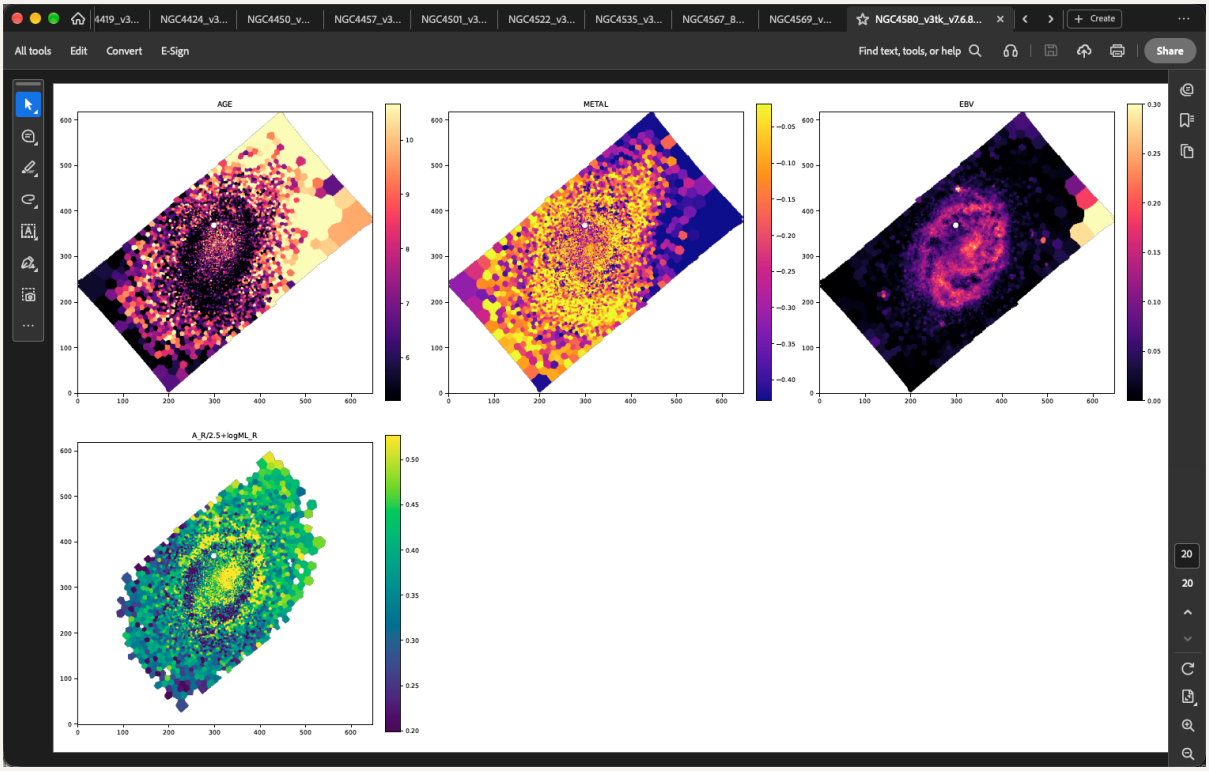

NGC4569: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=69006/62802
8900: R1/R2; Npix=69006/62802
7000: R1/R5; Npix=69006/67419
8900: R1/R5; Npix=69006/67419
7000: R2/R3; Npix=62802/62012
8900: R2/R3; Npix=62802/62012
7000: R2/R6; Npix=62802/59928
8900: R2/R6; Npix=62802/59928
7000: R3/R4; Npix=62012/70212
8900: R3/R4; Npix=62012/70212
7000: R3/R7; Npix=62012/60181
8900: R3/R7; Npix=62012/60181
7000: R4/R8; Npix=70212/66549
8900: R4/R8; Npix=70212/66549
7000: R5/R6; Npix=67419/59928
8900: R5/R6; Npix=67419/59928
7000: R6/R7; Npix=59928/60181
8900: R6/R7; Npix=59928/60181
7000: R7/R8; Npix=60181/66549
8900: R7/R8; Npix=60181/66549
Normalized ratio
Observed wavelength (Angstrom)



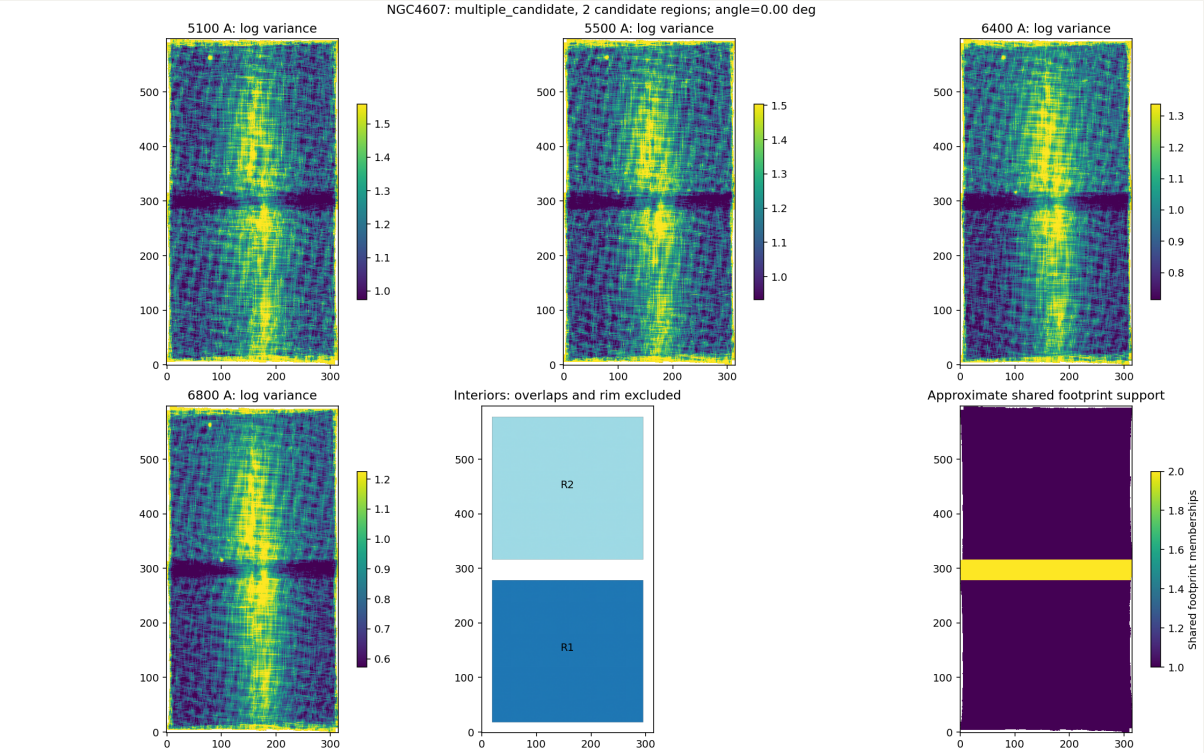
NGC4580 (fine)

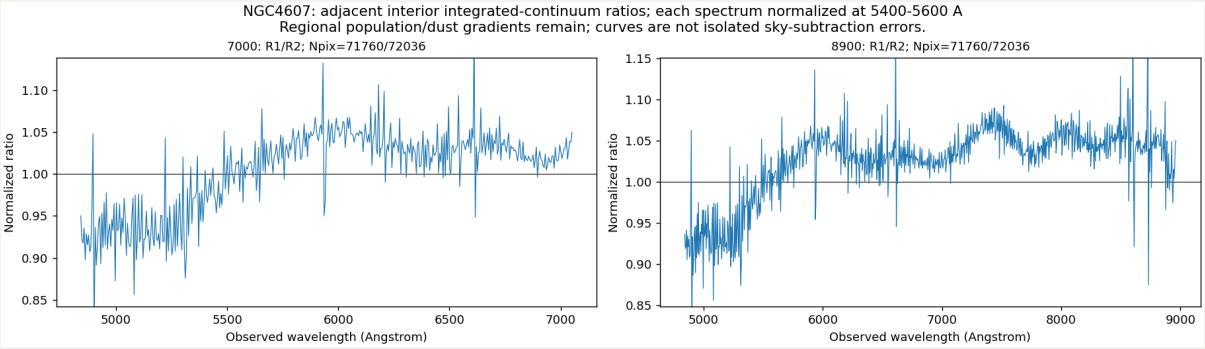


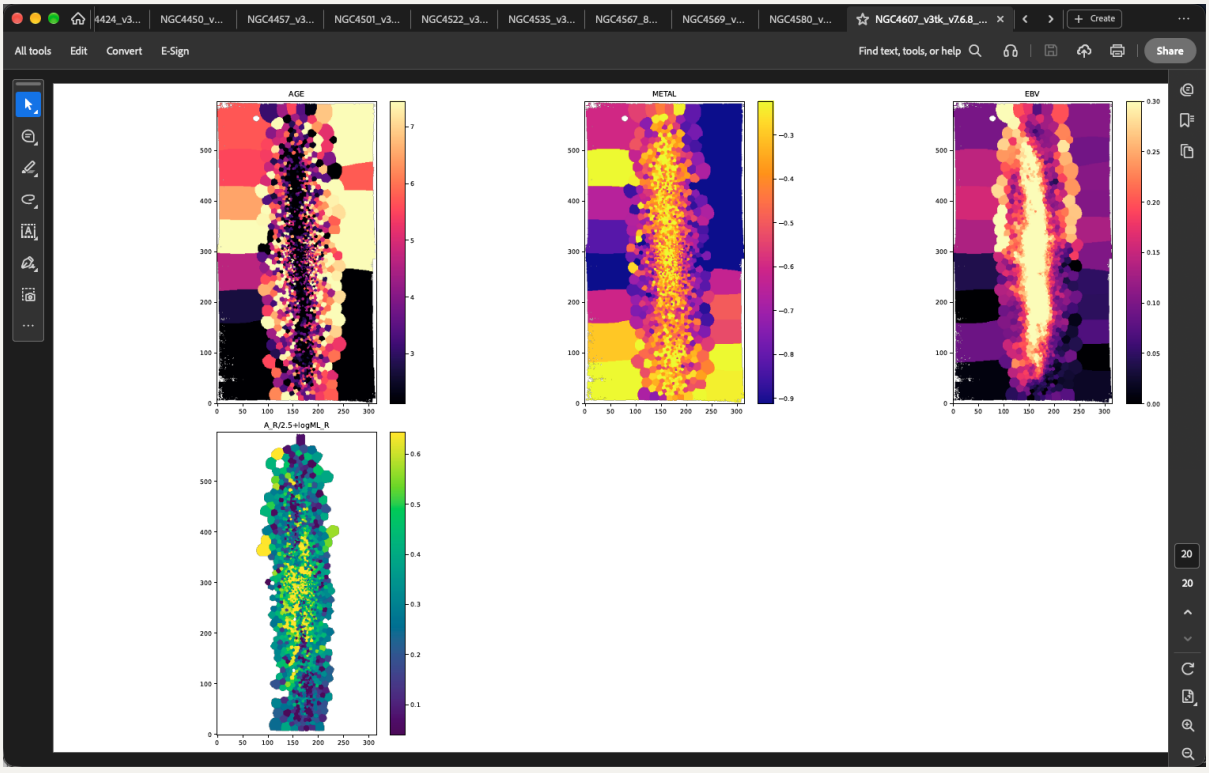




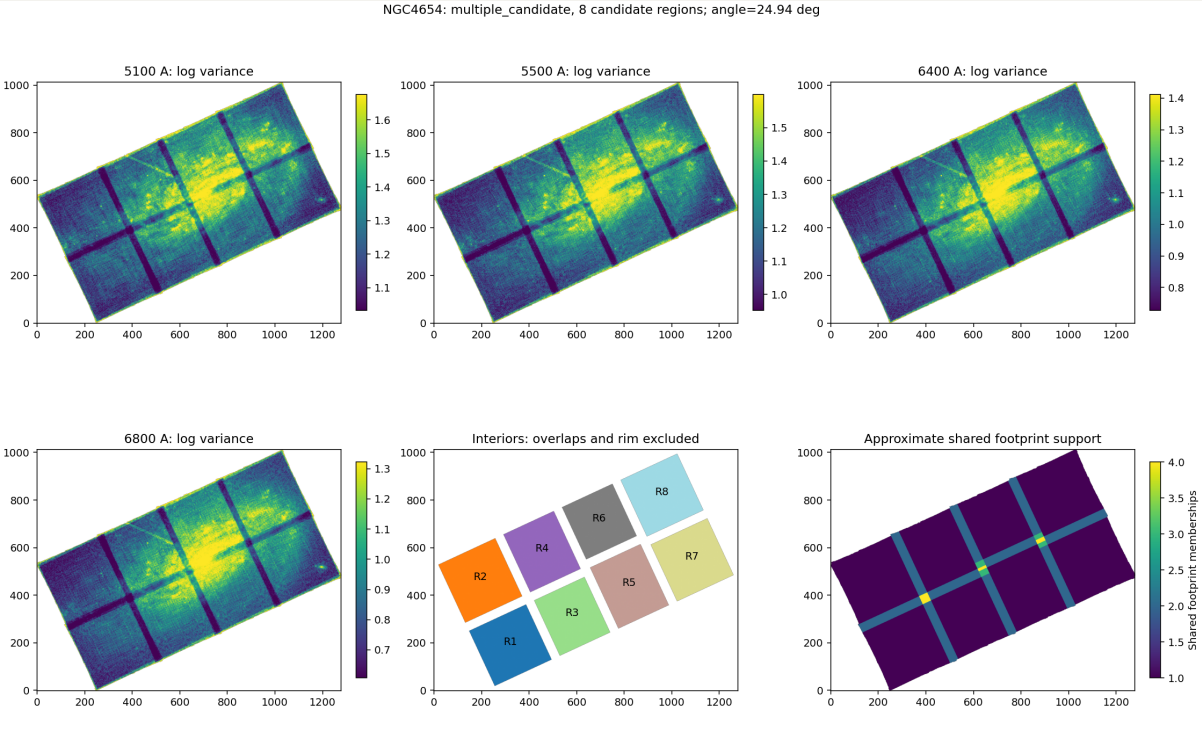
NGC4607 (fine)

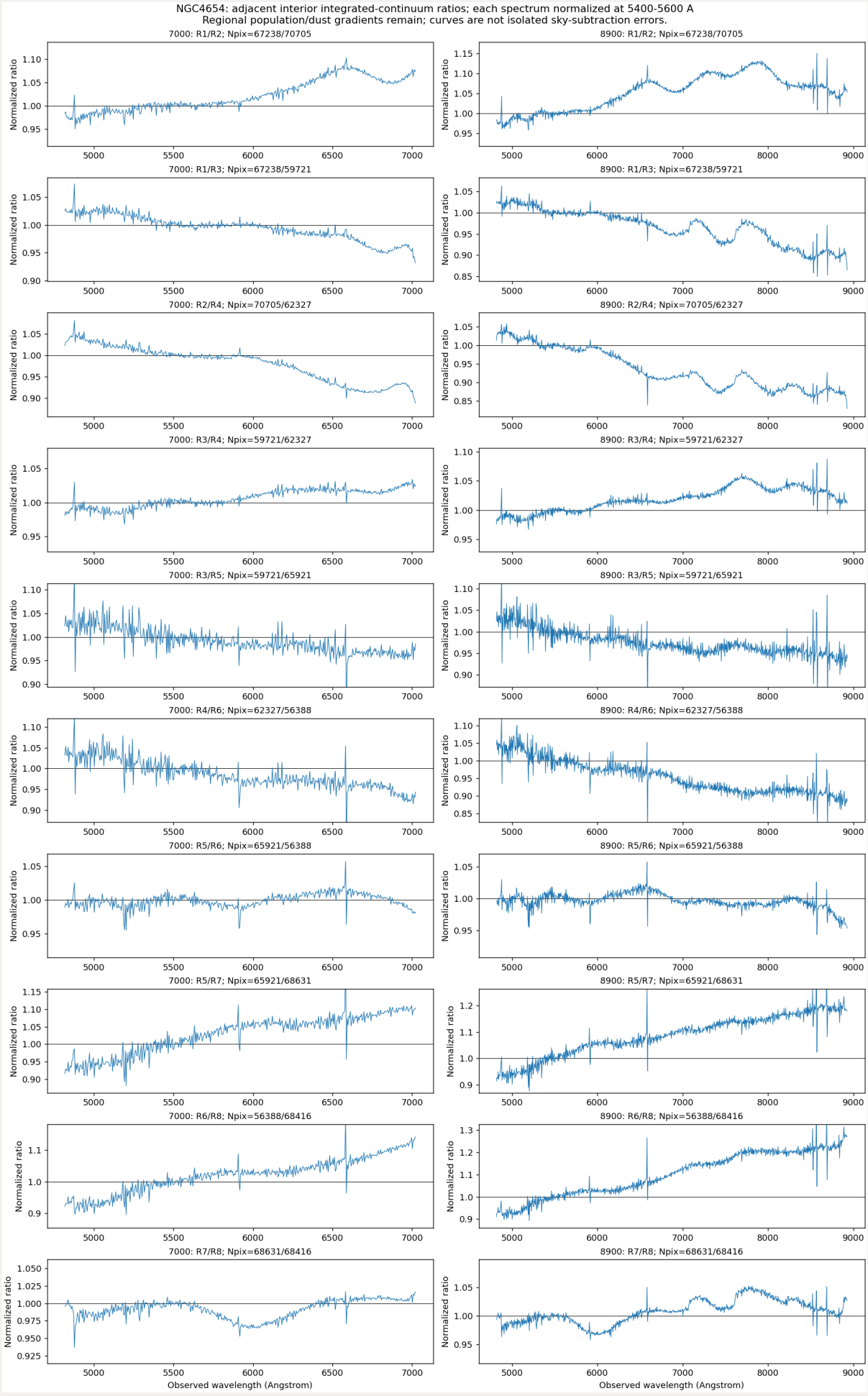

NGC4607: multiple_candidate, 2 candidate regions; angle=0.00 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R2
R1
Approximate shared footprint support
Shared footprint memberships

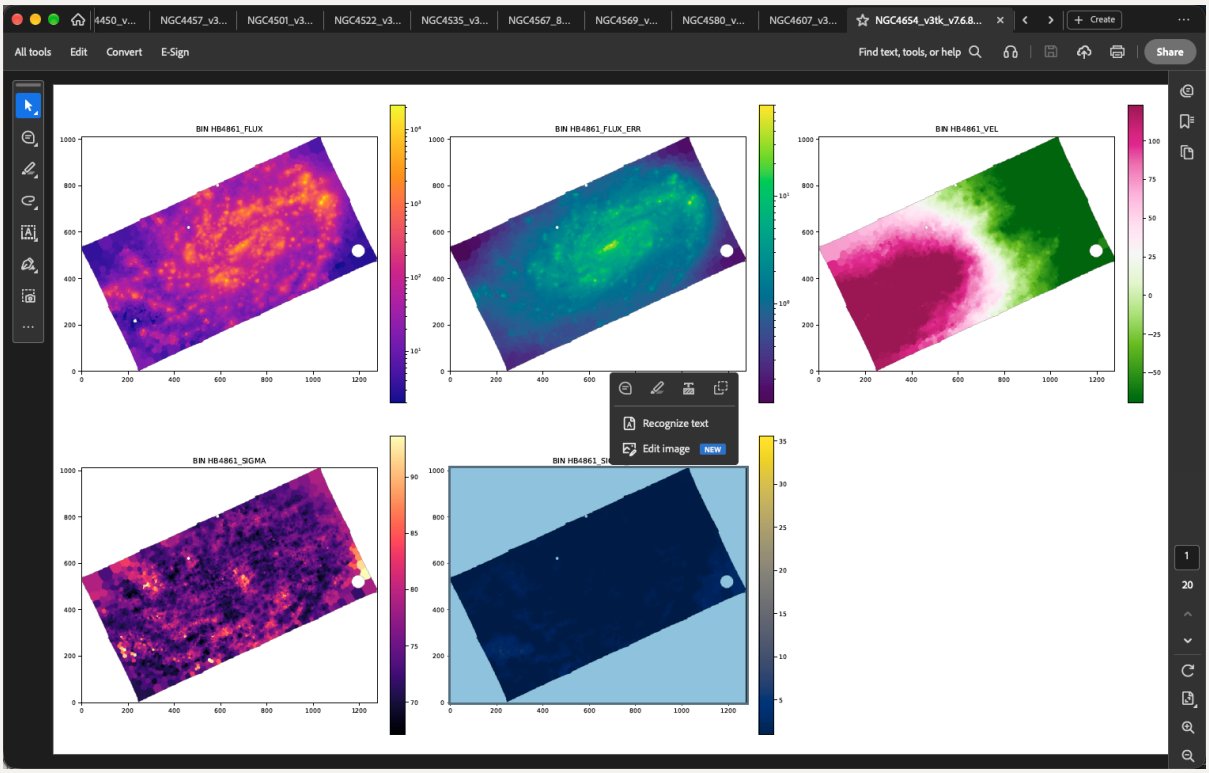

NGC4607: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=71760/72036
8900: R1/R2; Npix=71760/72036
Normalized ratio
Observed wavelength (Angstrom)


All tools
Edit
Convert
E-Sign
Share
AGE
METAL
EBV
A_R/2.5+logML_R

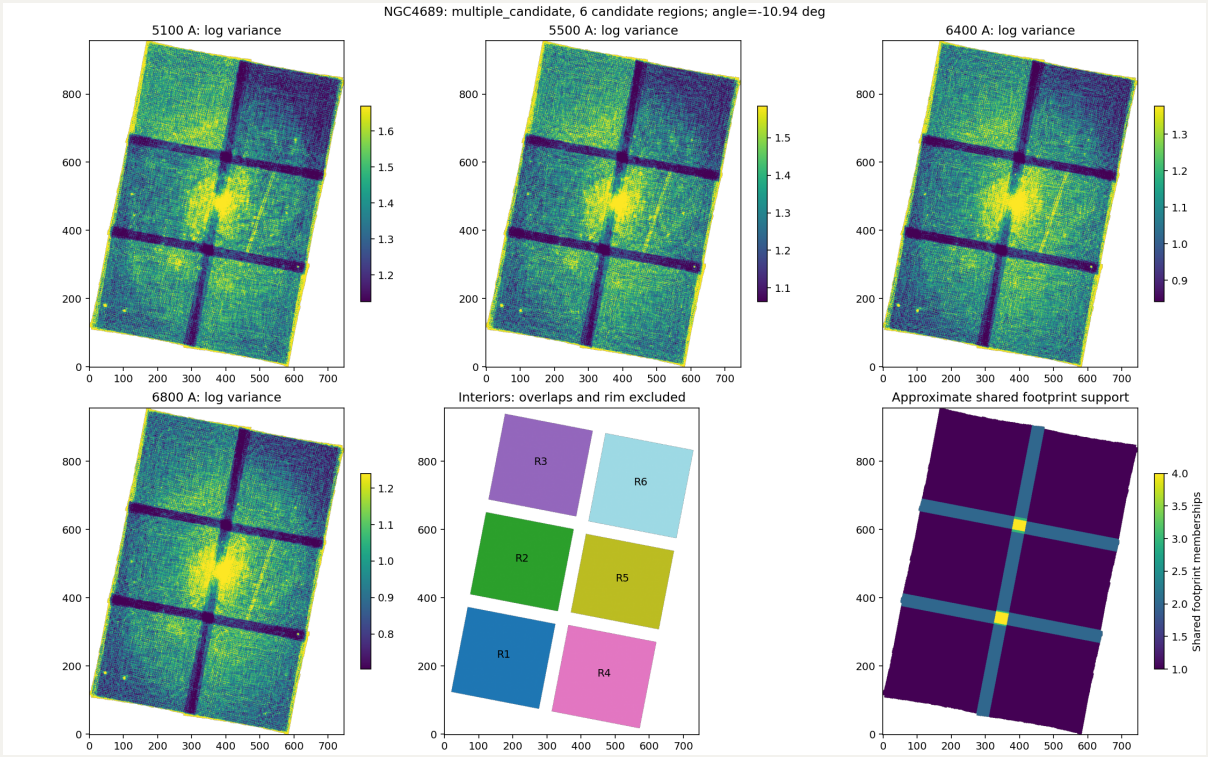
NGC4654 (needed)

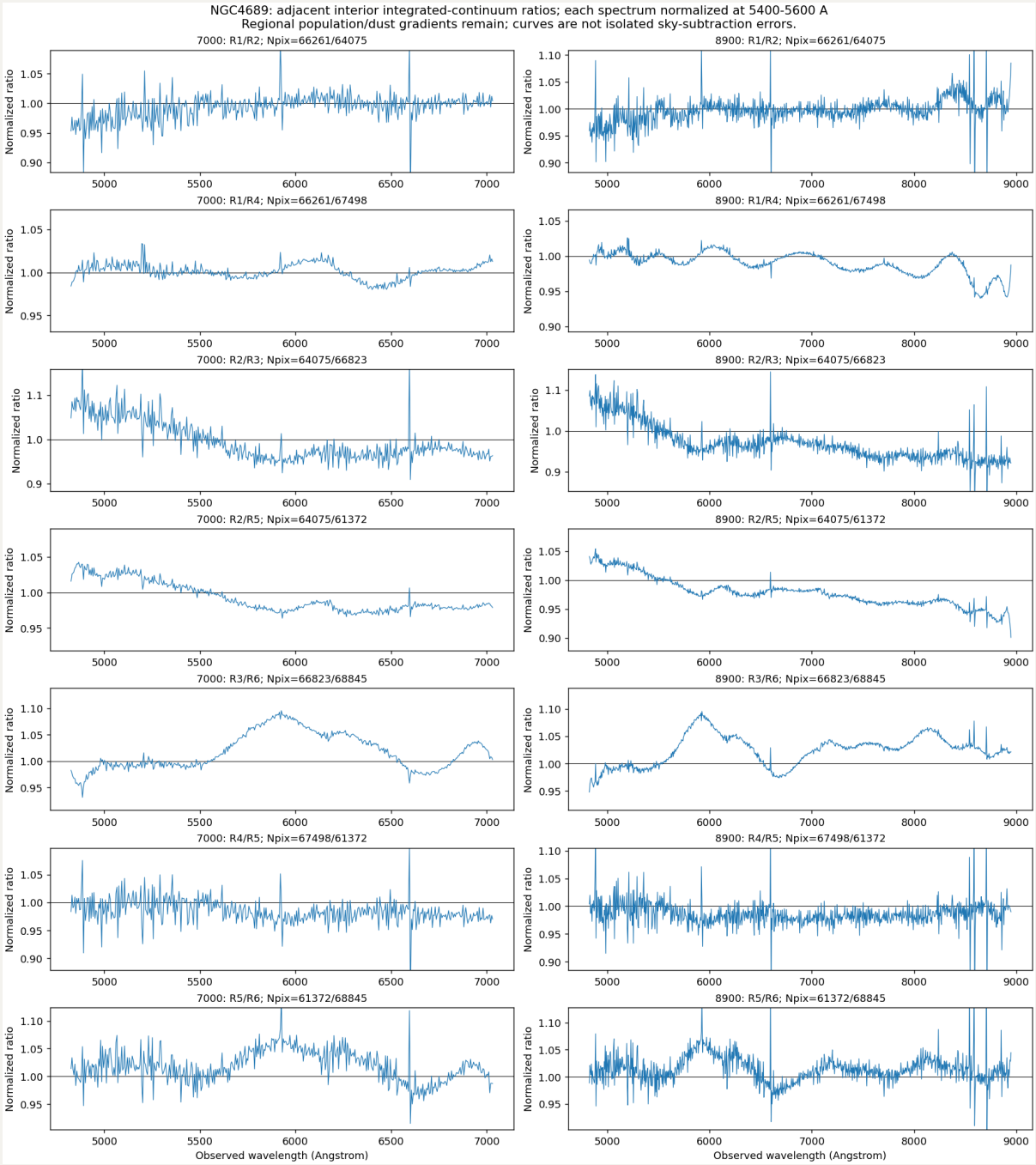


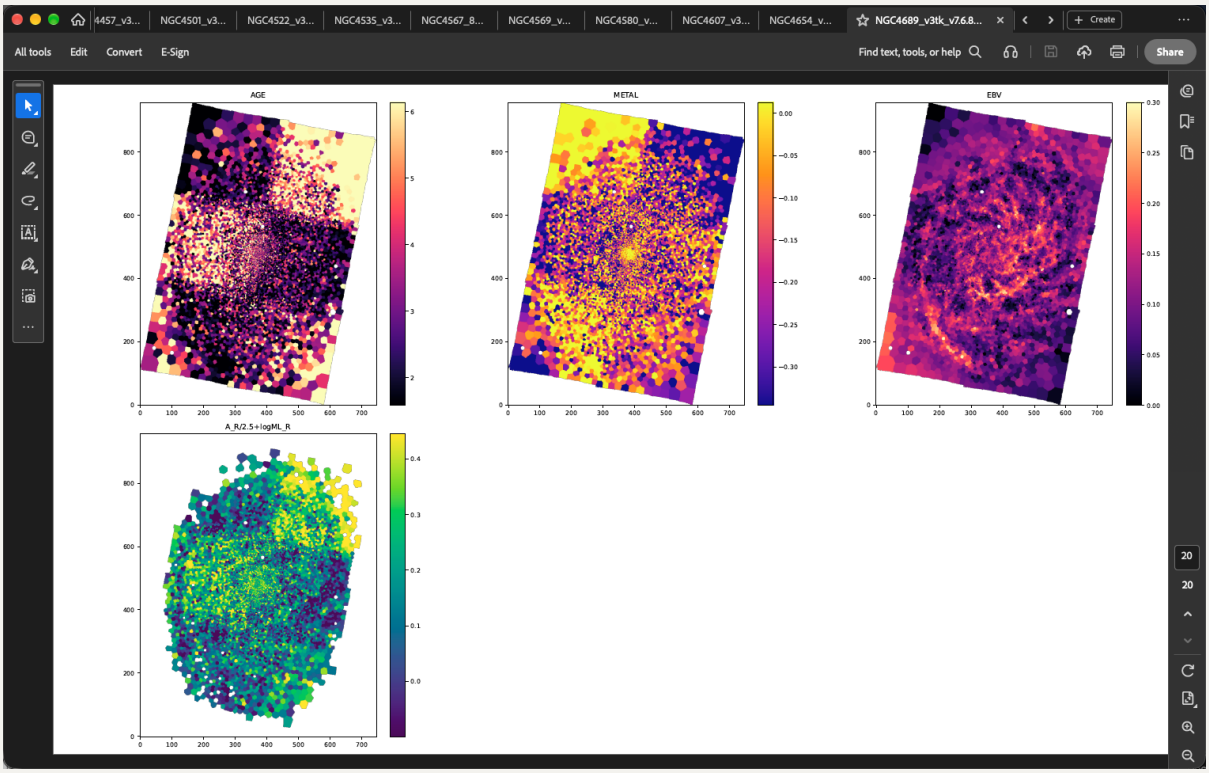

NGC4654: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=67238/70705
8900: R1/R2; Npix=67238/70705
7000: R1/R3; Npix=67238/59721
8900: R1/R3; Npix=67238/59721
7000: R2/R4; Npix=70705/62327
8900: R2/R4; Npix=70705/62327
7000: R3/R4; Npix=59721/62327
8900: R3/R4; Npix=59721/62327
7000: R3/R5; Npix=59721/65921
8900: R3/R5; Npix=59721/65921
7000: R4/R6; Npix=62327/56388
8900: R4/R6; Npix=62327/56388
7000: R5/R6; Npix=65921/56388
8900: R5/R6; Npix=65921/56388
7000: R5/R7; Npix=65921/68631
8900: R5/R7; Npix=65921/68631
7000: R6/R8; Npix=56388/68416
8900: R6/R8; Npix=56388/68416
7000: R7/R8; Npix=68631/68416
8900: R7/R8; Npix=68631/68416
Normalized ratio
Observed wavelength (Angstrom)

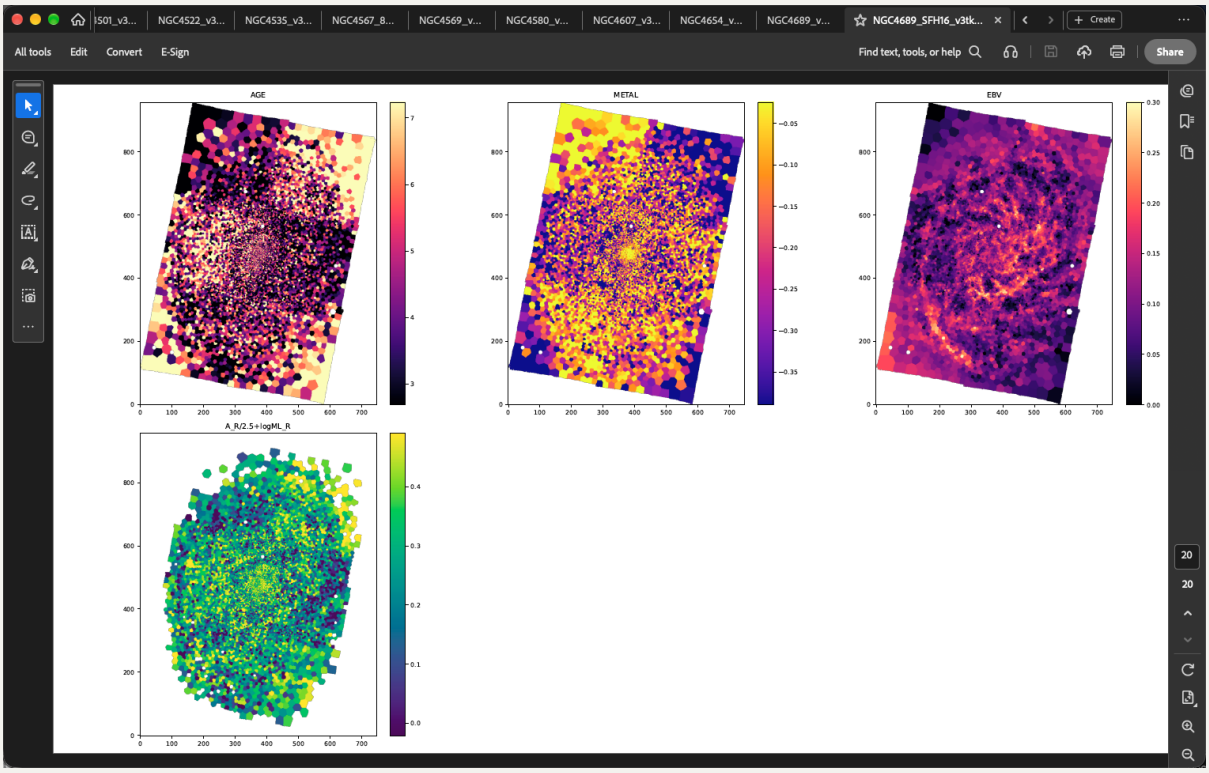

BIN HB4861_FLUX
BIN HB4861_FLUX_ERR
BIN HB4861_VEL
BIN HB4861_SIGMA
Recognize text
Edit image
NEW

NGC4689

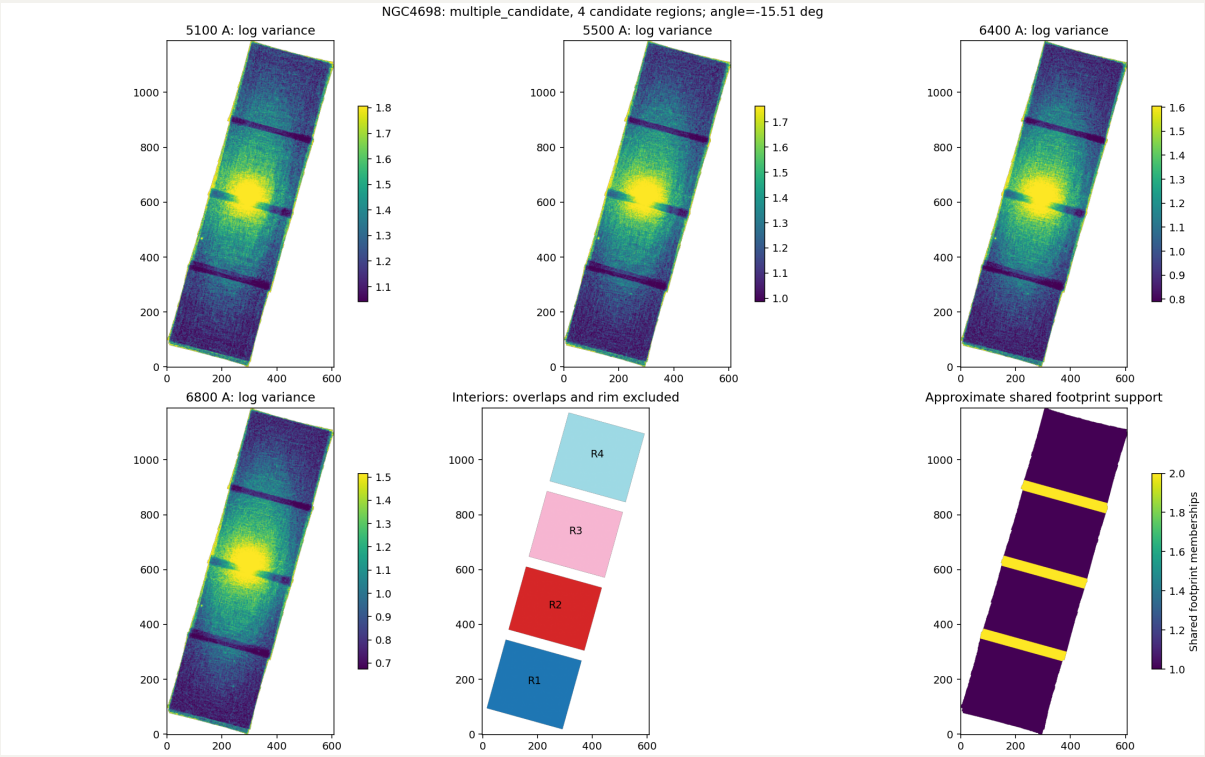

NGC4689: multiple_candidate, 6 candidate regions; angle=-10.94 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
Approximate shared footprint support
R1
R2
R3
R4
R5
R6
Shared footprint memberships

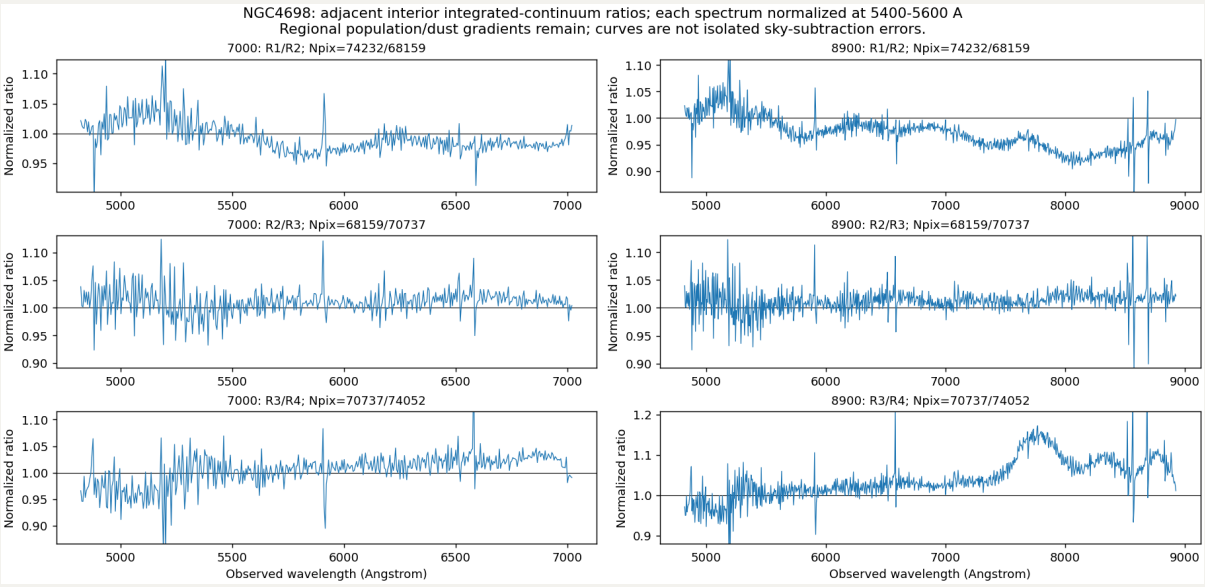

NGC4689: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=66261/64075
8900: R1/R2; Npix=66261/64075
7000: R1/R4; Npix=66261/67498
8900: R1/R4; Npix=66261/67498
7000: R2/R3; Npix=64075/66823
8900: R2/R3; Npix=64075/66823
7000: R2/R5; Npix=64075/61372
8900: R2/R5; Npix=64075/61372
7000: R3/R6; Npix=66823/68845
8900: R3/R6; Npix=66823/68845
7000: R4/R5; Npix=67498/61372
8900: R4/R5; Npix=67498/61372
7000: R5/R6; Npix=61372/68845
8900: R5/R6; Npix=61372/68845
Normalized ratio
Observed wavelength (Angstrom)

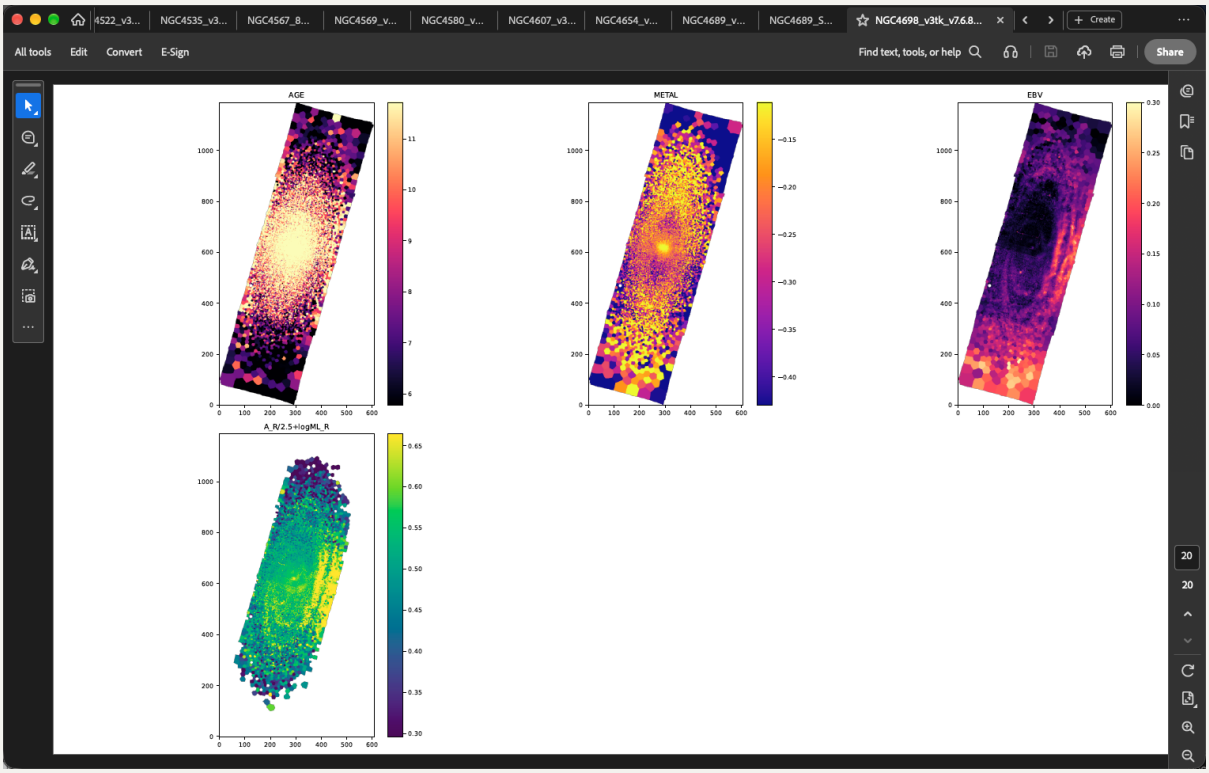




NGC4698 (maybe)


NGC4698: multiple_candidate, 4 candidate regions; angle=-15.51 deg
5100 A: log variance
5500 A: log variance
6400 A: log variance
6800 A: log variance
Interiors: overlaps and rim excluded
R4
R3
R2
R1
Approximate shared footprint support
Shared footprint memberships


NGC4698: adjacent interior integrated-continuum ratios; each spectrum normalized at 5400-5600 A
Regional population/dust gradients remain; curves are not isolated sky-subtraction errors.
7000: R1/R2; Npix=74232/68159
8900: R1/R2; Npix=74232/68159
7000: R2/R3; Npix=68159/70737
8900: R2/R3; Npix=68159/70737
7000: R3/R4; Npix=70737/74052
8900: R3/R4; Npix=70737/74052
Normalized ratio
Observed wavelength (Angstrom)


AGE
METAL
EBV
A_R/2.5+logML_R